8.0 Critical Reasoning

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Critical reasoning questions appear in the Verbal section of the GMAT® exam. The Verbal section uses multiple-choice questions to measure your ability to read and comprehend written material, to reason and to evaluate arguments, and to correct written material to conform to standard written English. Because the Verbal section includes content from a variety of topics, you may be generally familiar with some of the material; however, neither the passages nor the questions assume knowledge of the topics discussed. Critical reasoning questions are intermingled with reading comprehension and sentence correction questions throughout the Verbal section of the test.

You will have 65 minutes to complete the Verbal section, or about 1¾ minutes to answer each question. Although critical reasoning questions are based on written passages, these passages are shorter than reading comprehension passages. They tend to be less than 100 words in length and generally are followed by one or two questions. For these questions, you will see a split computer screen. The written passage will remain visible at the top of the screen. In turn, each associated question will appear either below or above the relevant passage. You will see only one question at a time.

Critical reasoning questions are designed to test the reasoning skills involved in (1) making arguments, (2) evaluating arguments, and (3) formulating or evaluating a plan of action. The materials on which questions are based are drawn from a variety of sources. The GMAT exam does not expect any familiarity with the subject matter of those materials.

In these questions, you are to analyze the situation on which each question is based, and then select the answer choice that most appropriately answers the question. Begin by reading the passages carefully, then reading the five answer choices. If the correct answer is not immediately obvious to you, see whether you can eliminate some of the wrong answers. Reading the passage a second time may be helpful in illuminating subtleties that were not immediately evident.

Answering critical reasoning questions requires no specialized knowledge of any particular field; you don't have to have knowledge of the terminology and conventions of formal logic. The sample critical reasoning questions in this chapter illustrate the variety of topics the test may cover, the kinds of questions it may ask, and the level of analysis it requires.

The following pages describe what critical reasoning questions are designed to measure and present the directions that will precede questions of this type. Sample questions and explanations of the correct answers follow.

8.1 What Is Measured

Critical reasoning questions are designed to provide one measure of your ability to reason effectively in the following areas:

- **Argument construction**
  Questions in this category may ask you to recognize such things as the basic structure of an argument, properly drawn conclusions, underlying assumptions, well-supported explanatory hypotheses, and parallels between structurally similar arguments.
- **Argument evaluation**
  These questions may ask you to analyze a given argument and to recognize such things as factors that would strengthen or weaken the given argument; reasoning errors committed in making that argument; and aspects of the method by which the argument proceeds.
- **Formulating and evaluating a plan of action**
  This type of question may ask you to recognize such things as the relative appropriateness, effectiveness, or efficiency of different plans of action, factors that would strengthen or weaken the prospects of success of a proposed plan of action, and assumptions underlying a proposed plan of action.

8.2 Test-Taking Strategies

1. **Read very carefully the set of statements on which a question is based.**

   Pay close attention to

   - what is put forward as factual information
   - what is not said but necessarily follows from what is said
   - what is claimed to follow from facts that have been put forward
   - how well substantiated are any claims that a particular conclusion follows from the facts that have been put forward

   In reading the arguments, it is important to pay attention to the logical reasoning used; the actual truth of statements portrayed as fact is not important.

2. **Identify the conclusion.**

   The conclusion does not necessarily come at the end of the text; it may come somewhere in the middle or even at the beginning. Be alert to clues in the text that an argument follows logically from another statement or statements in the text.

3. **Determine exactly what each question asks.**

   You might find it helpful to read the question first, before reading the material on which it is based; don't assume that you know what you will be asked about an argument. An argument may have obvious flaws, and one question may ask you to detect them. But another question may direct you to select the one answer choice that does NOT describe a flaw in the argument.

4. **Read all the answer choices carefully.**

   Do not assume that a given answer is the best without first reading all the choices.

8.3 The Directions

These are the directions you will see for critical reasoning questions when you take the GMAT exam. If you read them carefully and understand them clearly before going to sit for the test, you will not need to spend too much time reviewing them when you are at the test center and the test is under way.

For these questions, select the best of the answer choices given.

8.4 Practice Questions

Each of the critical reasoning questions is based on a short argument, a set of statements, or a plan of action. For each question, select the best answer of the choices given.

*CR09616

544. Neuroscientist: Memory evolved to help animals react appropriately to situations they encounter by drawing on the past experience of similar situations. But this does not require that animals perfectly recall every detail of all their experiences. Instead, to function well, memory should generalize from past experiences that are similar to the current one.

The neuroscientist's statements, if true, most strongly support which of the following conclusions?

(A) At least some animals perfectly recall every detail of at least some past experiences.

(B) Perfectly recalling every detail of all their past experiences could help at least some animals react more appropriately than they otherwise would to new situations they encounter.

(C) Generalizing from past experiences requires clear memories of most if not all the details of those experiences.

(D) Recalling every detail of all past experiences would be incompatible with any ability to generalize from those experiences.

(E) Animals can often react more appropriately than they otherwise would to situations they encounter if they draw on generalizations from past experiences of similar situations.

CR09994

545. Astronomer: Most stars are born in groups of thousands, each star in a group forming from the same parent cloud of gas. Each cloud has a unique, homogeneous chemical composition. Therefore, whenever two stars have the same chemical composition as each other, they must have originated from the same cloud of gas.

Which of the following, if true, would most strengthen the astronomer's argument?

(A) In some groups of stars, not every star originated from the same parent cloud of gas.

(B) Clouds of gas of similar or identical chemical composition may be remote from each other.

(C) Whenever a star forms, it inherits the chemical composition of its parent cloud of gas.

(D) Many stars in vastly different parts of the universe are quite similar in their chemical compositions.

(E) Astronomers can at least sometimes precisely determine whether a star has the same chemical composition as its parent cloud of gas.

CR08017

546. With employer-paid training, workers have the potential to become more productive not only in their present employment but also in any number of jobs with different employers. To increase the productivity of their workforce, many firms are planning to maintain or even increase their investments in worker training. But some training experts object that if a trained worker is hired away by another firm, the employer that paid for the training has merely subsidized a competitor. They note that such hiring has been on the rise in recent years.

Which of the following would, if true, contribute most to defeating the training experts' objection to the firms' strategy?

(A) Firms that promise opportunities for advancement to their employees get, on average, somewhat larger numbers of job applications from untrained workers than do firms that make no such promise.

*These numbers correlate with the online test bank question number. See the GMAT Official Guide Online Index in the back of this book.

(B) In many industries, employees who take continuing-education courses are more competitive in the job market.

(C) More and more educational and training institutions are offering reduced tuition fees to firms that subsidize worker training.

(D) Research shows that workers whose training is wholly or partially subsidized by their employer tend to get at least as much training as do workers who pay for all their own training.

(E) For most firms that invest in training their employees, the value added by that investment in employees who stay exceeds the value lost through other employees' leaving to work for other companies.

CR01107

547. Candle Corporation's television stations are likely to have more income from advertisers than previously. This is because advertisers prefer to reach people in the 18- to 49-year-old age group and the number of people in that group watching Candle television is increasing. Furthermore, among Candle viewers, the percentage of viewers 18 to 49 years old is increasing.

Which of the following, if true, would most strengthen the argument that Candle Corporation will receive more income from advertisers?

(A) Advertisers carefully monitor the demographic characteristics of television audiences and purchase advertising time to reach the audiences they prefer to reach.

(B) Among people over 49 years old, fewer viewers of Candle stations buy products advertised on television than do viewers of other stations.

(C) There will be increasingly more advertisements on television that are directed at viewers who are over 49 years old.

(D) Candle stations plan to show reruns of television shows during hours when other stations run shows for the first time.

(E) People 18 to 49 years old generally have less disposable income to spend than do people over 49 years old.

CR12584

548. A provincial government plans to raise the gasoline tax to give people an incentive to drive less, reducing traffic congestion in the long term. However, skeptics point out that most people in the province live in areas where cars are the only viable transportation to jobs and stores and therefore cannot greatly change their driving habits in response to higher gasoline prices.

In light of the skeptics' objection, which of the following, if true, would most logically support the prediction that the government's plan will achieve its goal of reducing traffic congestion?

(A) The revenue from the tax will be used to make public transportation a viable means of transportation to jobs and stores for far more people.

(B) The tax will encourage many residents to switch to more fuel-efficient cars, reducing air pollution and other problems.

(C) Because gasoline has been underpriced for decades, the province has many neighborhoods where cars are the only viable means of transportation.

(D) Most residents who cannot greatly change their driving habits could compensate for high gasoline prices by reducing other expenses.

(E) Traffic congestion is an especially serious problem for people for whom cars are the only viable means of transportation.

CR03940

549. Editorial: The roof of Northtown's municipal equipment-storage building collapsed under the weight of last week's heavy snowfall. The building was constructed recently and met local building-safety codes in every particular, except that the nails used for attaching roof supports to the building's columns were of a smaller size than the codes specify for this purpose. Clearly, this collapse exemplifies how even a single, apparently insignificant departure from safety standards can have severe consequences.

Which of the following, if true, most seriously weakens the editorial's argument?

(A) The only other buildings to suffer roof collapses from the weight of the snowfall were older buildings constructed according to less exacting standards than those in the codes.

(B) The amount of snow that accumulated on the roof of the equipment-storage building was greater than the predicted maximum that was used in drawing up the safety codes.

(C) Because the equipment-storage building was not intended for human occupation, some safety-code provisions that would have applied to an office building did not apply to it.

(D) The municipality of Northtown itself has the responsibility for ensuring that buildings constructed within its boundaries meet the provisions of the building-safety codes.

(E) Because the equipment-storage building was used for storing snow-removal equipment, the building was almost completely empty when the roof collapsed.

CR12078

550. Political theorist: Even with the best spies, area experts, and satellite surveillance, foreign policy assessments can still lack important information. In such circumstances intuitive judgment is vital. A national leader with such judgment can make good decisions about foreign policy even when current information is incomplete, since _____.

Which of the following, if true, most logically completes the argument?

(A) the central reason for failure in foreign policy decision making is the absence of critical information

(B) those leaders whose foreign policy decisions have been highly ranked have also been found to have good intuitive judgment

(C) both intuitive judgment and good information are required for sound decision making

(D) good foreign policy decisions often lead to improved methods of gathering information

(E) intuitive judgment can produce good decisions based on past experience, even when there are important gaps in current information

CR01295

551. During the earliest period of industrialization in Britain, steam engines were more expensive to build and operate than either windmills or water mills, the other practicable sources of power for factories. Yet despite their significant cost disadvantage, steam-powered factories were built in large numbers well before technical improvements brought their cost down. Furthermore, they were built even in regions where geographical conditions permitted the construction of wind- and water-powered factories close to major markets.

Which of the following, if true, most helps to explain the proliferation of steam-powered factories during the earliest period of industrialization in Britain?

(A) In many areas of Britain, there were fewer steam-powered factories than wind- or water-powered factories in the earliest period of industrialization.

(B) Unlike wind- or water-powered factories, steam-powered factories were fueled with coal, which sometimes had to be transported significant distances from the mine to the site of the factory.

(C) It was both difficult and expensive to convert a factory from wind power or water power to steam power.

(D) In the early period of industrialization, many goods sold in towns and cities could not be mass-produced in factories.

(E) In Britain, the number of sites where a wind- or water-powered factory could be built was insufficient to provide for all of the demand for factory-produced goods at the time.

CR03938

552. Snowmaking machines work by spraying a mist that freezes immediately on contact with cold air. Because the sudden freezing kills bacteria, QuickFreeze is planning to market a wastewater purification system

that works on the same principle. The process works only when temperatures are cold, however, so municipalities using it will still need to maintain a conventional system.

Which of the following, if true, provides the strongest grounds for a prediction that municipalities will buy QuickFreeze's purification system despite the need to maintain a conventional purification system as well?

(A) Bacteria are not the only impurities that must be removed from wastewater.

(B) Many municipalities have old wastewater purification systems that need to be replaced.

(C) Conventional wastewater purification systems have not been fully successful in killing bacteria at cold temperatures.

(D) During times of warm weather, when it is not in use, QuickFreeze's purification system requires relatively little maintenance.

(E) Places where the winters are cold rarely have a problem of water shortage.

CR05080

553. **Plant scientists have used genetic engineering on seeds to produce crop plants that are highly resistant to insect damage**. Unfortunately, the seeds themselves are quite expensive, and the plants require more fertilizer and water to grow well than normal ones. Accordingly, **for most farmers the savings on pesticides would not compensate for the higher seed costs and the cost of additional fertilizer**. However, since consumer demand for grains, fruits, and vegetables grown without the use of pesticides continues to rise, the use of genetically engineered seeds of this kind is likely to become widespread.

In the argument given, the two portions in **boldface** play which of the following roles?

(A) The first supplies a context for the argument; the second is the argument's main conclusion.

(B) The first introduces a development that the argument predicts will have a certain outcome; the second is a state of affairs that, according to the argument, contributes to bringing about that outcome.

(C) The first presents a development that the argument predicts will have a certain outcome; the second acknowledges a consideration that tends to weigh against that prediction.

(D) The first provides evidence to support a prediction that the argument seeks to defend; the second is that prediction.

(E) The first and the second each provide evidence to support the argument's main conclusion.

CR04159

554. Which of the following most logically completes the passage?

Leptin, a protein occurring naturally in the blood, appears to regulate how much fat the body carries by speeding up the metabolism and decreasing the appetite when the body has too much fat. Mice that do not naturally produce leptin have more fat than other mice, but lose fat rapidly when they are given leptin injections. Unfortunately, however, leptin cannot be used as a dietary supplement to control fat, since _____.

(A) the digestive system breaks down proteins before they can enter the bloodstream

(B) there are pharmaceuticals already available that can contribute to weight loss by speeding up the metabolism

(C) people with unusually low levels of leptin in their blood tend to have a high percentage of body fat

(D) the mice that do not naturally produce leptin were from a specially bred strain of mice

(E) mice whose bodies did produce leptin also lost some of their body fat when given leptin injections

CR05452

555. Suncorp, a new corporation with limited funds, has been clearing large sections of the tropical Amazon forest for cattle ranching. This practice continues even though greater profits can be made from rubber tapping, which does not destroy the forest, than from cattle ranching, which does destroy the forest.

Which of the following, if true, most helps to explain why Suncorp has been pursuing the less profitable of the two economic activities mentioned above?

(A) The soil of the Amazon forest is very rich in nutrients that are important in the development of grazing lands.

(B) Cattle-ranching operations that are located in tropical climates are more profitable than cattle-ranching operations that are located in cold-weather climates.

(C) In certain districts, profits made from cattle ranching are more heavily taxed than profits made from any other industry.

(D) Some of the cattle that are raised on land cleared in the Amazon are killed by wildcats.

(E) The amount of money required to begin a rubber-tapping operation is twice as high as the amount needed to begin a cattle ranch.

CR09963

556. Archaeologists use technology to analyze ancient sites. It is likely that this technology will advance considerably in the near future, allowing archaeologists to gather more information than is currently possible. If they study certain sites now, they risk contaminating or compromising them for future studies. Therefore, in order to maximize the potential for gathering knowledge in the long run, a team of archaeologists plans to delay the examination of a newly excavated site.

Which of the following would be most useful to investigate for the purpose of evaluating the plan's prospects for achieving its goal?

(A) Whether any of the contents of the site will significantly deteriorate before the anticipated technology is available

(B) Whether there will continue to be improvements on the relevant technology

(C) Whether the team can study a site other than the newly excavated site for the time being

(D) Whether the site was inhabited by a very ancient culture

(E) Whether the anticipated technology will damage objects under study

CR01102

557. More and more law firms specializing in corporate taxes are paid on a contingency-fee basis. Under this arrangement, if a case is won, the firm usually receives more than it would have received if it had been paid on the alternate hourly rate basis. If the case is lost, the firm receives nothing. Most firms are likely to make more under the contingency-fee arrangement.

Which of the following, if true, would most strengthen the prediction above?

(A) Firms that work exclusively under the hourly rate arrangement spend, on average, fewer hours on cases that are won than on cases that are lost.

(B) Some litigation can last for years before any decision is reached, and, even then, the decision may be appealed.

(C) Firms under the contingency-fee arrangement still pay their employees on an hourly basis.

(D) Since the majority of firms specialize in certain kinds of cases, they are able to assess accurately their chances of winning each potential case.

(E) Firms working under the contingency-fee arrangement take in fewer cases per year than do firms working under the hourly rate arrangement.

CR10671

558. A newly discovered painting seems to be the work of one of two seventeenth-century artists, either the northern German Johannes Drechen or the Frenchman Louis Birelle, who sometimes painted in the same style as Drechen. Analysis of the carved picture frame, which has been identified as the painting's original seventeenth-century frame, showed that it is made of wood found widely in northern Germany at the time, but rare in the part of France where Birelle lived. This shows that the painting is most likely the work of Drechen.

Which of the following is an assumption that the argument requires?

(A) The frame was made from wood local to the region where the picture was painted.

(B) Drechen is unlikely to have ever visited the home region of Birelle in France.

(C) Sometimes a painting so resembles others of its era that no expert is able to confidently decide who painted it.

(D) The painter of the picture chose the frame for the picture.

(E) The carving style of the picture frame is not typical of any specific region of Europe.

CR00766

559. Beginning in 1966 all new cars sold in Morodia were required to have safety belts and power steering. Previously, most cars in Morodia were without these features. Safety belts help to prevent injuries in collisions, and power steering helps to avoid collisions in the first place. But even though in 1966 one-seventh of the cars in Morodia were replaced with new cars, the number of car collisions and collision-related injuries did not decline.

Which of the following, if true about Morodia, most helps to explain why the number of collisions and collision-related injuries in Morodia failed to decline in 1966?

(A) Because of a driver-education campaign, most drivers and passengers in cars that did have safety belts used them in 1966.

(B) Most of the new cars bought in 1966 were bought in the months of January and February.

(C) In 1965, substantially more than one-seventh of the cars in Morodia were replaced with new cars.

(D) An excessive reliance on the new safety features led many owners of new cars to drive less cautiously in 1966 than before.

(E) The seat belts and power steering put into new cars sold in 1966 had to undergo strict quality-control inspections by manufacturers, whether the cars were manufactured in Morodia or not.

CR04882

560. Enterprise Bank currently requires customers with checking accounts to maintain a minimum balance or pay a monthly fee. Enterprise plans to offer accounts with no monthly fee and no minimum-balance requirement; to cover their projected administrative costs of $3 per account per month they plan to charge $30 for overdrawing an account. Since each month on average slightly more than 10 percent of Enterprise's customers overdraw their accounts, bank officials predict the new accounts will generate a profit.

Which of the following, if true, most strongly supports the bank officials' prediction?

(A) Some of Enterprise Bank's current checking account customers are expected to switch to the new accounts once they are offered.

(B) One third of Enterprise Bank's revenues are currently derived from monthly fees tied to checking accounts.

(C) Many checking account customers who occasionally pay a fee for not maintaining a minimum balance in their account generally maintain a balance well above the minimum.

(D) Customers whose checking accounts do not have a minimum-balance requirement are more likely than others to overdraw their checking accounts.

(E) Customers whose checking accounts do not have a minimum-balance requirement are more likely than others to write checks for small amounts.

CR05667

561. In virtually any industry, technological improvements increase labor productivity, which is the output of goods and services per person-hour worked. In Parland's industries, labor productivity is significantly higher than it is in Vergia's industries. Clearly, therefore, Parland's industries must, on the whole, be further advanced technologically than Vergia's are.

The argument is most vulnerable to which of the following criticisms?

(A) It offers a conclusion that is no more than a paraphrase of one of the pieces of information provided in its support.

(B) It presents as evidence in support of a claim information that is inconsistent with other evidence presented in support of the same claim.

(C) It takes one possible cause of a condition to be the actual cause of that condition without considering any other possible causes.

(D) It takes a condition to be the effect of something that happened only after the condition already existed.

(E) It makes a distinction that presupposes the truth of the conclusion that is to be established.

CR08471

562. Chaco Canyon, a settlement of the ancient Anasazi culture in North America, had massive buildings. **It must have been a major Anasazi center**. Analysis of wood samples shows that some of the timber for the buildings came from the Chuska and San Mateo mountains, 50 miles from Chaco Canyon. **Only a major cultural center would have the organizational power to import timber from 50 miles away**.

In the argument given, the two portions in boldface play which of the following roles?

(A) The first is a premise used to support the argument's main conclusion; the second is the argument's main conclusion.

(B) The first is the argument's main conclusion; the second is a premise used to support that conclusion.

(C) The first is one of two premises used to support the argument's main conclusion; the second is the other of those two premises.

(D) The first is a premise used to support the argument's main conclusion; the second is a premise used to support another conclusion drawn in the argument.

(E) The first is inferred from another statement in the argument; the second is inferred from the first.

CR04364

563. The Maxilux car company's design for its new luxury model, the Max 100, included a special design for the tires that was intended to complement the model's image. The winning bid for supplying these tires was submitted by Rubco. Analysts concluded that the bid would only just cover Rubco's costs on the tires, but Rubco executives claim that winning the bid will actually make a profit for the company.

Which of the following, if true, most strongly justifies the claim made by Rubco's executives?

(A) In any Maxilux model, the spare tire is exactly the same make and model as the tires that are mounted on the wheels.

(B) Rubco holds exclusive contracts to supply Maxilux with the tires for a number of other models made by Maxilux.

(C) The production facilities for the Max 100 and those for the tires to be supplied by Rubco are located very near each other.

(D) When people who have purchased a carefully designed luxury automobile need to replace a worn part of it, they almost invariably replace it with a part of exactly the same make and type.

(E) When Maxilux awarded the tire contract to Rubco, the only criterion on which Rubco's bid was clearly ahead of its competitors' bids was price.

CR05186

564. Which of the following most logically completes the passage?

Most bicycle helmets provide good protection for the top and back of the head, but little or no protection for the temple regions on the sides of the head. A study of head injuries resulting from bicycle accidents showed that a large proportion were caused by blows to the temple area. Therefore, if bicycle helmets protected this area, the risk of serious head injury in bicycle accidents would be greatly reduced, especially since _____.

(A) among the bicyclists included in the study's sample of head injuries, only a very small proportion had been wearing a helmet at the time of their accident

(B) even those bicyclists who regularly wear helmets have a poor understanding of the degree and kind of protection that helmets afford

(C) a helmet that included protection for the temples would have to be somewhat larger and heavier than current helmets

(D) the bone in the temple area is relatively thin and impacts in that area are thus very likely to cause brain injury

(E) bicyclists generally land on their arm or shoulder when they fall to the side, which reduces the likelihood of severe impacts on the side of the head

CR01867

565. Which of the following most logically completes the argument?

In a typical year, Innovair's airplanes are involved in 35 collisions while parked or being towed in airports, with a resulting yearly cost of $1,000,000 for repairs.

To reduce the frequency of ground collisions, Innovair will begin giving its ground crews additional training, at an annual cost of $500,000. Although this will cut the number of ground collisions by about half at best, the drop in repair costs can be expected to be much greater, since _____.

(A) most ground collisions happen when ground crews are rushing to minimize the time a delayed airplane spends on the ground

(B) a ground collision typically occurs when there are no passengers on the airplane

(C) the additional training will focus on helping ground crews avoid those kinds of ground collisions that cause the most costly damage

(D) the $500,000 cost figure for the additional training of ground crews includes the wages that those crews will earn during the time spent in actual training

(E) most ground collisions have been caused by the least experienced ground-crew members

CR12558

566. Many agriculturally intensive areas of the world are beginning to encounter water scarcity problems. As a result, many farmers in these areas are likely to reduce their output as the water supply they need in order to maintain production shrinks. However, one group of farmers in such a region plans to increase their production by implementing techniques for water conservation.

Which of the following, if true, would most strongly support the prediction that the group's plan will succeed?

(A) Farmers that can gain a larger share of the food market in their regions will be better positioned to control more water resources.

(B) Most agricultural practices in areas with water shortages are water-intensive.

(C) Other regions of the world not facing water shortages are likely to make up for the reduction in agricultural output.

(D) Demand for agricultural products in the group's region is not expected to decline.

(E) More than half the water used for agriculture in the farmers' region is lost to evaporation or leakage from irrigation channels.

CR03367

567. Hollywood restaurant is replacing some of its standard tables with tall tables and stools. The restaurant already fills every available seat during its operating hours, and the change in seating arrangements will not result in an increase in the restaurant's seating capacity. Nonetheless, the restaurant's management expects revenue to increase as a result of the seating change without any concurrent change in menu, prices, or operating hours.

Which of the following, if true, provides the best reason for the expectation?

(A) One of the taller tables takes up less floor space than one of the standard tables.

(B) Diners seated on stools typically do not linger over dinner as long as diners seated at standard tables.

(C) Since the restaurant will replace only some of its standard tables, it can continue to accommodate customers who do not care for the taller tables.

(D) Few diners are likely to avoid the restaurant because of the new seating arrangement.

(E) The standard tables being replaced by tall tables would otherwise have to be replaced with new standard tables at a greater expense.

CR07660

568. A major network news organization experienced a drop in viewership in the week following the airing of a controversial report on the economy. The network also received a very large number of complaints regarding the report. The network, however, maintains that negative reactions to the report had nothing to do with its loss of viewers.

Which of the following, if true, most strongly supports the network's position?

(A) The other major network news organizations reported similar reductions in viewership during the same week.

(B) The viewers who registered complaints with the network were regular viewers of the news organization's programs.

(C) Major network news organizations publicly attribute drops in viewership to their own reports only when they receive complaints about those reports.

(D) This was not the first time that this network news organization has aired a controversial report on the economy that has inspired viewers to complain to the network.

(E) Most network news viewers rely on network news broadcasts as their primary source of information regarding the economy.

CR04366

569. Only a reduction of 10 percent in the number of scheduled flights using Greentown's airport will allow the delays that are so common there to be avoided. Hevelia airstrip, 40 miles away, would, if upgraded and expanded, be an attractive alternative for fully 20 percent of the passengers using Greentown airport. Nevertheless, experts reject the claim that turning Hevelia into a full-service airport would end the chronic delays at Greentown.

Which of the following, if true, most helps to justify the experts' position?

(A) Turning Hevelia into a full-service airport would require not only substantial construction at the airport itself, but also the construction of new access highways.

(B) A second largely undeveloped airstrip close to Greentown airport would be a more attractive alternative than Hevelia for many passengers who now use Greentown.

(C) Hevelia airstrip lies in a relatively undeveloped area but would, if it became a full-service airport, be a magnet for commercial and residential development.

(D) If an airplane has to wait to land, the extra jet fuel required adds significantly to the airline's costs.

(E) Several airlines use Greentown as a regional hub, so that most flights landing at Greentown have many passengers who then take different flights to reach their final destinations.

CR07712

570. Farmer: Worldwide, just three grain crops—rice, wheat, and corn—account for most human caloric intake. To maintain this level of caloric intake and also keep pace with global population growth, yields per acre from each of these crops will have to increase at least 1.5 percent every year, given that the supply of cultivated land is diminishing. Therefore, the government should increase funding for research into new ways to improve yields.

Which of the following is an assumption on which the farmer's argument depends?

(A) It is solely the government's responsibility to ensure that the amount of rice, wheat, and corn produced worldwide keeps pace with global population growth.

(B) Increasing government funding for research into new ways to improve the yields per acre of rice, wheat, and corn crops would help to increase total worldwide annual production of food from these crops.

(C) Increasing the yields per acre of rice, wheat, and corn is more important than increasing the yields per acre of other crops.

(D) Current levels of funding for research into ways of improving grain crop yields per acre have enabled grain crop yields per acre to increase by more than 1.5 percent per year worldwide.

(E) In coming decades, rice, wheat, and corn will become a minor part of human caloric intake, unless there is government-funded research to increase their yields per acre.

CR08770

571. The air quality board recently informed Coffee Roast, a small coffee roasting firm, of a complaint regarding the smoke from its roaster. Recently enacted air quality regulations require machines roasting more than 10 pounds of coffee to be equipped with expensive smoke-dissipating afterburners. The firm, however, roasts only 8 pounds of coffee at a time. Nevertheless, the company has decided to purchase and install an afterburner.

Which of the following, if true, most strongly supports the firm's decision?

(A) Until settling on the new air quality regulations, the board had debated whether to require afterburners for machines roasting more than 5 pounds of coffee at a time.

(B) Coffee roasted in a machine equipped with an afterburner has its flavor subtly altered.

(C) The cost to the firm of an afterburner is less than the cost of replacing its roaster with a smaller one.

(D) Fewer complaints are reported in areas that maintain strict rules regarding afterburners.

(E) The firm has reason to fear that negative publicity regarding the complaints could result in lost sales.

CR03695

572. People who do regular volunteer work tend to live longer, on average, than people who do not. It has been found that "doing good," a category that certainly includes volunteer work, releases endorphins, the brain's natural opiates, which induce in people a feeling of well-being. Clearly, there is a connection: Regular releases of endorphins must in some way help to extend people's lives.

Which of the following, if true, most seriously undermines the force of the evidence given as support for the hypothesis that endorphins promote longevity?

(A) People who do regular volunteer work are only somewhat more likely than others to characterize the work they do for a living as "doing good."

(B) Although extremely high levels of endorphins could be harmful to health, such levels are never reached as a result of the natural release of endorphins.

(C) There are many people who have done some volunteer work but who do not do such work regularly.

(D) People tend not to become involved in regular volunteer work unless they are healthy and energetic to begin with.

(E) Releases of endorphins are responsible for the sense of well-being experienced by many long-distance runners while running.

CR04140

573. A study compared a sample of Swedish people older than 75 who needed in-home assistance with a similar sample of Israeli people. The people in the two samples received both informal assistance, provided by family and friends, and formal assistance, professionally provided. Although Sweden and Israel have equally well-funded and comprehensive systems for providing formal assistance, the study found that the people in the Swedish sample received more formal assistance, on average, than those in the Israeli sample.

Which of the following, if true, does most to explain the difference that the study found?

(A) A companion study found that among children needing special in-home care, the amount of formal assistance they received was roughly the same in Sweden as in Israel.

(B) More Swedish than Israeli people older than 75 live in rural areas where formal assistance services are sparse or nonexistent.

(C) Although in both Sweden and Israel much of the funding for formal assistance ultimately comes from the central government, the local structures through which assistance is delivered are different in the two countries.

(D) In recent decades, the increase in life expectancy of someone who is 75 years old has been greater in Israel than in Sweden.

(E) In Israel, people older than 75 tend to live with their children, whereas in Sweden people of that age tend to live alone.

CR05077

574. Film Director: It is true that certain characters and plot twists in my newly released film *The Big Heist* are similar to characters and plot twists in *Thieves*, a movie that came out last year. Pointing to these similarities, the film studio that produced *Thieves* is now accusing me of taking ideas from that film. The accusation is clearly without merit. All production work on *The Big Heist* was actually completed months before *Thieves* was released.

Which of the following, if true, provides the strongest support for the director's position?

(A) Before *Thieves* began production, its script had been circulating for several years among various film studios, including the studio that produced *The Big Heist*.

(B) The characters and plot twists that are most similar in the two films have close parallels in many earlier films of the same genre.

(C) The film studio that produced *Thieves* seldom produces films in this genre.

(D) The director of *Thieves* worked with the director of *The Big Heist* on several earlier projects.

(E) Production work on *Thieves* began before production work on *The Big Heist* was started.

CR05412

575. In Mernia commercial fossil hunters often sell important fossils they have found, not to universities or museums, but to individual collectors, who pay much better but generally do not allow researchers access to their collections. To increase the number of fossils available for research, some legislators propose requiring all fossils that are found in Mernia to be sold only to universities or museums.

Which of the following, if true, most strongly indicates that the legislators' proposal will fail to achieve its goal?

(A) Some fossil hunters in Mernia are not commercial fossil hunters, but rather are amateurs who keep the fossils that they find.

(B) Most fossils found in Mernia are common types that have little scientific interest.

(C) Commercial fossil hunters in Mernia currently sell some of the fossils they find to universities and museums.

(D) Many universities in Mernia do not engage in fossil research.

(E) Most fossils are found by commercial fossil hunters, and they would give up looking for fossils if they were no longer allowed to sell to individual collectors.

CR02702

576. Economist: Tropicorp, which constantly seeks profitable investment opportunities, has been buying and clearing sections of tropical forest for cattle ranching, although pastures newly created there become useless for grazing after just a few years. The company has not gone into rubber tapping, even though greater profits can be made from rubber tapping, which leaves the forest intact. Thus, some environmentalists argue that **Tropicorp's actions do not serve even its own economic interest**. However, the initial investment required for a successful rubber-tapping operation is larger than that needed for a cattle ranch; there is a shortage of workers employable in rubber-tapping operations; and taxes are higher on profits from rubber tapping than on profits from cattle ranching. Consequently, **the environmentalists' conclusion is probably wrong**.

In the economist's argument, the two **boldface** portions play which of the following roles?

(A) The first supports the conclusion of the economist's argument; the second calls that conclusion into question.

(B) The first states the conclusion of the economist's argument; the second supports that conclusion.

(C) The first supports the conclusion of the environmentalists' argument; the second states that conclusion.

(D) The first states the conclusion of the environmentalists' argument; the second states the conclusion of the economist's argument.

(E) Each supports the conclusion of the economist's argument.

CR08831

577. Although the school would receive financial benefits if it had soft drink vending machines in the cafeteria, we should not allow them. Allowing soft drink machines there would not be in our students' interest. If our students start drinking more soft drinks, they will be less healthy.

The argument depends on which of the following?

(A) If the soft drink vending machines were placed in the cafeteria, students would consume more soft drinks as a result.

(B) The amount of soft drinks that most students at the school currently drink is not detrimental to their health.

(C) Students are apt to be healthier if they do not drink soft drinks at all than if they just drink small amounts occasionally.

(D) Students will not simply bring soft drinks from home if the soft drink vending machines are not placed in the cafeteria.

(E) The school's primary concern should be to promote good health among its students.

CR01112

578. Many athletes inhale pure oxygen after exercise in an attempt to increase muscular reabsorption of oxygen. Measured continuously after exercise, however, the blood lactate levels of athletes who inhale pure oxygen are practically identical, on average, to those of athletes who breathe normal air. The lower the blood lactate level is, the higher the muscular reabsorption of oxygen is.

If the statements above are all true, they most strongly support which of the following conclusions?

(A) Athletes' muscular reabsorption of oxygen is not increased when they inhale pure oxygen instead of normal air.

(B) High blood lactate levels cannot be reduced.

(C) Blood lactate levels are a poor measure of oxygen reabsorption by muscles.

(D) The amount of oxygen reabsorbed by an athlete's muscles always remains constant.

(E) The inhaling of pure oxygen has no legitimate role in athletics.

CR02143

579. Boreal owls range over a much larger area than do other owls of similar size. Scientists have hypothesized that **it is scarcity of prey that leads the owls to range so widely**. This hypothesis would be hard to confirm directly, since it is not possible to produce a sufficiently accurate count of the populations of small mammals inhabiting the forests where boreal owls live. Careful study of owl behavior has, however, shown that **boreal owls do range over larger areas when they live in regions where food of the sort eaten by small mammals is comparatively sparse**. This indicates that the scientists' hypothesis is not sheer speculation.

In the argument given, the two **boldfaced** portions play which of the following roles?

(A) The first presents an explanatory hypothesis; the second states the main conclusion of the argument.

(B) The first presents an explanatory hypothesis; the second presents evidence tending to support this hypothesis.

(C) The first presents an explanatory hypothesis; the second presents evidence to support an alternative explanation.

(D) The first describes a position that the argument opposes; the second presents evidence to undermine the support for the position being opposed.

(E) The first describes a position that the argument opposes; the second states the main conclusion of the argument.

CR02888

580. Last year a record number of new manufacturing jobs were created. Will this year bring another record? Well, any new manufacturing job is created either within an existing company or by the start-up of a new company. **Within existing firms, new jobs have been created this year at well below last year's record pace**. At the same time, there is considerable evidence that the number of new companies starting up will be no higher this year than it was last year and **there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups**. So clearly, the number of new jobs created this year will fall short of last year's record.

In the argument given, the two portions in **boldface** play which of the following roles?

(A) The first is a claim that the argument challenges; the second is an explicit assumption on which that challenge is based.

(B) The first is a claim that the argument challenges; the second is a judgment advanced in support of the main conclusion of the argument.

(C) The first provides evidence in support of the main conclusion of the argument; the second is an objection that has been raised against that main conclusion.

(D) The first provides evidence in support of the main conclusion of the argument; the second is a judgment advanced in support of that main conclusion.

(E) The first and the second are each claims that have been advanced in support of a position that the argument opposes.

CR07809

581. A study of ticket sales at a summer theater festival found that people who bought tickets to individual plays had a no-show rate of less than 1 percent, while those who paid in advance for all ten plays being performed that summer had a no-show rate of nearly 30 percent. This may be at least in part because the greater the awareness customers retain about the cost of an item, the more likely they are to use it.

Which of the following would, if true, best serve as an alternative explanation of the results of the study?

(A) The price per ticket was slightly cheaper for those who bought all ten tickets in advance.

(B) Many people who attended the theater festival believed strongly that they should support it financially.

(C) Those who attended all ten plays became eligible for a partial refund.

(D) Usually, people who bought tickets to individual plays did so immediately prior to each performance that they attended.

(E) People who arrived just before the performance began could not be assured of obtaining seats in a preferred location.

CR12019

582. Although there is no record of poet Edmund Spenser's parentage, we do know that as a youth Spenser attended the Merchant Tailors' School in London for a period between 1560 and 1570. Records from this time indicate that the Merchant Tailors' Guild then had only three members named Spenser: Robert Spenser, listed as a gentleman; Nicholas Spenser, elected the Guild's Warden in 1568; and John Spenser, listed as a "journeyman cloth-maker." Of these, the last was likely the least affluent of the three—and most likely Edmund's father, since school accounting records list Edmund as a scholar who attended the school at a reduced fee.

Which of the following is an assumption on which the argument depends?

(A) Anybody in sixteenth-century London who made clothing professionally would have had to be a member of the Merchant Tailors' Guild.

(B) The fact that Edmund Spenser attended the Merchant Tailors' School did not necessarily mean that he planned to become a tailor.

(C) No member of the Guild could become Guild warden in sixteenth-century London unless he was a gentleman.

(D) Most of those whose fathers were members of the Merchant Tailors' Guild were students at the Merchant Tailors' School.

(E) The Merchant Tailors' School did not reduce its fees for the children of the more affluent Guild members.

CR03749

583. Rainwater contains hydrogen of a heavy form called deuterium. The deuterium content of wood reflects the deuterium content of rainwater available to trees during their growth. Wood from trees that grew between 16,000 and 24,000 years ago in North America contains significantly more deuterium than wood from trees growing today. But water trapped in several North American caves that formed during that same early period contains significantly less deuterium than rainwater in North America contains today.

Which of the following, if true, most helps to reconcile the two findings?

(A) There is little deuterium in the North American caves other than the deuterium in the water trapped there.

(B) Exposure to water after a tree has died does not change the deuterium content of the wood.

(C) Industrialization in North America over the past 100 years has altered the deuterium content of rain.

(D) Trees draw on shallow groundwater from rain that falls during their growth, whereas water trapped in caves may have fallen as rainwater thousands of years before the caves formed.

(E) Wood with a high deuterium content is no more likely to remain preserved for long periods than is wood with a low deuterium content.

CR04925

584. Enforcement of local speed limits through police monitoring has proven unsuccessful in the town of Ardane. In many nearby towns, speed humps (raised areas of pavement placed across residential streets, about 300 feet apart) have reduced traffic speeds on residential streets by 20 to 25 percent. In order to reduce traffic speed and thereby enhance safety in residential neighborhoods, Ardane's transportation commission plans to install multiple speed humps in those neighborhoods.

Which of the following, if true, identifies a potentially serious drawback to the plan for installing speed humps in Ardane?

(A) On residential streets without speed humps, many vehicles travel at speeds more than 25 percent above the posted speed limit.

(B) Because of their high weight, emergency vehicles such as fire trucks and ambulances must slow almost to a stop at speed humps.

(C) The residential speed limit in Ardane is higher than that of the nearby towns where speed humps were installed.

(D) Motorists who are not familiar with the streets in Ardane's residential districts would be likely to encounter the speed humps unawares unless warned by signs and painted indicators.

(E) Bicyclists generally prefer that speed humps be constructed so as to leave a space on the side of the road where bicycles can travel without going over the humps.

CR00748

585. Which of the choices most logically completes the following argument?

NowNews, although still the most popular magazine covering cultural events in Kalopolis, has recently suffered a significant drop in advertising revenue because of falling circulation. Many readers have begun buying a competing magazine that, at 50 cents per copy, costs less than *NowNews* at $1.50 per copy. In order to boost circulation and thus increase advertising revenue, *NowNews's* publisher has proposed making it available at no charge. However, this proposal has a serious drawback, since _____.

(A) those Kalopolis residents with the greatest interest in cultural events are regular readers of both magazines.

(B) one reason *NowNews's* circulation fell was that its competitor's reporting on cultural events was superior.

(C) the newsstands and stores that currently sell *NowNews* will no longer carry it if it is being given away for free.

(D) at present, 10 percent of the total number of copies of each issue of *NowNews* are distributed free to students on college campuses in the Kalopolis area.

(E) *NowNews's* competitor would begin to lose large amounts of money if it were forced to lower its cover price.

CR07304

586. Archaeologist: Researchers excavating a burial site in Cyprus found a feline skeleton lying near a human skeleton. Both skeletons were in the same sediment at the same depth and equally well-preserved, suggesting that the feline and human were buried together about 9,500 years ago. This shows that felines were domesticated around the time farming began, when they would have been useful in protecting stores of grain from mice.

Which of the following, if true, would most seriously weaken the archaeologist's argument?

(A) Archaeologists have not found any remains of stores of grain in the immediate vicinity of the burial site.

(B) The burial site in Cyprus is substantially older than any other known burial site in which a feline skeleton and a human skeleton appear to have been buried together.

(C) Paintings found near the burial site seem to show people keeping felines as domestic companions, but do not show felines hunting mice.

(D) In Cyprus, there are many burial sites dating from around 9,500 years ago in which the remains of wild animals appear to have been buried alongside human remains.

(E) Before felines were domesticated, early farmers had no effective way to protect stores of grain from mice.

CR09117

587. The heavy traffic in Masana is a growing drain on the city's economy—the clogging of the streets of the central business district alone cost the economy more than $1.2 billion over the past year. In order to address this problem, officials plan to introduce congestion pricing, by which drivers would pay to enter the city's most heavily trafficked areas during the busiest times of the day.

Which of the following, if true, would most strongly indicate that the plan will be a success?

(A) Approximately one-fifth of the vehicles in the central business district are in transit from one side of the city to the other.

(B) Planners expect that, without congestion pricing, traffic in Masana is likely to grow by 6 percent in the next five years.

(C) In other urban areas, congestion pricing has strongly encouraged carpooling (sharing of rides by private commuters).

(D) Several studies have shown that a reduction in traffic of 15 percent in Masana could result in 5,500 or more new jobs.

(E) Over 30 percent of the vehicles in the city's center are occupied by more than one person.

CR09151

588. Economist: The most economically efficient way to reduce emissions of air pollutants is to tax them in proportion to the damage they are likely to cause. But in Country Y, many serious pollutants are untaxed and unregulated, and policy makers strongly oppose new taxes. Therefore, the best way to achieve a reduction in air pollutant emissions in Country Y would be to institute fixed upper limits on them.

Which of the following is an assumption of the economist's argument?

(A) Policy makers in Country Y oppose all new taxes equally strongly, regardless of any benefits they may provide.

(B) Country Y's air pollutant emissions would not fall significantly if they were taxed in proportion to the damage they are likely to cause.

(C) Policy makers in Country Y strongly favor reductions in air pollutant emissions.

(D) Country Y's policy makers believe that air pollutant emissions should be reduced with maximum economic efficiency.

(E) Policy makers in Country Y do not oppose setting fixed upper limits on air pollutant emissions as strongly as they oppose new taxes.

CR04986

589. Humans get Lyme disease from infected ticks. Ticks get infected by feeding on animals with Lyme disease, but the ease of transmission from host animal to tick varies. With most species of host animal, transmission of Lyme disease to ticks is extremely rare, but white-footed mice are an exception, readily passing Lyme disease to ticks. And white-footed mouse populations greatly expand, becoming the main food source for ticks, in areas where biodiversity is in decline.

The information in the passage most strongly supports which of the following?

(A) In areas where many humans are infected with Lyme disease, the proportion of ticks infected with Lyme disease is especially high.

(B) Very few animals that live in areas where there are no white-footed mice are infected with Lyme disease.

(C) Humans are less at risk of contracting Lyme disease in areas where biodiversity is high.

(D) Ticks feed on white-footed mice only when other host species are not available to them.

(E) The greater the biodiversity of an area, the more likely any given host animal in that area is to pass Lyme disease to ticks.

CR04935

590. Many industrialized nations are trying to reduce atmospheric concentrations of carbon dioxide, a gas released by the burning of fossil fuels. One proposal is to replace conventional cement, which is made with calcium carbonate, by a new "eco-cement." This new cement, made with magnesium carbonate, absorbs large amounts of carbon dioxide when exposed to the atmosphere. Therefore, using eco-cement for new concrete building projects will significantly help reduce atmospheric concentrations of carbon dioxide.

Which of the following, if true, most strengthens the argument?

(A) The cost of magnesium carbonate, currently greater than the cost of calcium carbonate, probably will fall as more magnesium carbonate is used in cement manufacture.

(B) Eco-cement is strengthened when absorbed carbon dioxide reacts with the cement.

(C) Before the development of eco-cement, magnesium-based cement was considered too susceptible to water erosion to be of practical use.

(D) The manufacture of eco-cement uses considerably less fossil fuel per unit of cement than the manufacture of conventional cement does.

(E) Most building-industry groups are unaware of the development or availability of eco-cement.

CR00895

591. Advertisement: When your car's engine is running at its normal operating temperature, any major brand of motor oil will protect it about as well as Tuff does. When the engine is cold, it is a different story: Tuff motor oil flows better at lower temperatures than its major competitors do. So, if you want your car's engine to have maximum protection, you should use Tuff.

Which of the following, if true, most strengthens the argument in the advertisement?

(A) Tuff motor oil provides above-average protection for engines that happen to overheat.

(B) Tuff motor oil is periodically supplied free of charge to automobile manufacturers to use in factory-new cars.

(C) Tuff motor oil's share of the engine oil market peaked three years ago.

(D) Tuff motor oil, like any motor oil, is thicker and flows less freely at cold temperatures than at hot temperatures.

(E) Tuff motor oil is manufactured at only one refinery and shipped from there to all markets.

CR13108

592. *The Testament of William Thorpe* was published around 1530 as an appendix to Thorpe's longer *Examination*. Many scholars, however, doubt the attribution of the *Testament* to Thorpe because, whereas the *Examination* is dated 1406, the *Testament* is dated 1460. One scholar has recently argued that the 1460 date be amended to 1409, based on the observation that when these numbers are expressed as Roman numerals, MCCCCLX and MCCCCIX, it becomes easy to see how the dates might have become confused through scribal error.

Which of the following, if true, would most support the scholar's hypothesis concerning the date of the *Testament*?

(A) The sole evidence that historians have had that William Thorpe died no earlier than 1460 was the presumed date of publication of the *Testament*.

(B) In the preface to the 1530 publication, the editor attributes both works to William Thorpe.

(C) Few writers in fifteenth-century England marked dates in their works using only Roman numerals.

(D) The *Testament* alludes to a date, "Friday, September 20," as apparently contemporaneous with the writing of the *Testament*, and September 20 fell on a Friday in 1409 but not in 1460.

(E) The *Testament* contains few references to historical events that occurred later than 1406.

CR00777

593. A prominent investor who holds a large stake in the Burton Tool Company has recently claimed that the company is mismanaged, citing as evidence the company's failure to slow down production in response to a recent rise in its inventory of finished products. It is doubtful whether an investor's sniping at management can ever be anything other than counterproductive, but in this case, it is clearly not justified. It is true that **an increased inventory of finished products often indicates that production is outstripping demand**, but in Burton's case it indicates no such thing. Rather, **the increase in inventory is entirely attributable to products that have already been assigned to orders received from customers**.

In the argument given, the two **boldfaced** portions play which of the following roles?

(A) The first states a generalization that underlies the position that the argument as a whole opposes; the second provides evidence to show that the generalization does not apply in the case at issue.

(B) The first states a generalization that underlies the position that the argument as a whole opposes; the second clarifies the meaning of a specific phrase as it is used in that generalization.

(C) The first provides evidence to support the conclusion of the argument as a whole; the second is evidence that has been used to support the position that the argument as a whole opposes.

(D) The first provides evidence to support the conclusion of the argument as a whole; the second states that conclusion.

(E) The first and the second each provide evidence against the position that the argument as a whole opposes.

CR10028

594. To reduce productivity losses from employees calling in sick, Corporation X implemented a new policy requiring employees to come into work unless they were so sick that they had to go to a doctor. But a year after the policy was implemented, a study found that Corporation X's overall productivity losses due to reported employee illnesses had increased.

Which of the following, if true, would best explain why the policy produced the reverse of its intended effect?

(A) After the policy was implemented, employees more frequently went to the doctor when they felt sick.

(B) Before the policy was implemented, employees who were not sick at all often called in sick.

(C) Employees coming into work when sick often infect many of their coworkers.

(D) Unusually few employees became genuinely sick during the year after the policy was implemented.

(E) There are many other factors besides employee illness that can adversely affect productivity.

CR08443

595. Advertising by mail has become much less effective, with fewer consumers responding. Because consumers are increasingly overwhelmed by the sheer amount of junk mail they receive, most discard almost all offers without considering them. Thus, an effective way for corporations to improve response rates would be to more carefully target the individuals to whom they mail advertising, thereby cutting down on the amount of junk mail each consumer receives.

Which of the following, if true, would most support the recommendation above?

(A) There are cost-effective means by which corporations that currently advertise by mail could improve response rates.

(B) Many successful corporations are already carefully targeting the individuals to whom they mail advertising.

(C) Any consumer who, immediately after receiving an advertisement by mail, merely glances at it, is very likely to discard it.

(D) Improvements in the quality of the advertising materials used in mail that is carefully targeted to individuals can improve the response rate for such mail.

(E) Response rates to carefully targeted advertisements by mail are considerably higher, on average, than response rates to most other forms of advertising.

CR01905

596. Petrochemical industry officials have said that the extreme pressure exerted on plant managers during the last five years to improve profits by cutting costs has done nothing to impair the industry's ability to operate safely. However, environmentalists contend that the recent rash of serious oil spills and accidents at petrochemical plants is traceable to cost-cutting measures.

Which of the following, if true, would provide the strongest support for the position held by industry officials?

(A) The petrochemical industry benefits if accidents do not occur, since accidents involve risk of employee injury as well as loss of equipment and product.

(B) Petrochemical industry unions recently demanded that additional money be spent on safety and environmental protection measures, but the unions readily abandoned those demands in exchange for job security.

(C) Despite major cutbacks in most other areas of operation, the petrochemical industry has devoted more of its resources to environmental and safety measures in the last five years than in the preceding five years.

(D) There is evidence that the most damaging of the recent oil spills would have been prevented had cost-cutting measures not been instituted.

(E) Both the large fines and the adverse publicity generated by the most recent oil spill have prompted the petrochemical industry to increase the resources devoted to oil-spill prevention.

CR01368

597. A company has developed a new sensing device that, according to the company's claims, detects weak, ultralow-frequency electromagnetic signals associated with a beating heart. These signals, which pass through almost any physical obstruction, are purportedly detected by the device even at significant distances. Therefore, if the company's claims are true, their device will radically improve emergency teams' ability to locate quickly people who are trapped within the wreckage of collapsed buildings.

Which of the following, if true, most strengthens the argument?

(A) People trapped within the wreckage of collapsed buildings usually have serious injuries that require prompt medical treatment.

(B) The device gives a distinctive reading when the signals it detects come from human beings rather than from any other living beings.

(C) Most people who have survived after being trapped in collapsed buildings were rescued within two hours of the building's collapse.

(D) Ultralow-frequency signals are not the only electromagnetic signals that can pass through almost any physical obstruction.

(E) Extensive training is required in order to operate the device effectively.

CR11639

598. Economist: The price of tap water in our region should be raised drastically. **Supplies in local freshwater reservoirs have been declining for years** because water is being used faster than it can be replenished. Since the price of tap water has been low, **few users have bothered to adopt even easy conservation measures**.

The two sections in boldface play which of the following roles in the economist's argument?

(A) The first is a conclusion for which support is provided, and which in turn supports the main conclusion; the second is the main conclusion.

(B) The first is an observation for which the second provides an explanation; the second is the main conclusion but not the only conclusion.

(C) The first is a premise supporting the argument's main conclusion; so is the second.

(D) The first is the only conclusion; the second provides an explanation for the first.

(E) The first is the main conclusion; the second is a conclusion for which support is provided, and which in turn supports the first.

CR13127

599. Politician: Hybrid cars use significantly less fuel per kilometer than nonhybrids. And fuel produces air pollution, which contributes to a number of environmental problems. Motorists can save money by driving cars that are more fuel efficient, and they will be encouraged to drive hybrid cars if we make them aware of that fact. Therefore, we can help reduce the total amount of pollution emitted by cars in this country by highlighting this advantage of hybrid cars.

Which of the following, if true, would most indicate a vulnerability of the politician's argument?

(A) People with more fuel-efficient cars typically drive more than do those with less fuel-efficient cars.

(B) Not all air pollution originates from automobiles.

(C) Hybrid cars have already begun to gain popularity.

(D) Fuel-efficient alternatives to hybrid cars will likely become available in the future.

(E) The future cost of gasoline and other fuel cannot be predicted with absolute precision or certainty.

CR03862

600. Which of the following most logically completes the passage?

Pecan growers get a high price for their crop when pecans are comparatively scarce, but the price drops sharply when pecans are abundant. Thus, in high-yield years, growers often hold back part of their crop in refrigerated warehouses for one or two years, hoping for higher prices in the future. This year's pecan crop was the smallest in five years. It is nonetheless quite possible that a portion of this year's crop will be held back, since _____.

(A) each of the last two years produced recordbreaking pecan yields

(B) the quality of this year's pecan crop is no worse than the quality of the pecan crops of the previous five years

(C) pecan prices have not been subject to sharp fluctuations in recent years

(D) for some pecan growers, this year's crop was no smaller than last year's

(E) the practice of holding back part of one year's crop had not yet become widespread the last time the pecan crop was as small as it was this year

CR09534

601. Coffee shop owner: A large number of customers will pay at least the fair market value for a cup of coffee, even if there is no formal charge. Some will pay more than this out of appreciation of the trust that is placed in them. And our total number of customers is likely to increase. We could therefore improve our net cash flow by implementing an honor system in which customers pay what they wish for coffee by depositing money in a can.

Manager: We're likely to lose money on this plan. Many customers would cheat the system, paying a very small sum or nothing at all.

Which of the following, if true, would best support the owner's plan, in light of the manager's concern?

(A) The new system, if implemented, would increase the number of customers.

(B) By roasting its own coffee, the shop has managed to reduce the difficulties (and cost) of maintaining an inventory of freshly roasted coffee.

(C) Many customers stay in the cafe for long stretches of time.

(D) The shop makes a substantial profit from pastries and other food bought by the coffee drinkers.

(E) No other coffee shop in the area has such a system.

CR06749

602. Which of the following most logically completes the argument?

By competing with rodents for seeds, black ants help control rodent populations that pose a public health risk. However, a very aggressive species of black ant, the Loma ant, which has recently invaded a certain region, has a venomous sting that is often fatal to humans. Therefore, the planned introduction into that region of ant flies, which prey on Loma ants, would benefit public health, since _____.

(A) ant flies do not attack black ants other than Loma ants

(B) Loma ants are less effective than many bird species in competing with rodents for seeds

(C) certain other species of black ants are more effective than Loma ants in competing with rodents for seeds

(D) the sting of Loma ants can also be fatal to rodents

(E) some pesticides that could be used to control Loma ants are harmful to the environment

CR01900

603. The city of Workney, in raising bus fares from $1.00 to $1.25, proposed that 18 fare tokens be sold for $20.00 to alleviate the extra burden of the fare increase on the city's poor people. Critics suggested alternatively that 9 fare tokens be sold for $10.00, because a $20.00 outlay would be prohibitive for poor riders.

The alternative proposal depends on which of the following assumptions?

(A) Poor residents of Workney will continue to ride the buses in the same numbers despite the fare increase.

(B) Riders who are poor would be more likely to take advantage of the savings afforded by the 9-token offer than would riders who are not poor.

(C) The outlay of $10.00 for the purchase of 9 fare tokens would not be prohibitive for bus riders who are poor.

(D) The proposed fare increase is needed for the purchase of new buses for the city's bus system.

(E) Fewer riders would regularly purchase 18 fare tokens at once than would purchase only 9 fare tokens at once.

CR03272

604. Birds have been said to be descended from certain birdlike dinosaur species with which they share distinctive structural features. The fossil record, however, shows that this cannot be so, since there are bird fossils that are much older than the earliest birdlike dinosaur fossils that have been found.

Which of the following is an assumption on which the argument relies?

(A) The birdlike dinosaurs have no living descendants.

(B) There are no flightless dinosaur species that have the distinctive structural features shared by birds and birdlike dinosaurs.

(C) There are no birdlike dinosaur fossils that are older than the bird fossils but have not yet been unearthed.

(D) It could not have been the case that some birds were descended from one of the birdlike dinosaur species and other birds from another.

(E) Birds cannot have been descended from dinosaur species with which the birds do not share the distinctive structural features.

CR08239

605. City council member: Demand for electricity has been increasing by 1.5 percent a year, and there simply is no more space to build additional power plants to meet future demand increases. We must therefore begin to curtail usage, which is why I propose passing ordinances requiring energy-conservation measures in all city departments.

The city council member's proposal assumes which of the following?

(A) Existing power plants do not have the capacity to handle all of the projected increase in demand for electricity.

(B) No city departments have implemented energy-conservation measures voluntarily.

(C) Passing ordinances designed to curtail electricity usage will not have negative economic consequences for the city.

(D) Residential consumers are not responsible for the recent increases in demand for electricity.

(E) City departments that successfully conserve energy will set a good example for residential and industrial consumers of electricity.

CR00862

606. Which of the following most logically completes the argument below?

Using broad-spectrum weed killers on weeds that are competing with crops for sunlight, water, and nutrients presents a difficulty: how to keep the crop from being killed along with the weeds. For at least some food crops, specially treated seed that produces plants resistant to weed killers is under development. This resistance wears off as the plants mature. Therefore, the special seed treatment will be especially useful for plants that _____.

(A) produce their crop over an extended period of time, as summer squash does

(B) produce large seeds that are easy to treat individually, as corn and beans do

(C) provide, as they approach maturity, shade dense enough to keep weeds from growing

(D) are typically grown in large tracts devoted to a single crop

(E) are cultivated specifically for the seed they produce rather than for their leaves or roots

CR00713

607. Previously, Autoco designed all of its cars itself and then contracted with specialized parts suppliers to build parts according to its specifications. Now it plans to include its suppliers in designing the parts they are to build. Since many parts suppliers have more designers with specialized experience than Autoco has, Autoco expects this shift to reduce the overall time and cost of the design of its next new car.

Which of the following, if true, most strongly supports Autoco's expectation?

(A) When suppliers provide their own designs, Autoco often needs to modify its overall design.

(B) In order to provide designs for Autoco, several of the parts suppliers will have to add to their existing staffs of designers.

(C) Parts and services provided by outside suppliers account for more than 50 percent of Autoco's total costs.

(D) When suppliers built parts according to specifications provided by Autoco, the suppliers competed to win contracts.

(E) Most of Autoco's suppliers have on hand a wide range of previously prepared parts designs that can readily be modified for a new car.

CR02830

608. In response to viral infection, the immune systems of mice typically produce antibodies that destroy the virus by binding to proteins on its surface. Mice infected with the herpesvirus generally develop keratitis, a degenerative disease affecting part of the eye. Since proteins on the surface of cells in this part of the eye closely resemble those on the herpesvirus surface, scientists hypothesize that these cases of keratitis are caused by antibodies to the herpesvirus.

Which of the following, if true, most helps to support the scientists' reasoning?

(A) Other types of virus have surface proteins that closely resemble proteins found in various organs of mice.

(B) Mice that are infected with the herpesvirus but do not develop keratitis produce as many antibodies as infected mice that do develop keratitis.

(C) Mice infected with a new strain of the herpesvirus that has different surface proteins did not develop keratitis.

(D) Mice that have never been infected with the herpesvirus can sometimes develop keratitis.

(E) There are mice that are unable to form antibodies in response to herpes infections, and these mice contract herpes at roughly the same rate as other mice.

CR02885

609. Last year a record number of new manufacturing jobs were created. Will this year bring another record? Well, **any new manufacturing job is created either within an existing company or by the start-up of a new company**. Within existing firms, new jobs have been created this year at well below last year's record pace. At the same time, there is considerable evidence that the number of new companies starting up this year will be no higher than it was last year and **there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups**. So clearly, the number of new jobs created this year will fall short of last year's record.

In the argument given, the two portions in **boldface** play which of the following roles?

(A) The first provides evidence in support of the main conclusion of the argument; the second is a claim that the argument challenges.

(B) The first is a generalization that the argument seeks to establish; the second is a conclusion that the argument draws in order to support that generalization.

(C) The first is a generalization that the argument seeks to establish; the second is a judgment that has been advanced in order to challenge that generalization.

(D) The first is presented as an obvious truth on which the argument is based; the second is a claim that has been advanced in support of a position that the argument opposes.

(E) The first is presented as an obvious truth on which the argument is based; the second is a judgment advanced in support of the main conclusion of the argument.

CR02886

610. Last year a record number of new manufacturing jobs were created. Will this year bring another record? Well, **any new manufacturing job is created either within an existing company or by the start-up of a new company**. Within existing firms, new jobs have been created this year at well below last year's record pace. At the same time, there is considerable evidence that the number of new companies starting up will be no higher this year than it was last year and there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups. So clearly, **the number of new jobs created this year will fall short of last year's record**.

In the argument given, the two portions in **boldface** play which of the following roles?

(A) The first is presented as an obvious truth on which the argument is based; the second is the main conclusion of the argument.

(B) The first is presented as an obvious truth on which the argument is based; the second is a conclusion drawn in order to support the main conclusion of the argument.

(C) The first and the second each provide evidence in support of the main conclusion of the argument.

(D) The first is a generalization that the argument seeks to establish; the second is the main conclusion of the argument.

(E) The first is a generalization that the argument seeks to establish; the second is a conclusion that has been drawn in order to challenge that generalization.

CR00827

611. In Stenland, many workers have been complaining that they cannot survive on minimum wage, the lowest wage an employer is permitted to pay. The government is proposing to raise the minimum wage. Many employers who pay their workers the current minimum wage argue that if it is raised, unemployment will increase because they will no longer be able to afford to employ as many workers.

Which of the following, if true in Stenland, most strongly supports the claim that raising the minimum wage there will not have the effects that the employers predict?

(A) For any position with wages below a living wage, the difficulty of finding and retaining employees adds as much to employment costs as would raising wages.

(B) Raising the minimum wage does not also increase the amount employers have to contribute in employee benefits.

(C) When inflation is taken into account, the proposed new minimum wage is not as high as the current one was when it was introduced.

(D) Many employees currently being paid wages at the level of the proposed new minimum wage will demand significant wage increases.

(E) Many employers who pay some workers only the minimum wage also pay other workers wages that are much higher than the minimum.

CR07810

612. Biologists with a predilection for theory have tried—and largely failed—to define what it is that makes something a living thing. Organisms take in energy-providing materials and excrete waste products, but so do automobiles. Living things replicate and take part in evolution, but so do some computer programs. We must be open to the possibility that there are living things on other planets. Therefore, we will not be successful in defining what it is that makes something a living thing merely by examining living things on Earth—the only ones we know. Trying to do so is analogous to trying to specify _____.

Which of the following most logically completes the passage?

(A) the laws of physics by using pure mathematics

(B) what a fish is by listing its chemical components

(C) what an animal is by examining a plant

(D) what a machine is by examining a sketch of it

(E) what a mammal is by examining a zebra

CR01145

613. When trying to identify new technologies that promise to transform the marketplace, market researchers survey the managers of those companies that are developing new technologies. Such managers have an enormous stake in succeeding, so they invariably overstate the potential of their new technologies. Surprisingly, however, market researchers typically do not survey a new technology's potential buyers, even though it is the buyers—not the producers—who will ultimately determine a technology's commercial success.

Which of the following, if true, best accounts for the typical survey practices among market researchers?

(A) If a new technology succeeds, the commercial benefits accrue largely to the producers, not to the buyers, of that technology.

(B) People who promote the virtues of a new technology typically fail to consider that the old technology that is currently in use continues to be improved, often substantially.

(C) Investors are unlikely to invest substantial amounts of capital in a company whose own managers are skeptical about the commercial prospects of a new technology they are developing.

(D) The potential buyers for not-yet-available technologies can seldom be reliably identified.

(E) The developers of a new technology are generally no better positioned than its potential buyers to gauge how rapidly the new technology can be efficiently mass-produced.

CR02829

614. Sammy: For my arthritis, I am going to try my aunt's diet: large amounts of wheat germ and garlic. She was able to move more easily right after she started that diet.

Pat: When my brother began that diet, his arthritis got worse. But he has been doing much better since he stopped eating vegetables in the nightshade family, such as tomatoes and peppers.

Which of the following, if true, would provide a basis for explaining the fact that Sammy's aunt and Pat's brother had contrasting experiences with the same diet?

(A) A change in diet, regardless of the nature of the change, frequently brings temporary relief from arthritis symptoms.

(B) The compounds in garlic that can lessen the symptoms of arthritis are also present in tomatoes and peppers.

(C) Arthritis is a chronic condition whose symptoms improve and worsen from time to time without regard to diet.

(D) In general, men are more likely to have their arthritis symptoms alleviated by avoiding vegetables in the nightshade family than are women.

(E) People who are closely related are more likely to experience the same result from adopting a particular diet than are people who are unrelated.

CR05756

615. Infotek, a computer manufacturer in Katrovia, has just introduced a new personal computer model that sells for significantly less than any other model. Market research shows, however, that very few Katrovian households without personal computers would buy a computer, regardless of its price. Therefore, introducing the new model is unlikely to increase the number of computers in Katrovian homes.

Which of the following is an assumption on which the argument depends?

(A) Infotek achieved the lower price of the new model by using components of lower quality than those used by other manufacturers.

(B) The main reason cited by consumers in Katrovia for replacing a personal computer is the desire to have an improved model.

(C) Katrovians in households that already have computers are unlikely to purchase the new Infotek model as an additional computer for home use.

(D) The price of other personal computers in Katrovia is unlikely to drop below the price of Infotek's new model in the near future.

(E) Most personal computers purchased in Katrovia are intended for home use.

CR05501

616. Fast-food restaurants make up 45 percent of all restaurants in Canatria. Customers at these restaurants tend to be young; in fact, studies have shown that the older people get, the less likely they are to eat in fast-food restaurants. Since the average age of the Canatrian population is gradually rising and will continue to do so, the number of fast-food restaurants is likely to decrease.

Which of the following, if true, most seriously weakens the argument?

(A) Fast-food restaurants in Canatria are getting bigger, so each one can serve more customers.

(B) Some older people eat at fast-food restaurants more frequently than the average young person.

(C) Many people who rarely eat in fast-food restaurants nevertheless eat regularly in restaurants.

(D) The overall population of Canatria is growing steadily.

(E) As the population of Canatria gets older, more people are eating at home.

CR04805

617. Last year a chain of fast-food restaurants, whose menu had always centered on hamburgers, added its first vegetarian sandwich, much lower in fat than the chain's other offerings. Despite heavy marketing, the new sandwich accounts for a very small proportion of the chain's sales. The sandwich's sales would have to quadruple to cover the costs associated with including it on the menu. Since such an increase is unlikely, the chain would be more profitable if it dropped the sandwich.

Which of the following, if true, most seriously weakens the argument?

(A) Although many of the chain's customers have never tried the vegetarian sandwich, in a market research survey most of those who had tried it reported that they were very satisfied with it.

(B) Many of the people who eat at the chain's restaurants also eat at the restaurants of competing chains and report no strong preference among the competitors.

(C) Among fast-food chains in general, there has been little or no growth in hamburger sales over the past several years as the range of competing offerings at other restaurants has grown.

(D) When even one member of a group of diners is a vegetarian or has a preference for low-fat food, the group tends to avoid restaurants that lack vegetarian or low-fat menu options.

(E) An attempt by the chain to introduce a lower-fat hamburger failed several years ago, since it attracted few new customers and most of the chain's regular customers greatly preferred the taste of the regular hamburgers.

CR03727

618. Transportation expenses accounted for a large portion of the total dollar amount spent on trips for pleasure by residents of the United States in 1997, and about half of the total dollar amount spent on transportation was for airfare. However, the large majority of United States residents who took trips for pleasure in 1997 did not travel by airplane but used other means of transportation.

If the statements above are true, which of the following must also be true about United States residents who took trips for pleasure in 1997?

(A) Most of those who traveled by airplane did so because the airfare to their destination was lower than the cost of other available means of transportation.

(B) Most of those who traveled by airplane did so because other means of transportation to their destination were unavailable.

(C) Per mile traveled, those who traveled by airplane tended to spend more on transportation to their destination than did those who used other means of transportation.

(D) Overall, people who did not travel by airplane had lower average transportation expenses than people who did.

(E) Those who traveled by airplane spent about as much, on average, on other means of transportation as they did on airfare.

CR12051

619. Voters commonly condemn politicians for being insincere, but politicians often must disguise their true feelings when they make public statements. If they expressed their honest views—about, say, their party's policies—then achieving politically necessary compromises would be much more difficult. Clearly, the very insincerity that people decry shows that our government is functioning well.

Which of the following, if true, most seriously undermines this reasoning?

(A) Achieving political compromises is not all that is necessary for the proper functioning of a government.

(B) Some political compromises are not in the best long-term interest of the government.

(C) Voters often judge politicians by criteria other than the sincerity with which they express their views.

(D) A political party's policies could turn out to be detrimental to the functioning of a government.

(E) Some of the public statements made by politicians about their party's policies could in fact be sincere.

CR06728

620. To reduce waste of raw materials, the government of Sperland is considering requiring household appliances to be broken down for salvage when discarded. To cover the cost of salvage, the government is planning to charge a fee, which would be imposed when the appliance is first sold. Imposing the fee at the time of salvage would reduce waste more effectively, however, because consumers tend to keep old appliances longer if they are faced with a fee for discarding them.

Which of the following, if true, most seriously weakens the argument?

(A) Increasing the cost of disposing of an appliance properly increases the incentive to dispose of it improperly.

(B) The fee provides manufacturers with no incentive to produce appliances that are more durable.

(C) For people who have bought new appliances recently, the salvage fee would not need to be paid for a number of years.

(D) People who sell their used, working appliances to others would not need to pay the salvage fee.

(E) Many nonfunctioning appliances that are currently discarded could be repaired at relatively little expense.

CR02866

621. When there is less rainfall than normal, the water level of Australian rivers falls and the rivers flow more slowly. Because algae whose habitat is river water grow best in slow-moving water, the amount of algae per unit of water generally increases when there has been little rain. By contrast, however, following a period of extreme drought, algae levels are low even in very slow-moving river water.

Which of the following, if true, does most to explain the contrast described above?

(A) During periods of extreme drought, the populations of some of the species that feed on algae tend to fall.

(B) The more slowly water moves, the more conducive its temperature is to the growth of algae.

(C) When algae populations reach very high levels, conditions within the river can become toxic for some of the other species that normally live there.

(D) Australian rivers dry up completely for short intervals in periods of extreme drought.

(E) Except during periods of extreme drought, algae levels tend to be higher in rivers in which the flow has been controlled by damming than in rivers that flow freely.

CR04924

622. Increased use of incineration is sometimes advocated as a safe way to dispose of chemical waste. But opponents of incineration point to the 40 incidents involving unexpected releases of dangerous chemical agents that were reported just last year at two existing incinerators commissioned to destroy a quantity of chemical waste material. Since designs for proposed new incinerators include no additional means of preventing such releases, leaks will only become more prevalent if use of incineration increases.

Which of the following, if true, most seriously weakens the argument?

(A) At the two incinerators at which leaks were reported, staff had had only cursory training on the proper procedures for incinerating chemical waste.

(B) Other means of disposing of chemical waste, such as chemical neutralization processes, have not been proven safer than incineration.

(C) The capacity of existing incinerators is sufficient to allow for increased incineration of chemical waste without any need for new incinerators.

(D) The frequency of reports of unexpected releases of chemical agents at newly built incinerators is about the same as the frequency at older incinerators.

(E) In only three of the reported incidents of unexpected chemical leaks did the releases extend outside the property on which the incinerators were located.

CR10049

623. Public health expert: **Increasing the urgency of a public health message may be counterproductive.** In addition to irritating the majority who already behave responsibly, **it may undermine all government pronouncements on health by convincing people that such messages are overly cautious.** And there is no reason to believe that those who ignore measured voices will listen to shouting.

The two sections in boldface play which of the following roles in the public health expert's argument?

(A) The first is a conclusion for which support is provided, but is not the argument's main conclusion; the second is an unsupported premise supporting the argument's main conclusion.

(B) The first is a premise supporting the only explicit conclusion; so is the second.

(C) The first is the argument's main conclusion; the second supports that conclusion and is itself a conclusion for which support is provided.

(D) The first is a premise supporting the argument's only conclusion; the second is that conclusion.

(E) The first is the argument's only explicit conclusion; the second is a premise supporting that conclusion.

CR01163

624. Several industries have recently switched at least partly from older technologies powered by fossil fuels to new technologies powered by electricity. It is thus evident that less fossil fuel is being used as a result of the operations of these industries than would have been used if these industries had retained their older technologies.

Which of the following, if true, most strengthens the argument above?

(A) Many of the industries that have switched at least partly to the new technologies have increased their output.

(B) Less fossil fuel was used to manufacture the machinery employed in the new technologies than was originally used to manufacture the machinery employed in the older technologies.

(C) More electricity is used by those industries that have switched at least partly to the new technologies than by those industries that have not switched.

(D) Some of the industries that have switched at least partly to the new technologies still use primarily technologies that are powered by fossil fuels.

(E) The amount of fossil fuel used to generate the electricity needed to power the new technologies is less than the amount that would have been used to power the older technologies.

CR00792

625. The difference in average annual income in favor of employees who have college degrees, compared with those who do not have such degrees, doubled between 1980 and 1990. Some analysts have hypothesized that increased competition between employers for employees with college degrees drove up income for such employees.

Which of the following, if true, most seriously undermines the explanation described above?

(A) During the 1980s a growing percentage of college graduates, unable to find jobs requiring a college degree, took unskilled jobs.

(B) The average age of all employees increased slightly during the 1980s.

(C) The unemployment rate changed very little throughout most of the 1980s.

(D) From 1980 to 1990 the difference in average income between employees with advanced degrees and those with bachelor's degrees also increased.

(E) During the 1980s there were some employees with no college degree who earned incomes comparable to the top incomes earned by employees with a college degree.

CR01239

626. Which of the following most logically completes the passage?

According to the last pre-election poll in Whippleton, most voters believe that the three problems government needs to address, in order of importance, are pollution, crime, and unemployment. Yet in the election, candidates from parties perceived as strongly against pollution were defeated, while those elected were all from parties with a history of opposing legislation designed to reduce pollution. These results should not be taken to indicate that the poll was inaccurate, however, since _____.

(A) some voters in Whippleton do not believe that pollution needs to be reduced

(B) every candidate who was defeated had a strong antipollution record

(C) there were no issues other than crime, unemployment, and pollution on which the candidates had significant differences of opinion

(D) all the candidates who were elected were perceived as being stronger against both crime and unemployment than the candidates who were defeated

(E) many of the people who voted in the election refused to participate in the poll

CR01153

627. Manufacturing plants in Arundia have recently been acquired in substantial numbers by investors from abroad. Arundian politicians are proposing legislative action to stop such investment, justifying the proposal by arguing that foreign investors, opportunistically exploiting a recent fall in the value of the Arundian currency, were able to buy Arundian assets at less than their true value.

Which of the following, if true, casts the most serious doubt on the adequacy of the Arundian politicians' justification for the proposed legislation?

(A) The Arundian government originally welcomed the fall in the value of the Arundian currency because the fall made Arundian exports more competitive on international markets.

(B) Foreign investors who acquired Arundian manufacturing plants generally did so with no intention of keeping and running those plants over the long term.

(C) Without the recent fall in the value of the Arundian currency, many of the Arundian assets bought by foreign investors would have been beyond the financial reach of those investors.

(D) In Concordia, a country broadly similar to Arundia, the share of manufacturing assets that is foreign-controlled is 60 percent higher than it is in Arundia.

(E) The true value of an investment is determined by the value of the profits from it, and the low value of the Arundian currency has depressed the value of any profits earned by foreign investors from Arundian assets.

CR04964

628. Proposed new safety rules for Beach City airport would lengthen considerably the minimum time between takeoffs from the airport. In consequence, the airport would be able to accommodate 10 percent fewer flights than currently use the airport daily. The city's operating budget depends heavily on taxes generated by tourist spending, and most of the tourists come by plane. Therefore, the proposed new safety rules, if adopted, will reduce the revenue available for the operating budget.

The argument depends on assuming which of the following?

(A) There are no periods of the day during which the interval between flights taking off from the airport is significantly greater than the currently allowed minimum.

(B) Few, if any, of the tourists who use Beach City airport do so when their main destination is a neighboring community and not Beach City itself.

(C) If the proposed safety rules are adopted, the reduction in tourist numbers will not result mainly from a reduction in the number of tourists who spend relatively little in Beach City.

(D) Increasing the minimum time between takeoffs is the only way to achieve necessary safety improvements without a large expenditure by the city government on airport enhancements.

(E) The response to the adoption of the new safety rules would not include a large increase in the number of passengers per flight.

CR01096

629. The introduction of new drugs into the market is frequently prevented by a shortage of human subjects for the clinical trials needed to show that the drugs are safe and effective. Since the lives and health of people in future generations may depend on treatments that are currently experimental, practicing physicians are morally in the wrong when, in the absence of any treatment proven to be effective, they fail to encourage suitable patients to volunteer for clinical trials.

Which of the following, if true, casts most doubt on the conclusion of the argument?

(A) Many drugs undergoing clinical trials are intended for the treatment of conditions for which there is currently no effective treatment.

(B) Patients do not share the physician's professional concern for public health, but everyone has a moral obligation to alleviate suffering when able to do so.

(C) Usually, half the patients in a clinical trial serve as a control group and receive a nonactive drug in place of the drug being tested.

(D) An experimental drug cannot legally be made available to patients unless those patients are subjects in clinical trials of the drug.

(E) Physicians have an overriding moral and legal duty to care for the health and safety of their current patients.

CR01285

630. As a construction material, bamboo is as strong as steel and sturdier than concrete. Moreover, in tropical areas bamboo is a much less expensive construction material than either steel or concrete and is always readily available. In tropical areas, therefore, building with bamboo makes better economic sense than building with steel or concrete, except where land values are high.

Which of the following, if true, most helps to explain the exception noted above?

(A) Buildings constructed of bamboo are less likely to suffer earthquake damage than are steel and concrete buildings.

(B) Bamboo is unsuitable as a building material for multistory buildings.

(C) In order to protect it from being damaged by termites and beetles, bamboo must be soaked, at some expense, in a preservative.

(D) In some tropical areas, bamboo is used to make the scaffolding that is used during large construction projects.

(E) Bamboo growing in an area where land values are increasing is often cleared to make way for construction.

CR00788

631. Newspaper editors should not allow reporters to write the headlines for their own stories. The reason for this is that, while the headlines that reporters themselves write are often clever, what typically makes them clever is that they allude to little-known information that is familiar to the reporter but that never appears explicitly in the story itself.

Which of the following, if true, most strengthens the argument?

(A) The reporter who writes a story is usually better placed than the reporter's editor is to judge what the story's most newsworthy features are.

(B) To write a headline that is clever, a person must have sufficient understanding of the story that the headline accompanies.

(C) Most reporters rarely bother to find out how other reporters have written stories and headlines about the same events that they themselves have covered.

(D) For virtually any story that a reporter writes, there are at least a few people who know more about the story's subject matter than does the reporter.

(E) The kind of headlines that newspaper editors want are those that anyone who has read a reporter's story in its entirety will recognize as clever.

CRO3251

632. Scientists have modified feed corn genetically, increasing its resistance to insect pests. Farmers who tried out the genetically modified corn last season applied less insecticide to their corn fields and still got yields comparable to those they would have gotten with ordinary corn. Ordinary corn seed, however, costs less, and what these farmers saved on insecticide rarely exceeded their extra costs for seed. Therefore, for most feed-corn farmers, switching to genetically modified seed would be unlikely to increase profits.

Which of the following would it be most useful to know in order to evaluate the argument?

(A) Whether there are insect pests that sometimes reduce feed-corn yields, but against which commonly used insecticides and the genetic modification are equally ineffective

(B) Whether the price that farmers receive for feed corn has remained steady over the past few years

(C) Whether the insecticides typically used on feed corn tend to be more expensive than insecticides typically used on other crops

(D) Whether most of the farmers who tried the genetically modified corn last season applied more insecticide than was actually necessary

(E) Whether, for most farmers who plant feed corn, it is their most profitable crop

CRO7318

633. Debater: The average amount of overtime per month worked by an employee in the manufacturing division of the Haglut Corporation is 14 hours. Most employees of the Haglut Corporation work in the manufacturing division. Furthermore, the average amount of overtime per month worked by any employee in the company generally does not fluctuate much from month to month. Therefore, each month, most employees of the Haglut Corporation almost certainly work at least some overtime.

The debater's argument is most vulnerable to criticism on which of these grounds?

(A) It takes for granted that the manufacturing division is a typical division of the corporation with regard to the average amount of overtime its employees work each month.

(B) It takes for granted that if a certain average amount of overtime is worked each month by each employee of the Haglut Corporation, then approximately the same amount of overtime must be worked each month by each employee of the manufacturing division.

(C) It confuses a claim from which the argument's conclusion about the Haglut Corporation would necessarily follow with a claim that would follow from the argument's conclusion only with a high degree of probability.

(D) It overlooks the possibility that even if, on average, a certain amount of overtime is worked by the members of some group, many members of that group may work no overtime at all.

(E) It overlooks the possibility that even if most employees of the corporation work some overtime each month, any one corporate employee may, in some months, work no overtime.

CR05446

634. Proponents of the recently introduced tax on sales of new luxury boats had argued that a tax of this sort would be an equitable way to increase government revenue because the admittedly heavy tax burden would fall only on wealthy people and neither they nor anyone else would suffer any economic hardship. In fact, however, 20 percent of the workers employed by manufacturers of luxury boats have lost their jobs as a direct result of this tax.

The information given, if true, most strongly supports which of the following?

(A) The market for luxury boats would have collapsed even if the new tax on luxury boats had been lower.

(B) The new tax would produce a net gain in tax revenue for the government only if the yearly total revenue that it generates exceeds the total of any yearly tax-revenue decrease resulting from the workers' loss of jobs.

(C) Because many people never buy luxury items, imposing a sales tax on luxury items is the kind of legislative action that does not cost incumbent legislators much popular support.

(D) Before the tax was instituted, luxury boats were largely bought by people who were not wealthy.

(E) Taxes can be equitable only if their burden is evenly distributed over the entire population.

CR05191

635. In Wareland last year, 16 percent of licensed drivers under 21 and 11 percent of drivers ages 21–24 were in serious accidents. By contrast, only 3 percent of licensed drivers 65 and older were involved in serious accidents. These figures clearly show that the greater experience and developed habits of caution possessed by drivers in the 65-and-older group make them far safer behind the wheel than the younger drivers are.

Which of the following is an assumption on which the argument depends?

(A) Drivers 65 and older do not, on average, drive very many fewer miles per year than drivers 24 and younger.

(B) Drivers 65 and older do not constitute a significantly larger percentage of licensed drivers in Wareland than drivers ages 18–24 do.

(C) Drivers 65 and older are less likely than are drivers 24 and younger to drive during weather conditions that greatly increase the risk of accidents.

(D) The difference between the accident rate of drivers under 21 and of those ages 21–24 is attributable to the greater driving experience of those in the older group.

(E) There is no age bracket for which the accident rate is lower than it is for licensed drivers 65 and older.

CR05614

636. In the past the country of Malvernia has relied heavily on imported oil. Malvernia recently implemented a program to convert heating systems from oil to natural gas. Malvernia currently produces more natural gas each year than it uses, and oil production in Malvernian oil fields is increasing at a steady pace. If these trends in fuel production and usage continue, therefore, Malvernian reliance on foreign sources for fuel is likely to decline soon.

Which of the following would it be most useful to establish in evaluating the argument?

(A) When, if ever, will production of oil in Malvernia outstrip production of natural gas?

(B) Is Malvernia among the countries that rely most on imported oil?

(C) What proportion of Malvernia's total energy needs is met by hydroelectric, solar, and nuclear power?

(D) Is the amount of oil used each year in Malvernia for generating electricity and fuel for transportation increasing?

(E) Have any existing oil-burning heating systems in Malvernia already been converted to natural-gas-burning heating systems?

CR03618

637. Exposure to certain chemicals commonly used in elementary schools as cleaners or pesticides causes allergic reactions in some children. Elementary school nurses in Renston report that the proportion of schoolchildren sent to them for treatment of allergic reactions to those chemicals has increased significantly over the past ten years. Therefore, either Renston's schoolchildren have been exposed to greater quantities of the chemicals, or they are more sensitive to them than schoolchildren were ten years ago.

Which of the following is an assumption on which the argument depends?

(A) The number of school nurses employed by Renston's elementary schools has not decreased over the past ten years.

(B) Children who are allergic to the chemicals are no more likely than other children to have allergies to other substances.

(C) Children who have allergic reactions to the chemicals are not more likely to be sent to a school nurse now than they were ten years ago.

(D) The chemicals are not commonly used as cleaners or pesticides in houses and apartment buildings in Renston.

(E) Children attending elementary school do not make up a larger proportion of Renston's population now than they did ten years ago.

CR01854

638. Normally, the pineal gland governs a person's sleep-wake cycle by secreting melatonin in response to the daily cycle of light and darkness as detected by the eye. Nonetheless, many people who are totally blind due to lesions in the visual cortex of the brain easily maintain a 24-hour sleep-wake cycle. So the neural pathway by which the pineal gland receives information from the eye probably does not pass through the visual cortex.

For purposes of evaluating the argument it would be most useful to establish which of the following?

(A) Whether melatonin supplements help people who have difficulty maintaining a 24-hour sleep cycle to establish such a pattern

(B) Whether the melatonin levels of most totally blind people who successfully maintain a 24-hour sleep-wake cycle change in response to changes in exposure to light and darkness

(C) Whether melatonin is the only substance secreted by the pineal gland

(D) Whether most people who do not have a 24-hour sleep-wake cycle nevertheless have a cycle of consistent duration

(E) Whether there are any people with normal vision whose melatonin levels respond abnormally to periods of light and darkness

CR00942

639. **In countries where automobile insurance includes compensation for whiplash injuries sustained in automobile accidents, reports of having suffered such injuries are twice as frequent as they are in countries where whiplash is not covered.** Presently, no objective test for whiplash exists, so it is true that spurious reports of whiplash injuries cannot be readily identified. Nevertheless, these facts do not warrant the conclusion drawn by some commentators that in the countries with the higher rates of reported whiplash injuries, half of the reported cases are spurious. Clearly, **in countries where automobile insurance does not include compensation for whiplash, people often have little incentive to report whiplash injuries that they actually have suffered.**

In the argument given, the two boldfaced portions play which of the following roles?

(A) The first is a claim that the argument disputes; the second is a conclusion that has been based on that claim.

(B) The first is a claim that has been used to support a conclusion that the argument accepts; the second is that conclusion.

(C) The first is evidence that has been used to support a conclusion for which the argument provides further evidence; the second is the main conclusion of the argument.

(D) The first is a finding whose implications are at issue in the argument; the second is a claim presented in order to argue against deriving certain implications from that finding.

(E) The first is a finding whose accuracy is evaluated in the argument; the second is evidence presented to establish that the finding is accurate.

CR03859

640. Last year Comfort Airlines had twice as many delayed flights as the year before, but the number of complaints from passengers about delayed flights went up three times. It is unlikely that this disproportionate increase in complaints was rooted in an increase in overall dissatisfaction with the service Comfort Airlines provides, since the airline made a special effort to improve other aspects of its service last year.

Which of the following, if true, most helps to explain the disproportionate increase in customer complaints?

(A) Comfort Airlines had more flights last year than the year before.

(B) Last year a single period of unusually bad weather caused a large number of flights to be delayed.

(C) Some of the improvements that Comfort Airlines made in its service were required by new government regulations.

(D) The average length of a flight delay was greater last year than it was the year before.

(E) The average number of passengers per flight was no higher last year than the year before.

CR01337

641. Last year a global disturbance of weather patterns disrupted harvests in many of the world's important agricultural areas. Worldwide production of soybeans, an important source of protein for people and livestock alike, was not adversely affected, however. Indeed, last year's soybean crop was actually slightly larger than average. Nevertheless, the weather phenomenon is probably responsible for a recent increase in the world price of soybeans.

Which of the following, if true, provides the strongest justification for the attribution of the increase in soybean prices to the weather phenomenon?

(A) Last year's harvest of anchovies, which provide an important protein source for livestock, was disrupted by the effects of the weather phenomenon.

(B) Most countries that produce soybeans for export had above-average harvests of a number of food crops other than soybeans last year.

(C) The world price of soybeans also rose several years ago, immediately after an earlier occurrence of a similar global weather disturbance.

(D) Heavy rains attributable to the weather phenomenon improved grazing pastures last year, allowing farmers in many parts of the world to reduce their dependence on supplemental feed.

(E) Prior to last year, soybean prices had been falling for several years.

CR03541

642. Most of the year, the hermit thrush, a North American songbird, eats a diet consisting mainly of insects, but in autumn, as the thrushes migrate to their Central and South American wintering grounds, they feed almost exclusively on wild berries. Wild berries, however, are not as rich in calories as insects, yet thrushes need to consume plenty of calories in order to complete their migration. One possible explanation is that berries contain other nutrients that thrushes need for migration and that insects lack.

Which of the following, if true, most seriously calls into question the explanation given for the thrush's diet during migration?

(A) Hermit thrushes, if undernourished, are unable to complete their autumn migration before the onset of winter.

(B) Insect species contain certain nutrients that are not found in wild berries.

(C) For songbirds, catching insects requires the expenditure of significantly more calories than eating wild berries does.

(D) Along the hermit thrushes' migration routes, insects are abundant throughout the migration season.

(E) There are some species of wild berries that hermit thrushes generally do not eat, even though these berry species are exceptionally rich in calories.

CR01879

643. The kinds of hand and wrist injuries that result from extended use of a computer while maintaining an incorrect posture are common among schoolchildren in Harnville. Computers are important to the school curriculum there, so instead of reducing the amount their students use computers, teachers plan to bring about a sharp reduction in the number of these injuries by carefully monitoring their students' posture when using computers in the classroom.

Which of the following would it be most useful to know in order to assess the likelihood that the teachers' plan will be successful?

(A) Whether extended use of a computer while maintaining incorrect posture can cause injuries other than hand and wrist injuries

(B) Whether hand and wrist injuries not caused by computer use are common among schoolchildren in Harnville

(C) What proportion of schoolchildren in Harnville with hand and wrist injuries use computers extensively outside the classroom

(D) Whether changes in the curriculum could reduce the schools' dependence on computers

(E) What proportion of schoolchildren in Harnville already use correct posture while using a computer

CR04718

644. A certain cultivated herb is one of a group of closely related plants that thrive in soil with high concentrations of metals that are toxic to most other plants. Agronomists studying the growth of this herb have discovered that it produces large amounts of histidine, an amino acid that, in test-tube solutions, renders these metals chemically inert. Hence, the herb's high histidine production must be the key feature that allows it to grow in metal-rich soils.

In evaluating the argument, it would be most important to determine which of the following?

(A) Whether the herb can thrive in soil that does not have high concentrations of the toxic metals

(B) Whether others of the closely related group of plants also produce histidine in large quantities

(C) Whether the herb's high level of histidine production is associated with an unusually low level of production of some other amino acid

(D) Whether growing the herb in soil with high concentrations of the metals will, over time, reduce their concentrations in the soil

(E) Whether the concentration of histidine in the growing herb declines as the plant approaches maturity

CR01293

645. Many people suffer an allergic reaction to certain sulfites, including those that are commonly added to wine as preservatives. However, since there are several winemakers who add sulfites to none of the wines they produce, people who would like to drink wine but are allergic to sulfites can drink wines produced by these winemakers without risking an allergic reaction to sulfites.

Which of the following is an assumption on which the argument depends?

(A) These winemakers have been able to duplicate the preservative effect produced by adding sulfites by means that do not involve adding any potentially allergenic substances to their wine.

(B) Not all forms of sulfite are equally likely to produce the allergic reaction.

(C) Wine is the only beverage to which sulfites are commonly added.

(D) Apart from sulfites, there are no substances commonly present in wine that give rise to an allergic reaction.

(E) Sulfites are not naturally present in the wines produced by these winemakers in amounts large enough to produce an allergic reaction in someone who drinks these wines.

CR11447

646. A new law gives ownership of patents—documents providing exclusive right to make and sell an invention—to universities, not the government, when those patents result from government-sponsored university research. Administrators at Logos University plan to sell any patents they acquire to corporations in order to fund programs to improve undergraduate teaching.

Which of the following, if true, would cast the most doubt on the viability of the college administrators' plan described above?

(A) Profit-making corporations interested in developing products based on patents held by universities are likely to try to serve as exclusive sponsors of ongoing university research projects.

(B) Corporate sponsors of research in university facilities are entitled to tax credits under new federal tax-code guidelines.

(C) Research scientists at Logos University have few or no teaching responsibilities and participate little if at all in the undergraduate programs in their field.

(D) Government-sponsored research conducted at Logos University for the most part duplicates research already completed by several profitmaking corporations.

(E) Logos University is unlikely to attract corporate sponsorship of its scientific research.

CR01848

647. Since it has become known that **several of a bank's top executives have been buying shares in their own bank,** the bank's depositors, who had been worried by rumors that the bank faced impending financial collapse, have been greatly relieved. They reason that, since top executives evidently have faith in the bank's financial soundness, those worrisome rumors must be false. Such reasoning might well be overoptimistic, however, since **corporate executives have been known to buy shares in their own company in a calculated attempt to dispel negative rumors about the company's health.**

In the argument given, the two boldfaced portions play which of the following roles?

(A) The first describes evidence that has been taken as supporting a conclusion; the second gives a reason for questioning that support.

(B) The first describes evidence that has been taken as supporting a conclusion; the second states a contrary conclusion that is the main conclusion of the argument.

(C) The first provides evidence in support of the main conclusion of the argument; the second states that conclusion.

(D) The first describes the circumstance that the argument as a whole seeks to explain; the second gives the explanation that the argument seeks to establish.

(E) The first describes the circumstance that the argument as a whole seeks to explain; the second provides evidence in support of the explanation that the argument seeks to establish.

CR03814

648. Between 1980 and 2000 the sea otter population of the Aleutian Islands declined precipitously. There were no signs of disease or malnutrition, so there was probably an increase in the number of otters being eaten by predators. Orcas will eat otters when seals, their normal prey, are unavailable, and the Aleutian Islands seal population declined dramatically in the 1980s. Therefore, orcas were most likely the immediate cause of the otter population decline.

Which of the following, if true, most strengthens the argument?

(A) The population of sea urchins, the main food of sea otters, has increased since the sea otter population declined.

(B) Seals do not eat sea otters, nor do they compete with sea otters for food.

(C) Most of the surviving sea otters live in a bay that is inaccessible to orcas.

(D) The population of orcas in the Aleutian Islands has declined since the 1980s.

(E) An increase in commercial fishing near the Aleutian Islands in the 1980s caused a slight decline in the population of the fish that seals use for food.

CR05960

649. Studies in restaurants show that the tips left by customers who pay their bill in cash tend to be larger when the bill is presented on a tray that bears a credit-card logo. Consumer psychologists hypothesize that simply seeing a credit-card logo makes many credit-card holders willing to spend more because it reminds them that their spending power exceeds the cash they have immediately available.

Which of the following, if true, most strongly supports the psychologists' interpretation of the studies?

(A) The effect noted in the studies is not limited to patrons who have credit cards.

(B) Patrons who are under financial pressure from their credit-card obligations tend to tip less when presented with a restaurant bill on a tray with a credit-card logo than when the tray has no logo.

(C) In virtually all of the cases in the studies, the patrons who paid bills in cash did not possess credit cards.

(D) In general, restaurant patrons who pay their bills in cash leave larger tips than do those who pay by credit card.

(E) The percentage of restaurant bills paid with a given brand of credit card increases when that credit card's logo is displayed on the tray with which the bill is presented.

CR11633

650. In an experiment, each volunteer was allowed to choose between an easy task and a hard task and was told that another volunteer would do the other task. Each volunteer could also choose to have a computer assign the two tasks randomly. Most volunteers chose the easy task for themselves and under questioning

later said they had acted fairly. But when the scenario was described to another group of volunteers, almost all said choosing the easy task would be unfair. This shows that most people apply weaker moral standards to themselves than to others.

Which of the following is an assumption required by this argument?

(A) At least some volunteers who said they had acted fairly in choosing the easy task would have said that it was unfair for someone else to do so.

(B) The most moral choice for the volunteers would have been to have the computer assign the two tasks randomly.

(C) There were at least some volunteers who were assigned to do the hard task and felt that the assignment was unfair.

(D) On average, the volunteers to whom the scenario was described were more accurate in their moral judgments than the other volunteers were.

(E) At least some volunteers given the choice between assigning the tasks themselves and having the computer assign them felt that they had made the only fair choice available to them.

CR08527

651. Country X's recent stock-trading scandal should not diminish investors' confidence in the country's stock market. For one thing, **the discovery of the scandal confirms that Country X has a strong regulatory system**, as the following considerations show. In any stock market, some fraudulent activity is inevitable. If a stock market is well regulated, any significant stock-trading fraud in it will very likely be discovered. This deters potential perpetrators and facilitates improvement in regulatory processes.

In the argument, the portion in boldface plays which of the following roles?

(A) It is the argument's only conclusion.

(B) It is a conclusion for which the argument provides support and which itself is used to support the argument's main conclusion.

(C) It is the argument's main conclusion and is supported by another explicitly stated conclusion for which further support is provided.

(D) It is an assumption for which no explicit support is provided and is used to support the argument's only conclusion.

(E) It is a compound statement containing both the argument's main conclusion and an assumption used to support that conclusion.

CR05644

652. **Delta Products Inc. has recently switched at least partly from older technologies using fossil fuels to new technologies powered by electricity.** The question has been raised whether it can be concluded that **for a given level of output Delta's operation now causes less fossil fuel to be consumed than it did formerly.** The answer, clearly, is yes, since the amount of fossil fuel used to generate the electricity needed to power the new technologies is less than the amount needed to power the older technologies, provided level of output is held constant.

In the argument given, the two boldfaced portions play which of the following roles?

(A) The first identifies the content of the conclusion of the argument; the second provides support for that conclusion.

(B) The first provides support for the conclusion of the argument; the second identifies the content of that conclusion.

(C) The first states the conclusion of the argument; the second calls that conclusion into question.

(D) The first provides support for the conclusion of the argument; the second calls that conclusion into question.

(E) Each provides support for the conclusion of the argument.

CR00907

653. Theater Critic: The play *La Finestrina*, now at Central Theater, was written in Italy in the eighteenth century. The director claims that this production is as similar to the original production as is possible in a modern theater. Although the actor who plays Harlequin the clown gives a performance very reminiscent of the twentieth-century American comedian Groucho Marx, Marx's comic style was very much within the comic acting tradition that had begun in sixteenth-century Italy.

The considerations given best serve as part of an argument that

(A) modern audiences would find it hard to tolerate certain characteristics of a historically accurate performance of an eighteenth-century play

(B) Groucho Marx once performed the part of the character Harlequin in *La Finestrina*

(C) in the United States the training of actors in the twentieth century is based on principles that do not differ radically from those that underlay the training of actors in eighteenth-century Italy

(D) the performance of the actor who plays Harlequin in *La Finestrina* does not serve as evidence against the director's claim

(E) the director of *La Finestrina* must have advised the actor who plays Harlequin to model his performance on comic performances of Groucho Marx

CR07257

654. Although the discount stores in Goreville's central shopping district are expected to close within five years as a result of competition from a SpendLess discount department store that just opened, those locations will not stay vacant for long. In the five years since the opening of Colson's, a nondiscount department store, a new store has opened at the location of every store in the shopping district that closed because it could not compete with Colson's.

Which of the following, if true, most seriously weakens the argument?

(A) Many customers of Colson's are expected to do less shopping there than they did before the SpendLess store opened.

(B) Increasingly, the stores that have opened in the central shopping district since Colson's opened have been discount stores.

(C) At present, the central shopping district has as many stores operating in it as it ever had.

(D) Over the course of the next five years, it is expected that Goreville's population will grow at a faster rate than it has for the past several decades.

(E) Many stores in the central shopping district sell types of merchandise that are not available at either SpendLess or Colson's.

CR05685

655. Last year all refuse collected by Shelbyville city services was incinerated. This incineration generated a large quantity of residual ash. In order to reduce the amount of residual ash Shelbyville generates this year to half of last year's total, the city has revamped its collection program. This year city services will separate for recycling enough refuse to reduce the number of truckloads of refuse to be incinerated to half of last year's number.

Which of the following is required for the revamped collection program to achieve its aim?

(A) This year, no materials that city services could separate for recycling will be incinerated.

(B) Separating recyclable materials from materials to be incinerated will cost Shelbyville less than half what it cost last year to dispose of the residual ash.

(C) Refuse collected by city services will contain a larger proportion of recyclable materials this year than it did last year.

(D) The refuse incinerated this year will generate no more residual ash per truckload incinerated than did the refuse incinerated last year.

(E) The total quantity of refuse collected by Shelbyville city services this year will be no greater than that collected last year.

CR01801

656. Veterinarians generally derive some of their income from selling several manufacturers' lines of pet-care products. Knowing that pet owners rarely throw away mail from their pet's veterinarian unread, one manufacturer of pet-care products offered free promotional materials on its products to veterinarians for mailing to their clients. Very few veterinarians accepted the offer, however, even though the manufacturer's products are of high quality.

Which of the following, if true, most helps to explain the veterinarians' reaction to the manufacturer's promotional scheme?

(A) Most of the veterinarians to whom the free promotional materials were offered were already selling the manufacturer's pet-care products to their clients.

(B) The special promotional materials were intended as a supplement to the manufacturer's usual promotional activities rather than as a replacement for them.

(C) The manufacturer's products, unlike most equally good competing products sold by veterinarians, are also available in pet stores and in supermarkets.

(D) Many pet owners have begun demanding quality in products they buy for their pets that is as high as that in products they buy for themselves.

(E) Veterinarians sometimes recommend that pet owners use products formulated for people when no suitable product specially formulated for animals is available.

CR00778

657. The average hourly wage of television assemblers in Vernland has long been significantly lower than that in neighboring Borodia. Since Borodia dropped all tariffs on Vernlandian televisions three years ago, the number of televisions sold annually in Borodia has not changed. However, recent statistics show a drop in the number of television assemblers in Borodia. Therefore, updated trade statistics will probably indicate that the number of televisions Borodia imports annually from Vernland has increased.

Which of the following is an assumption on which the argument depends?

(A) The number of television assemblers in Vernland has increased by at least as much as the number of television assemblers in Borodia has decreased.

(B) Televisions assembled in Vernland have features that televisions assembled in Borodia do not have.

(C) The average number of hours it takes a Borodian television assembler to assemble a television has not decreased significantly during the past three years.

(D) The number of televisions assembled annually in Vernland has increased significantly during the past three years.

(E) The difference between the hourly wage of television assemblers in Vernland and the hourly wage of television assemblers in Borodia is likely to decrease in the next few years.

CR05725

658. Guidebook writer: I have visited hotels throughout the country and have noticed that in those built before 1930 the quality of the original carpentry work is generally superior to that in hotels built afterward. Clearly carpenters working on hotels before 1930 typically worked with more skill, care, and effort than carpenters who have worked on hotels built subsequently.

Which of the following, if true, most seriously weakens the guidebook writer's argument?

(A) The quality of original carpentry in hotels is generally far superior to the quality of original carpentry in other structures, such as houses and stores.

(B) Hotels built since 1930 can generally accommodate more guests than those built before 1930.

(C) The materials available to carpenters working before 1930 were not significantly different in quality from the materials available to carpenters working after 1930.

(D) The better the quality of original carpentry in a building, the less likely that building is to fall into disuse and be demolished.

(E) The average length of apprenticeship for carpenters has declined significantly since 1930.

CR02997

659. Scientists typically do their most creative work before the age of forty. It is commonly thought that this happens because aging by itself brings about a loss of creative capacity. However, studies show that **of scientists who produce highly creative work beyond the age of forty, a disproportionately large number entered their field at an older age than is usual**. Since by the age of forty the large majority of scientists have been working in their field for at least fifteen years, the studies' finding strongly suggests that the real reason why scientists over forty rarely produce highly creative work is not that they have aged but rather that **scientists over forty have generally spent too long in their field**.

In the argument given, the two portions in boldface play which of the following roles?

(A) The first is a claim, the accuracy of which is at issue in the argument; the second is a conclusion drawn on the basis of that claim.

(B) The first is an objection that has been raised against a position defended in the argument; the second is that position.

(C) The first is evidence that has been used to support an explanation that the argument challenges; the second is that explanation.

(D) The first is evidence that has been used to support an explanation that the argument challenges; the second is a competing explanation that the argument favors.

(E) The first provides evidence to support an explanation that the argument favors; the second is that explanation.

CR03818

660. NorthAir charges low fares for its economy-class seats, but it provides very cramped seating and few amenities. Market research shows that economy passengers would willingly pay more for wider seating and better service, and additional revenue provided by these higher ticket prices would more than cover the additional cost of providing these amenities. Even though NorthAir is searching for ways to improve its profitability, it has decided not to make these improvements.

Which of the following, if true, would most help to explain NorthAir's decision in light of its objectives?

(A) None of NorthAir's competitors offers significantly better seating and service to economy-class passengers than NorthAir does.

(B) On many of the routes that NorthAir flies, it is the only airline to offer direct flights.

(C) A few of NorthAir's economy-class passengers are satisfied with the service they receive, given the low price they pay.

(D) Very few people avoid flying on NorthAir because of the cramped seating and poor service offered in economy class.

(E) The number of people who would be willing to pay the high fares NorthAir charges for its business-class seats would decrease if its economy-class seating were more acceptable.

CR00774

661. Which of the following most logically completes the argument given?

Asthma, a chronic breathing disorder, is significantly more common today among adult competitive swimmers than it is among competitive athletes who specialize in other sports. Although chlorine is now known to be a lung irritant and swimming pool water is generally chlorinated, it would be rash to assume that frequent exposure to chlorine is the explanation of the high incidence of asthma among these swimmers, since _____.

(A) young people who have asthma are no more likely to become competitive athletes than are young people who do not have asthma

(B) competitive athletes who specialize in sports other than swimming are rarely exposed to chlorine

(C) competitive athletes as a group have a significantly lower incidence of asthma than do people who do not participate in competitive athletics

(D) until a few years ago, physicians routinely recommended competitive swimming to children with asthma, in the belief that this form of exercise could alleviate asthma symptoms

(E) many people have asthma without knowing they have it and thus are not diagnosed with the condition until they begin engaging in very strenuous activities, such as competitive athletics

CR01289

662. In the country of Marut, the Foreign Trade Agency's records were reviewed in 1994 in light of information then newly available about neighboring Goro. The review revealed that in every year since 1963, the agency's projection of what Goro's gross national product (GNP) would be five years later was a serious underestimate. The review also revealed that in every year since 1963, the agency estimated Goro's GNP for the previous year—a Goro state secret—very accurately.

Of the following claims, which is most strongly supported by the statements given?

(A) Goro's GNP fluctuated greatly between 1963 and 1994.

(B) Prior to 1995, Goro had not released data intended to mislead the agency in making its five-year projections.

(C) The amount by which the agency underestimated the GNP it projected for Goro tended to increase over time.

(D) Even before the new information came to light, the agency had reason to think that at least some of the five-year projections it had made were inaccurate.

(E) The agency's five-year projections of Goro's GNP had no impact on economic planning in Marut.

CR05082

663. Vargonia has just introduced a legal requirement that student-teacher ratios in government-funded schools not exceed a certain limit. All Vargonian children are entitled to education, free of charge, in these schools. When a recession occurs and average incomes fall, the number of children enrolled in government-funded schools tends to increase. Therefore, though most employment opportunities contract in economic recessions, getting a teaching job in Vargonia's government-funded schools will not be made more difficult by a recession.

Which of the following would be most important to determine in order to evaluate the argument?

(A) Whether in Vargonia there are any schools not funded by the government that offer children an education free of charge

(B) Whether the number of qualified applicants for teaching positions in government-funded schools increases significantly during economic recessions

(C) What the current student-teacher ratio in Vargonia's government-funded schools is

(D) What proportion of Vargonia's workers currently hold jobs as teachers in government-funded schools

(E) Whether in the past a number of government-funded schools in Vargonia have had student-teacher ratios well in excess of the new limit

CR09951

664. In Colorado subalpine meadows, nonnative dandelions co-occur with a native flower, the larkspur. Bumblebees visit both species, creating the potential for interactions between the two species with respect to pollination. In a recent study, researchers selected 16 plots containing both species; all dandelions were removed from eight plots; the remaining eight control plots were left undisturbed. The control plots yielded significantly more larkspur seeds than the dandelion-free plots, leading the researchers to conclude that the presence of dandelions facilitates pollination (and hence seed production) in the native species by attracting more pollinators to the mixed plots.

Which of the following, if true, most seriously undermines the researchers' reasoning?

(A) Bumblebees preferentially visit dandelions over larkspurs in mixed plots.

(B) In mixed plots, pollinators can transfer pollen from one species to another to augment seed production.

(C) If left unchecked, nonnative species like dandelions quickly crowd out native species.

(D) Seed germination is a more reliable measure of a species' fitness than seed production.

(E) Soil disturbances can result in fewer blooms, and hence lower seed production.

CR11453

665. An experiment was done in which human subjects recognize a pattern within a matrix of abstract designs and then select another design that completes that pattern. The results of the experiment were surprising. The lowest expenditure of energy in neurons in the brain was found in those subjects who performed most successfully in the experiments.

Which of the following hypotheses best accounts for the findings of the experiment?

(A) The neurons of the brain react less when a subject is trying to recognize patterns than when the subject is doing other kinds of reasoning.

(B) Those who performed best in the experiment experienced more satisfaction when working with abstract patterns than did those who performed less well.

(C) People who are better at abstract pattern recognition have more energy-efficient neural connections.

(D) The energy expenditure of the subjects' brains increases when a design that completes the initially recognized pattern is determined.

(E) The task of completing a given design is more capably performed by athletes, whose energy expenditure is lower when they are at rest.

CR01202

666. With seventeen casinos, Moneyland operates the most casinos in a certain state. Although intent on expanding, it was outmaneuvered by Apex Casinos in negotiations to acquire the Eldorado chain. To complete its acquisition of Eldorado, Apex must sell five casinos to comply with a state law forbidding any owner to operate more than one casino per county. Since Apex will still be left operating twenty casinos in the state, it will then have the most casinos in the state.

Which of the following, if true, most seriously undermines the prediction?

(A) Apex, Eldorado, and Moneyland are the only organizations licensed to operate casinos in the state.

(B) The majority of Eldorado's casinos in the state will need extensive renovations if they are to continue to operate profitably.

(C) Some of the state's counties do not permit casinos.

(D) Moneyland already operates casinos in the majority of the state's counties.

(E) Apex will use funds it obtains from the sale of the five casinos to help fund its acquisition of the Eldorado chain.

CR05093

667. It is widely assumed that people need to engage in intellectual activities such as solving crossword puzzles or mathematics problems in order to maintain mental sharpness as they age. In fact, however, simply talking to other people—that is, participating in social interaction, which engages many mental and perceptual skills—suffices. Evidence to this effect comes from a study showing that the more social contact people report, the better their mental skills.

Which of the following, if true, most seriously weakens the force of the evidence cited?

(A) As people grow older, they are often advised to keep exercising their physical and mental capacities in order to maintain or improve them.

(B) Many medical conditions and treatments that adversely affect a person's mental sharpness also tend to increase that person's social isolation.

(C) Many people are proficient both in social interactions and in solving mathematical problems.

(D) The study did not itself collect data but analyzed data bearing on the issue from prior studies.

(E) The tasks evaluating mental sharpness for which data were compiled by the study were more akin to mathematics problems than to conversation.

8.5 Answer Key

544. E
545. C
546. E
547. A
548. A
549. B
550. E
551. E
552. C
553. C
554. A
555. E
556. A
557. D
558. B
559. D
560. D
561. C
562. B
563. D
564. D
565. C
566. E
567. B
568. A
569. E
570. B
571. E
572. D
573. E
574. B
575. E
576. D
577. A
578. A
579. B
580. D
581. D
582. E
583. D
584. B
585. C
586. D
587. C
588. E
589. C
590. D
591. A
592. D
593. A
594. C
595. E
596. C
597. B
598. C
599. A
600. A
601. D
602. A
603. C
604. C
605. A
606. C
607. E
608. C
609. E
610. A
611. A
612. E
613. D
614. C
615. C
616. D
617. D
618. D
619. A
620. A
621. D
622. A
623. E
624. E
625. A
626. D
627. E
628. E
629. E
630. B
631. E
632. D
633. D
634. B
635. A
636. D
637. C
638. B
639. D
640. D
641. A
642. C
643. C
644. B
645. E
646. D
647. A
648. C
649. B
650. A
651. B
652. B
653. D
654. B
655. D
656. C
657. C
658. D
659. E
660. E
661. D
662. D
663. B
664. E
665. C
666. A
667. B

8.6 Answer Explanations

The following discussion is intended to familiarize you with the most efficient and effective approaches to critical reasoning questions. The particular questions in this chapter are generally representative of the kinds of critical reasoning questions you will encounter on the GMAT exam. Remember that it is the problem solving strategy that is important, not the specific details of a particular question.

*CR09616

544. Neuroscientist: Memory evolved to help animals react appropriately to situations they encounter by drawing on the past experience of similar situations. But this does not require that animals perfectly recall every detail of all their experiences. Instead, to function well, memory should generalize from past experiences that are similar to the current one.

The neuroscientist's statements, if true, most strongly support which of the following conclusions?

(A) At least some animals perfectly recall every detail of at least some past experiences.

(B) Perfectly recalling every detail of all their past experiences could help at least some animals react more appropriately than they otherwise would to new situations they encounter.

(C) Generalizing from past experiences requires clear memories of most if not all the details of those experiences.

(D) Recalling every detail of all past experiences would be incompatible with any ability to generalize from those experiences.

(E) Animals can often react more appropriately than they otherwise would to situations they encounter if they draw on generalizations from past experiences of similar situations.

Argument Construction

Situation A neuroscientist claims that memory evolved to help animals learn how to react appropriately by generalizing from past experiences but that this does not require animals to remember all details of those experiences.

Reasoning *What conclusion would the neuroscientist's theory about memory most strongly support?* The neuroscientist asserts that the evolutionary function of memory is to help animals learn to react appropriately by drawing on generalizations from similar experiences they have had. If memory is to serve this function, drawing on generalizations must actually help animals learn to react more appropriately than they otherwise would, even when they do not remember all the details of past experiences.

A Even if no animal ever recalls all the details of any past experience, animals could still learn through generalizations, as the neuroscientist claims.

B This statement could be false even if all of what the neuroscientist says is true. Even if it were never helpful for any animal to recall every detail of all its past experiences, animals could still benefit by learning through generalizations.

C Generalizations from experiences might be made while the experiences are occurring, so that only the generalizations and not the details need to be remembered.

D The neuroscientist only claims that remembering perfect details is not required for memory to serve its function, not that such perfect recall is incompatible with memory serving its function.

E **Correct.** If the evolutionary function of memory is to help animals react more appropriately by drawing on generalizations from past experiences, it follows that animal memories can often successfully serve this function in this manner.

The correct answer is E.

*These numbers correlate with the online test bank question number. See the GMAT Official Guide Online Index in the back of this book.

CR09994

545. Astronomer: Most stars are born in groups of thousands, each star in a group forming from the same parent cloud of gas. Each cloud has a unique, homogeneous chemical composition. Therefore, whenever two stars have the same chemical composition as each other, they must have originated from the same cloud of gas.

Which of the following, if true, would most strengthen the astronomer's argument?

(A) In some groups of stars, not every star originated from the same parent cloud of gas.

(B) Clouds of gas of similar or identical chemical composition may be remote from each other.

(C) Whenever a star forms, it inherits the chemical composition of its parent cloud of gas.

(D) Many stars in vastly different parts of the universe are quite similar in their chemical compositions.

(E) Astronomers can at least sometimes precisely determine whether a star has the same chemical composition as its parent cloud of gas.

Argument Evaluation

Situation Most stars are born in groups, any one of which forms from a parent gas cloud with a unique, homogenous chemical composition.

Reasoning What would be additional evidence that any two stars with the same chemical composition originated from the same gas cloud? The implicit reasoning is that since the chemical composition of each gas cloud is unique and homogenous, any two stars that formed from gas with the same chemical composition must have originated from the same cloud. The astronomer then infers that if two stars have the same composition now, they must have originated from the same cloud. This inference requires the assumption that the composition each star has now depends only on the composition of the cloud in which it originated. Any evidence that supports this assumption will strengthen the argument.

A Whether or not stars born in different clouds of gas are ever in the same "group" is not clearly relevant to whether or not they ever have the same chemical composition.

B How remote clouds of similar compositions are from each other is not clearly relevant to whether stars that have the same chemical composition may have formed from different clouds of gas. Also, the suggestion that different gas clouds may have identical compositions conflicts with the astronomer's premise that the composition of each cloud from which stars form is unique.

C **Correct.** If each star's composition is identical to that of its parent cloud, and each cloud's composition is unique, then any two stars identical in composition must have formed from the same parent cloud.

D If anything, this would suggest that stars with the same composition might have formed from different clouds, so it would weaken rather than strengthen the argument.

E If astronomers could do this, they might be able to obtain additional evidence for or against the position taken in the argument, but this, in itself, provides no reason to suppose that the evidence would support, rather than weaken, that position. They might find that the stars' compositions do not precisely correlate with the compositions of the stars' parent gas clouds.

The correct answer is C.

CR08017

546. With employer-paid training, workers have the potential to become more productive not only in their present employment but also in any number of jobs with different employers. To increase the productivity of their workforce, many firms are planning to maintain or even increase their investments in worker training. But some training experts object that if a trained worker is hired away by another firm, the employer that paid for the training has merely subsidized a competitor. They note that such hiring has been on the rise in recent years.

Which of the following would, if true, contribute most to defeating the training experts' objection to the firms' strategy?

(A) Firms that promise opportunities for advancement to their employees get, on average, somewhat larger numbers of job applications from untrained workers than do firms that make no such promise.

(B) In many industries, employees that take continuing-education courses are more competitive in the job market.

(C) More and more educational and training institutions are offering reduced tuition fees to firms that subsidize worker training.

(D) Research shows that workers whose training is wholly or partially subsidized by their employer tend to get at least as much training as do workers who pay for all their own training.

(E) For most firms that invest in training their employees, the value added by that investment in employees who stay exceeds the value lost through other employees' leaving to work for other companies.

Evaluation of a Plan

Situation Many firms pay to train their workers in order to increase their workforces' productivity. But in recent years firms have been increasingly hiring away from each other workers who have had such training.

Reasoning *What would most help address the concern that firms that pay to train workers are thereby subsidizing competitors that hire away those workers?* In order for the employer-paid training to be worthwhile for a given firm despite the risk of subsidizing competitors that may hire away the trained workers, that firm has to gain more benefits from the training than it loses by subsidizing such competitors. Any evidence that this is true for most firms would help to address the experts' concern.

A A typical firm does not necessarily want larger numbers of applications from unqualified workers. And if hired, those workers can still be hired away by competitors after the firm has paid to train them, just as the experts warned.

B This suggests that in many industries, companies, rather than investing in employee training, prefer to hire employees who already have specifically relevant training (perhaps funded by other companies). If anything, this slightly supports, rather than defeats, the training experts' view. No firm has an interest in making its own employees more competitive in the job market unless the firm is likely to benefit from their being so.

C Even firms that pay reduced tuition fees for worker training may lose the money they pay for those fees and effectively subsidize competitors that hire the trained employees away. So this does not defeat the training experts' objection.

D The more highly trained workers—regardless of whether their training was company subsidized or not—would presumably be prime targets for recruitment by competing firms, just as the experts warned. The research finding in question does not help defeat the experts' objection.

E **Correct.** This explicitly indicates that most firms gain more than they lose from the general practice of firms paying to train their workers.

The correct answer is E.

CR01107

547. Candle Corporation's television stations are likely to have more income from advertisers than previously. This is because advertisers prefer to reach people in the 18- to 49-year-old age group and the number of people in that group watching Candle television is increasing. Furthermore, among Candle viewers, the percentage of viewers 18 to 49 years old is increasing.

Which of the following, if true, would most strengthen the argument that Candle Corporation will receive more income from advertisers?

(A) Advertisers carefully monitor the demographic characteristics of television audiences and purchase advertising time to reach the audiences they prefer to reach.

(B) Among people over 49 years old, fewer viewers of Candle stations buy products advertised on television than do viewers of other stations.

(C) There will be increasingly more advertisements on television that are directed at viewers who are over 49 years old.

(D) Candle stations plan to show reruns of television shows during hours when other stations run shows for the first time.

(E) People 18 to 49 years old generally have less disposable income to spend than do people over 49 years old.

Argument Evaluation

Situation Both the number and the percentage of Candle television viewers who are 18 to 49 years old are increasing. Advertisers prefer to reach people in this age group.

Reasoning *What evidence, when combined with the cited facts, would most support the prediction that Candle will receive more income from advertisers?* The argument assumes that the increasing number and percentage of Candle viewers in the age group that advertisers prefer to reach will probably encourage advertisers to spend more on advertising with Candle. This assumption could be supported by evidence that the advertisers realize that Candle is getting more viewers in that preferred age range or by evidence that this awareness will influence the advertisers' purchase of advertising time.

A **Correct.** Advertisers monitoring demographics will probably realize that Candle has increasing numbers of viewers in their preferred age range. If they purchase advertising to reach viewers in that age range, then they will probably purchase more advertising time with Candle.

B This gives advertisers less reason to advertise on Candle to reach viewers over 49 years old. Other things being equal, that makes Candle likely to receive less income from advertisers, not more income.

C Since the percentage of Candle viewers 18 to 49 years old is growing, the percentage over 49 years old is probably shrinking. This could make advertisers seeking to reach older viewers less inclined to advertise on Candle even as they increase their overall television advertising.

D Advertisers are not necessarily inclined to purchase more advertising during showings of reruns than during original airings of television shows and may even be inclined to purchase less advertising during such showings.

E This gives advertisers less incentive to try to reach audiences between 18 and 49 years old and hence less reason to purchase advertising on Candle.

The correct answer is A.

CR12584

548. A provincial government plans to raise the gasoline tax to give people an incentive to drive less, reducing traffic congestion in the long term. However, skeptics point out that most people in the province live in areas where cars are the only viable transportation to jobs and stores and therefore cannot greatly change their driving habits in response to higher gasoline prices.

In light of the skeptics' objection, which of the following, if true, would most logically support the prediction that the government's plan will achieve its goal of reducing traffic congestion?

(A) The revenue from the tax will be used to make public transportation a viable means of transportation to jobs and stores for far more people.

(B) The tax will encourage many residents to switch to more fuel-efficient cars, reducing air pollution and other problems.

(C) Because gasoline has been underpriced for decades, the province has many neighborhoods where cars are the only viable means of transportation.

(D) Most residents who cannot greatly change their driving habits could compensate for high gasoline prices by reducing other expenses.

(E) Traffic congestion is an especially serious problem for people for whom cars are the only viable means of transportation.

Evaluation of a Plan

Situation A provincial government plans to raise the gasoline tax in order to reduce traffic congestion by discouraging people from driving. But skeptics point out that most people in the province have no viable form of transportation other than driving.

Reasoning *What would suggest that raising the gasoline tax will reduce traffic congestion even though most people in the province have no viable form of transportation other than driving?* The skeptics point out that since most people in the province have no way to reach jobs or stores except by car, they will not be able to reduce their driving much even if the gasoline tax increases. Any evidence that raising the gasoline tax would reduce traffic congestion despite this obstacle would help to support the plan in light of the skeptics' objection.

A **Correct.** If the tax will fund these public transit improvements, then far more people will have a viable means of transportation other than driving, undermining the basis of the skeptics' objection.

B People switching to fuel-efficient cars would not reduce traffic congestion.

C This essentially only tends to support the skeptics' objection. Unless the plan somehow helps to alleviate the necessity of driving (by, for example, making alternative transportation available), the information provided gives no reason to suppose that the higher costs would significantly reduce traffic congestion.

D If residents cannot greatly change their driving habits, then the tax will not reduce traffic congestion.

E This suggests that many residents in the province could benefit if the plan did reduce traffic congestion, but it does not provide a reason to believe the plan will have that effect.

The correct answer is A.

CR03940

549. Editorial: The roof of Northtown's municipal equipment-storage building collapsed under the weight of last week's heavy snowfall. The building was constructed recently and met local building-safety codes in every particular, except that the nails used for attaching roof supports to the building's columns were of a smaller size than the codes specify for this purpose. Clearly, this collapse exemplifies how even a single, apparently insignificant departure from safety standards can have severe consequences.

Which of the following, if true, most seriously weakens the editorial's argument?

(A) The only other buildings to suffer roof collapses from the weight of the snowfall were older buildings constructed according to less exacting standards than those in the codes.

(B) The amount of snow that accumulated on the roof of the equipment-storage building was greater than the predicted maximum that was used in drawing up the safety codes.

(C) Because the equipment-storage building was not intended for human occupation, some safety-code provisions that would have applied to an office building did not apply to it.

(D) The municipality of Northtown itself has the responsibility for ensuring that buildings constructed within its boundaries meet the provisions of the building-safety codes.

(E) Because the equipment-storage building was used for storing snow-removal equipment, the building was almost completely empty when the roof collapsed.

Argument Evaluation

Situation The roof of a recently constructed building collapsed under heavy snowfall. The only way the building did not meet safety standards was that some nails for the roof supports were smaller than prescribed by the building codes.

Reasoning *What would make it less likely that the building's collapse resulted from a single, apparently minor departure from safety standards?* The building met safety standards except for the size of the nails. So if the collapse exemplifies how a departure from safety standards can have severe consequences, as the conclusion claims, then the size of the nails had to be responsible for the collapse. Thus, evidence that a factor other than the size of the nails could fully account for the collapse would weaken the argument.

A This suggests that the snow would not have been heavy enough to collapse the roof if the construction had completely met the safety standards, so it strengthens, rather than weakens, the argument.

B **Correct.** This suggests that the snow could have collapsed the roof even if the nails had met the safety standards, thus casting doubt on the assumption that the nails' inadequacy was responsible for the collapse.

C The claim that the safety requirements for this building were weaker than some others tends slightly to strengthen, rather than weaken, the hypothesis that the bad consequences resulted partly from a failure to comply. Even if safety-code provisions for an equipment-storage building differ from those for an office building, they may still be adequate to ensure the roof's stability.

D The question of who was responsible for ensuring compliance with the safety codes is irrelevant to whether a failure to comply was responsible for the roof's collapse.

E This suggests that the alleged consequences of failing to meet safety standards were less severe than they could have been, but it is irrelevant to determining the cause of the collapse.

The correct answer is B.

CR12078

550. Political theorist: Even with the best spies, area experts, and satellite surveillance, foreign policy assessments can still lack important information. In such circumstances intuitive judgment is vital. A national leader with such judgment can make good decisions about foreign policy even when current information is incomplete, since _____.

Which of the following, if true, most logically completes the argument?

(A) the central reason for failure in foreign policy decision making is the absence of critical information

(B) those leaders whose foreign policy decisions have been highly ranked have also been found to have good intuitive judgment

(C) both intuitive judgment and good information are required for sound decision making

(D) good foreign policy decisions often lead to improved methods of gathering information

(E) intuitive judgment can produce good decisions based on past experience, even when there are important gaps in current information

Argument Construction

Situation National leaders sometimes must make foreign policy decisions while lacking important information.

Reasoning *What would most help support the claim that a national leader with intuitive judgment can make good foreign policy decisions without complete information?* The word *since* preceding the blank indicates that the blank should be filled with a premise supporting the statement immediately before the blank. So an observation that supports the claim that a national leader with intuitive judgment can make good foreign policy decisions without complete information would logically complete the argument.

A This gives us no reason to suppose that intuitive judgment helps national leaders avoid such failures.

B This does not specify who ranked the foreign policy decisions, nor how they determined the rankings, so it gives us no reason to accept those rankings. For all we know, the anonymous rankers may have used the dubious rankings they created as the sole evidence for their so-called findings about which leaders have good intuitive judgment.

C This implies that intuitive judgment alone is inadequate without good information, so it undermines rather than supports the claim that national leaders can make good foreign policy decisions with intuitive judgment while lacking complete information.

D This gives us no reason to suppose that good foreign policy decisions can be made in the first place by leaders lacking important information.

E **Correct.** This suggests that national leaders can make good foreign policy decisions using intuitive judgment based on their past foreign policy experience, even without complete information about the current situations they're facing.

The correct answer is E.

CR01295

551. During the earliest period of industrialization in Britain, steam engines were more expensive to build and operate than either windmills or water mills, the other practicable sources of power for factories. Yet despite their significant cost disadvantage, steam-powered factories were built in large numbers well before technical improvements brought their cost down. Furthermore, they were built even in regions where geographical conditions permitted the construction of wind- and water-powered factories close to major markets.

Which of the following, if true, most helps to explain the proliferation of steam-powered factories during the earliest period of industrialization in Britain?

(A) In many areas of Britain, there were fewer steam-powered factories than wind- or water-powered factories in the earliest period of industrialization.

(B) Unlike wind- or water-powered factories, steam-powered factories were fueled with coal, which sometimes had to be transported significant distances from the mine to the site of the factory.

(C) It was both difficult and expensive to convert a factory from wind power or water power to steam power.

(D) In the early period of industrialization, many goods sold in towns and cities could not be mass-produced in factories.

(E) In Britain, the number of sites where a wind- or water-powered factory could be built was insufficient to provide for all of the demand for factory-produced goods at the time.

Argument Construction

Situation Although steam engines were more expensive than windmills and water mills in early industrial Britain, many steam-powered factories were built even in regions where the construction of wind- and water-powered factories was geographically feasible.

Reasoning *Why might steam-powered factories have proliferated despite their cost disadvantage?* Early industrialists would have needed some positive reason to choose steam over less expensive power sources for their factories. For example, steam engines might have operated faster or more effectively than windmills or water mills. Or steam engines might have received government subsidies. Or conditions restricting the number or locations of windmills and water mills might have forced industrialists to use steam power instead.

A This suggests that the steam-powered factories did not initially proliferate as widely as they might have, but it does not explain why they proliferated to the extent that they did.

B The inconvenience of transporting coal for steam-powered factories would have made those factories less likely to proliferate, not more likely.

C The difficulty of converting factories to steam power would have made steam-powered factories less likely to proliferate, not more likely.

D The technological inability to mass-produce popular products in factories would have made factories in general less likely to proliferate, including steam-powered factories.

E **Correct.** The inadequate number of sites for wind- and water-powered factories might have encouraged early industrialists to build steam-powered factories instead, since the high demand for factory-produced goods could have made these factories profitable despite their cost disadvantage.

The correct answer is E.

CR03938

552. Snowmaking machines work by spraying a mist that freezes immediately on contact with cold air. Because the sudden freezing kills bacteria, QuickFreeze is planning to market a wastewater purification system that works on the same principle. The process works only when temperatures are cold, however, so municipalities using it will still need to maintain a conventional system.

Which of the following, if true, provides the strongest grounds for a prediction that municipalities will buy QuickFreeze's purification system despite the need to maintain a conventional purification system as well?

(A) Bacteria are not the only impurities that must be removed from wastewater.

(B) Many municipalities have old wastewater purification systems that need to be replaced.

(C) Conventional wastewater purification systems have not been fully successful in killing bacteria at cold temperatures.

(D) During times of warm weather, when it is not in use, QuickFreeze's purification system requires relatively little maintenance.

(E) Places where the winters are cold rarely have a problem of water shortage.

Evaluation of a Plan

Situation QuickFreeze is planning to market wastewater purification systems that work by spraying a mist that freezes on contact with cold air. The sudden freezing kills bacteria. Because the system works only at cold temperatures, municipalities using it will still need to maintain a conventional system.

Reasoning *Which statement provides the strongest grounds for thinking that at least some municipalities will buy the purification system despite the need to maintain a conventional purification system as well?* The passage tells us why a municipality using a QuickFreeze wastewater purification system would still need a conventional system. But why would a municipality want the QuickFreeze system *in addition* to a conventional system? If conventional systems are not fully effective at cold temperatures, the QuickFreeze system would allow municipalities that sometimes experience cold temperatures to purify their wastewater more effectively.

A There is no basis in the passage for determining whether the QuickFreeze system will help remove impurities other than bacteria from wastewater. If it does not, this answer choice implies that the QuickFreeze system would not be sufficient for purifying wastewater. This would actually undermine the prediction.

B The passage states that municipalities using the QuickFreeze system would still need a conventional system. Thus, the old conventional wastewater systems would still need to be replaced with new conventional systems. This answer choice provides no reason to think municipalities would buy the QuickFreeze system.

C **Correct.** This statement, if true, would strengthen the prediction, because it provides a valid reason why the QuickFreeze system could be needed alongside conventional ones: it is more effective in cold weather.

D Although this claim does undercut one reason for thinking municipalities might *not* be likely to purchase the QuickFreeze system, it provides little reason to think that they will purchase such a system. Perhaps in times of cold weather, the QuickFreeze system is very expensive to maintain.

E The issue of whether or not there are water shortages in places where winters are cold is not directly relevant. If conventional wastewater systems are sufficient to purify water in such places, municipalities would not need the QuickFreeze system (as they would still need to maintain a conventional purification system).

The correct answer is C.

CR05080

553. **Plant scientists have used genetic engineering on seeds to produce crop plants that are highly resistant to insect damage**. Unfortunately, the seeds themselves are quite expensive, and the plants require more fertilizer and water to grow well than normal ones. Accordingly, **for most farmers the savings on pesticides would not compensate for the higher seed costs and the cost of additional fertilizer**. However, since consumer demand for grains, fruits, and vegetables grown without the use of pesticides continues to rise, the use of genetically engineered seeds of this kind is likely to become widespread.

In the argument given, the two portions in **boldface** play which of the following roles?

(A) The first supplies a context for the argument; the second is the argument's main conclusion.

(B) The first introduces a development that the argument predicts will have a certain outcome; the second is a state of affairs that, according to the argument, contributes to bringing about that outcome.

(C) The first presents a development that the argument predicts will have a certain outcome; the second acknowledges a consideration that tends to weigh against that prediction.

(D) The first provides evidence to support a prediction that the argument seeks to defend; the second is that prediction.

(E) The first and the second each provide evidence to support the argument's main conclusion.

Argument Construction

Situation Seeds genetically engineered by plant scientists produce crops highly resistant to insect damage and require less use of pesticides. The seeds would be costly to use and would not help most farmers increase their profitability. Nonetheless, consumer demand for pesticide-free food materials is increasing, so genetically engineered seeds are likely to become widely used.

Reasoning *What function is served by the statement that plant scientists have used genetic engineering on seeds to produce insect-resistant crop plants? What function is served by the statement that for most farmers the savings on pesticides would not outweigh other associated costs?* The first describes an innovation—genetically engineered seeds—that allows crops to be grown with little or no use of pesticides. The second notes that despite savings on pesticide use, most farmers would not increase their profits by using the new seeds. The argument's main conclusion, however, is the prediction that use of such genetically engineered seeds will become widespread.

A The second statement is a conclusion—but not the main conclusion—that has the first statement as partial support.

B This correctly characterizes the first statement but not the second. The second statement is not meant to indicate a factor that contributes to the predicted outcome; it indicates, rather, a factor that somewhat counts against the argument's prediction.

C **Correct.** This correctly characterizes both statements.

D The first statement provides partial support for the prediction stated in the argument's main conclusion; the second statement does not state that prediction.

E The second cannot accurately be described as giving evidence to support the prediction stated in the argument's main conclusion.

The correct answer is C.

CR04159

554. Which of the following most logically completes the passage?

Leptin, a protein occurring naturally in the blood, appears to regulate how much fat the body carries by speeding up the metabolism and decreasing the appetite when the body has too much fat. Mice that do not naturally produce leptin have more fat than other mice, but lose fat rapidly when they are given leptin injections. Unfortunately, however, leptin cannot be used as a dietary supplement to control fat, since _____.

(A) the digestive system breaks down proteins before they can enter the bloodstream

(B) there are pharmaceuticals already available that can contribute to weight loss by speeding up the metabolism

(C) people with unusually low levels of leptin in their blood tend to have a high percentage of body fat

(D) the mice that do not naturally produce leptin were from a specially bred strain of mice

(E) mice whose bodies did produce leptin also lost some of their body fat when given leptin injections

Argument Construction

Situation Leptin, a protein naturally occurring in the bloodstream, speeds up metabolism to induce loss of excessive fat. Mice that lack leptin have more fat than other mice, but lose fat when given leptin injections. However, leptin cannot be used as a dietary supplement to control fat.

Reasoning *What would explain the fact that a dietary supplement of leptin will not help to control fat?* Leptin injected into the bloodstream—but not leptin taken as a dietary supplement—helps control fat. So leptin taken as a dietary supplement is either inactivated in the gastrointestinal system or for some other reason fails to enter the bloodstream.

A **Correct.** The digestive system breaks down proteins and would therefore break down leptin, which is a protein. This means that leptin given as a dietary supplement would never reach the bloodstream.

B The question concerns leptin alone, and this new information fails to explain why leptin cannot help control fat if administered as a dietary supplement.

C It is unsurprising that this would be so, but this information does nothing to explain why leptin consumed as a supplement would fail to control fat.

D Presumably leptin administered as a dietary supplement was first tested on mice bred to lack leptin. However, the question about leptin does not concern only mice, but presumably humans and other mammals.

E This suggests that boosting existing normal leptin levels with injections can induce further fat loss. However, this has no obvious relevance to the question raised about why dietary supplements of leptin fail to produce fat loss.

The correct answer is A.

CR05452

555. Suncorp, a new corporation with limited funds, has been clearing large sections of the tropical Amazon forest for cattle ranching. This practice continues even though greater profits can be made from rubber tapping, which does not destroy the forest, than from cattle ranching, which does destroy the forest.

Which of the following, if true, most helps to explain why Suncorp has been pursuing the less profitable of the two economic activities mentioned above?

(A) The soil of the Amazon forest is very rich in nutrients that are important in the development of grazing lands.

(B) Cattle-ranching operations that are located in tropical climates are more profitable than cattle-ranching operations that are located in cold-weather climates.

(C) In certain districts, profits made from cattle ranching are more heavily taxed than profits made from any other industry.

(D) Some of the cattle that are raised on land cleared in the Amazon are killed by wildcats.

(E) The amount of money required to begin a rubber-tapping operation is twice as high as the amount needed to begin a cattle ranch.

Argument Construction

Situation Suncorp is a new corporation with limited funds. It has been clearing large sections of the tropical Amazon forest for ranching, even though rubber-tapping would be more profitable.

Reasoning *What would explain why Suncorp is clearing sections of the rain forest for ranching, even though rubber tapping would be more profitable?* Because Suncorp has limited funds, if rubber tapping has much higher start-up costs, Suncorp might not have enough money to start rubber-tapping operations. If cattle ranching has much lower start-up costs than rubber tapping, Suncorp might be able to afford such an operation.

A This statement gives a reason why cattle ranching in the Amazon might be more profitable than one might otherwise think it would be. However, we already know from the passage that rubber tapping would be more profitable than cattle ranching. So, this answer choice does not help explain why cattle ranching might be preferable to rubber tapping.

B The comparison between the profitableness of cattle ranching in tropical climates and in cold-weather climates is irrelevant. The passage only covers cattle ranching in the tropical Amazon forest. This answer choice would at most explain why Suncorp is undertaking cattle ranching in the Amazon rather than in some cold-weather location.

C This statement makes what needs to be explained harder to understand, for it indicates that cattle ranching in the Amazon might be less profitable than one would otherwise think.

D Like answer choice (C), this statement indicates a disadvantage of cattle ranching in the Amazon. So, it does not explain why cattle ranching would be preferred to some other economic activity.

E **Correct.** Because it costs less to begin cattle ranching than it does to begin rubber tapping, Suncorp—which has limited funds—would have a reason to pursue cattle ranching over a potentially more profitable activity.

The correct answer is E.

CR09963

556. Archaeologists use technology to analyze ancient sites. It is likely that this technology will advance considerably in the near future, allowing archaeologists to gather more information than is currently possible. If they study certain sites now, they risk contaminating or compromising them for future studies. Therefore, in order to maximize the potential for gathering knowledge in the long run, a team of archaeologists plans to delay the examination of a newly excavated site.

Which of the following would be most useful to investigate for the purpose of evaluating the plan's prospects for achieving its goal?

(A) Whether any of the contents of the site will significantly deteriorate before the anticipated technology is available

(B) Whether there will continue to be improvements on the relevant technology

(C) Whether the team can study a site other than the newly excavated site for the time being

(D) Whether the site was inhabited by a very ancient culture

(E) Whether the anticipated technology will damage objects under study

Evaluation of a Plan

Situation To avoid prematurely compromising a newly excavated site, an archaeological team plans to postpone examining it until more advanced technology is developed that will let them gather more information from it. Their goal is to maximize the potential for gathering knowledge.

Reasoning *What would be most helpful to investigate in order to assess how likely it is that delaying examination of the site will maximize the potential for gathering knowledge from it?* In order to maximize (or even increase) the potential for gathering knowledge from the site by delaying its examination, the risk of compromising the site by examining it now has to be greater than the risk that the site will be compromised as much or more by delaying the examination. The delay might also increase the risk that the site will never be examined at all—for example, the team might lose its funding while it delays, or changes in local political conditions might prevent the site's future examination. Investigating any of these risks could be helpful in assessing the likelihood that the team's plan will achieve its goal.

A **Correct.** If any of the site's contents will significantly deteriorate before the technology becomes available, that could reduce the ability to gather future information from the site even more than examining and compromising the site now would.

B The passage already tells us that it is likely the technology *will advance considerably in the near future.* Given this information, further inquiry into whether there will be any ongoing (perhaps minor) improvements is somewhat redundant and probably of minimal value with respect to evaluating the plan's likelihood of success.

C Even if the team can study a second site in the meanwhile, they might maximize the overall potential for gathering knowledge by delaying the examination of either site, both sites, or neither site until more advanced technology is available.

D The age of the culture that inhabited the site is irrelevant to assessing the risks of delaying the site's examination until more advanced technology is available.

E Even if the anticipated technology will damage or destroy the objects under study, it might still maximize the amount of knowledge that can be gathered from those objects. Without any comparison between the damage risk that would be incurred by proceeding with the current technology and the damage risk that would be incurred by waiting, the mere fact that some damage would occur is irrelevant.

The correct answer is A.

CR01102

557. More and more law firms specializing in corporate taxes are paid on a contingency-fee basis. Under this arrangement, if a case is won, the firm usually receives more than it would have received if it had been paid on the alternate hourly rate basis. If the case is lost, the firm receives nothing. Most firms are likely to make more under the contingency-fee arrangement.

Which of the following, if true, would most strengthen the prediction above?

(A) Firms that work exclusively under the hourly rate arrangement spend, on average, fewer hours on cases that are won than on cases that are lost.

(B) Some litigation can last for years before any decision is reached, and, even then, the decision may be appealed.

(C) Firms under the contingency-fee arrangement still pay their employees on an hourly basis.

(D) Since the majority of firms specialize in certain kinds of cases, they are able to assess accurately their chances of winning each potential case.

(E) Firms working under the contingency-fee arrangement take in fewer cases per year than do firms working under the hourly rate arrangement.

Argument Evaluation

Situation Law firms of a certain type are increasingly working on a contingency-fee basis, whereby the firm is only paid if the case won. For the individual cases that are thus taken and won, the payments are generally greater than the total payments would have been if the firms had been paid on an hourly basis. Furthermore, although cases taken on a contingency-fee basis present a significant risk of working for many hours on a case and not being paid, the passage claims that most firms are likely to make more money, on average, than they would if they took their cases on an hourly basis.

Reasoning *What would most strongly indicate that, despite the risks, the law firms working on a contingency fee basis are likely to make more money, on average, than they would have otherwise?* Our task is to find the statement that would most strongly support this prediction.

A Supposing that the firms mentioned in this option changed from working on an hourly-rate basis to working on a contingency-fee basis, we would not have enough information to predict what the results would be. For example, we may have no reason to expect that the firm would accept the same cases that they would have accepted if they were working on an hourly-rate basis. As such, patterns of work on cases taken on an hourly basis may be irrelevant for determining how much the firms would make if they were to take their cases on a contingency-fee basis.

B This option indicates that firms taking cases on a contingency-fee basis can work on the cases for years without payment. Rather than supporting the point that firms would make more money if they worked on a contingency-fee basis, the option illustrates an aspect of the risks associated with this payment arrangement.

C This option also helps to illustrate the risk associated with taking cases on a contingency-fee basis. Even though firms working on such cases will only be paid if they win and not until then, they will still incur significant costs when working on their cases.

D **Correct.** This option suggests that firms working on a contingency-fee basis would be able to select cases that they would be likely to win and therefore be paid for. This makes it more likely that the firms would make more money working on a contingency-fee basis than they would if working on an hourly basis.

E Although the difference in numbers of cases described in this option could, given certain possible facts, be relevant to the prediction made by the argument, we have not been given any such facts.

The correct answer is D.

CR10671

558. A newly discovered painting seems to be the work of one of two seventeenth-century artists, either the northern German Johannes Drechen or the Frenchman Louis Birelle, who sometimes painted in the same style as Drechen. Analysis of the carved picture frame, which has been identified as the painting's original seventeenth-century frame, showed that it is made of wood found widely in northern Germany at the time, but rare in the part of France where Birelle lived. This shows that the painting is most likely the work of Drechen.

Which of the following is an assumption that the argument requires?

(A) The frame was made from wood local to the region where the picture was painted.

(B) Drechen is unlikely to have ever visited the home region of Birelle in France.

(C) Sometimes a painting so resembles others of its era that no expert is able to confidently decide who painted it.

(D) The painter of the picture chose the frame for the picture.

(E) The carving style of the picture frame is not typical of any specific region of Europe.

Argument Construction

Situation The original frame of a seventeenth-century painting that seems to be by either the northern German Johannes Drechen or the Frenchman Louis Birelle is made of a type of wood much more common in northern Germany than in France, suggesting that Drechen was the painter.

Reasoning *What must be true in order for the facts presented to support the conclusion that the painting is by Drechen?* The argument is that in the seventeenth century, the type of wood in the frame was more common in northern Germany where Drechen was from than in France where Birelle was from, so probably Drechen painted the picture. In order for this inference to be plausible, the argument must implicitly assume that the frame's wood was from the same region the painter was from. And in order to justify this assumed connection between the wood and the region the painter was from, the argument must also assume that the painter painted the picture in that region.

A **Correct.** If the frame were not made of wood local to the region where the picture was painted, or if that region in turn were not where the painter was from, then the cited fact about where the wood was more common would be irrelevant to the conclusion about who painted the picture.

B The argument is compatible with the plausible hypothesis that Drechen visited France at some point during his lifetime but did not frame this or any painting with French wood.

C Even if experts always felt confident in deciding who painted any picture, examining the wood in the picture frame might help them decide correctly.

D The argument is compatible with the plausible hypothesis that the picture was sold to a local customer who then chose a frame of local wood.

E The argument would be even stronger if this were false and the carving style of the frame were typical of northern Germany specifically.

The correct answer is A.

CR00766

559. Beginning in 1966 all new cars sold in Morodia were required to have safety belts and power steering. Previously, most cars in Morodia were without these features. Safety belts help to prevent injuries in collisions, and power steering helps to avoid collisions in the first place. But even though in 1966 one-seventh of the cars in Morodia were replaced with new cars, the number of car collisions and collision-related injuries did not decline.

Which of the following, if true about Morodia, most helps to explain why the number of collisions and collision-related injuries in Morodia failed to decline in 1966?

(A) Because of a driver-education campaign, most drivers and passengers in cars that did have safety belts used them in 1966.

(B) Most of the new cars bought in 1966 were bought in the months of January and February.

(C) In 1965, substantially more than one-seventh of the cars in Morodia were replaced with new cars.

(D) An excessive reliance on the new safety features led many owners of new cars to drive less cautiously in 1966 than before.

(E) The seat belts and power steering put into new cars sold in 1966 had to undergo strict quality-control inspections by manufacturers, whether the cars were manufactured in Morodia or not.

Argument Construction

Situation Starting in 1966, new cars sold in Morodia were required to have safety belts and power steering. But the numbers of car collisions and collision-related injuries did not decline that year.

Reasoning *What could explain why the newly required safety features did not reduce the numbers of collisions and collision-related injuries in 1966?* The passage says that power steering helps to prevent collisions and that safety belts help to prevent collision-related injuries. Since most Morodian cars previously lacked these features, and one-seventh of them were replaced with new cars in 1966, the proportion of cars with these features must have increased that year. This should have reduced the numbers of collisions and collision-related injuries unless some other factor counteracted the reductions. Evidence of any such countervailing factor would help to explain why the numbers did not decrease.

A Increased usage of safety belts should have reduced the number of collision-related injuries, so it would not help explain why this number did not decrease.

B If the new cars bought in 1966 were mostly purchased early in the year, the increased proportion of cars with the newly required safety features should have started more significantly reducing the numbers of collisions and collision-related injuries early in the year, producing greater reductions for the year as a whole.

C However, many cars were replaced in the year before the safety features were required, in 1966 the replacement of one-seventh of all Morodian cars should still have increased the overall proportion of Morodian cars with the safety features and thus reduced the numbers of collisions and collision-related injuries.

D **Correct.** If many owners of the cars with the new safety features drove less cautiously, their recklessness could have increased the overall numbers of collisions and collision-related injuries despite any benefits from the safety features.

E Strict quality-control inspections should have made the safety features more reliable, further reducing the numbers of collisions and collision-related injuries.

The correct answer is D.

CR04882

560. Enterprise Bank currently requires customers with checking accounts to maintain a minimum balance or pay a monthly fee. Enterprise plans to offer accounts with no monthly fee and no minimum-balance requirement; to cover their projected administrative costs of $3 per account per month they plan to charge $30 for overdrawing an account. Since each month on average slightly more than 10 percent of Enterprise's customers overdraw their accounts, bank officials predict the new accounts will generate a profit.

Which of the following, if true, most strongly supports the bank officials' prediction?

(A) Some of Enterprise Bank's current checking account customers are expected to switch to the new accounts once they are offered.

(B) One third of Enterprise Bank's revenues are currently derived from monthly fees tied to checking accounts.

(C) Many checking account customers who occasionally pay a fee for not maintaining a minimum balance in their account generally maintain a balance well above the minimum.

(D) Customers whose checking accounts do not have a minimum-balance requirement are more likely than others to overdraw their checking accounts.

(E) Customers whose checking accounts do not have a minimum-balance requirement are more likely than others to write checks for small amounts.

Evaluation of a Plan

Situation Enterprise Bank gives customers checking accounts with no monthly fee provided they maintain a certain minimum balance. However, the bank plans to offer accounts with no minimum-balance requirement and no monthly fee. It plans to cover the bank's $3 per account per month administrative cost by charging a $30 penalty for overdrafts. Only slightly more than 10 percent of customers, on average, overdraw their accounts in a month. The bank officials predict the new accounts will generate a profit.

Reasoning *What new information, if accurate, would most strongly support the prediction?* If about only one customer in ten, on average, currently has an overdraft in a month, and if this trend continues among customers who sign up for the new account, then the proposed $30 penalty per overdraft will cover the $30 cost of maintaining checking accounts for 10 customers per month. Would removing the minimum-balance requirement significantly increase the 10 percent overdraft rate? If so, then significantly more than one in ten customers, on average, would pay a $30 penalty. If this were so, then the new plan would yield a profit, as predicted.

A "Some" might mean only a few, and this would probably not be sufficient to make the new plan significantly profitable.

B This suggests that many customers with the current minimum-balance no-monthly-fee account do not maintain the minimum-balance requirement and pay fees instead. However, this information by itself seems to have little bearing on the new plan.

C Such customers would be likely to overdraw their accounts less frequently. This suggests that if a preponderance of the customers for the proposed new account were such customers, the overdraft rate would decrease, and the proposed new account would be less profitable, or even unprofitable.

D **Correct.** This information provides strong support for the bank officials' prediction. It indicates that the currently roughly 10 percent overdraft rate might increase drastically with the no-minimum-balance account and, on average, cause the imposition of a $30 penalty on significantly more than 10 percent of customers per month. This would make the new account significantly profitable.

E This suggests that a check written by one of these customers is more likely to be for a small amount and is therefore somewhat less likely to cause an overdraft (unless such customers typically have small checking balances, which we are not told). If there were many such customers for the proposed new account, the overdraft rate might be less than 10 percent; this would indicate that the new account might not turn out to be profitable.

The correct answer is D.

CR05667

561. In virtually any industry, technological improvements increase labor productivity, which is the output of goods and services per person-hour worked. In Parland's industries, labor productivity is significantly higher than it is in Vergia's industries. Clearly, therefore, Parland's industries must, on the whole, be further advanced technologically than Vergia's are.

The argument is most vulnerable to which of the following criticisms?

(A) It offers a conclusion that is no more than a paraphrase of one of the pieces of information provided in its support.

(B) It presents as evidence in support of a claim information that is inconsistent with other evidence presented in support of the same claim.

(C) It takes one possible cause of a condition to be the actual cause of that condition without considering any other possible causes.

(D) It takes a condition to be the effect of something that happened only after the condition already existed.

(E) It makes a distinction that presupposes the truth of the conclusion that is to be established.

Argument Evaluation

Situation Technological improvements in nearly every industry increase labor productivity, which is the output of goods and services per person-hour worked. Because labor productivity is significantly higher in Parland than Vergia, Parland's industries are, in general, more technologically advanced than Vergia's.

Reasoning *To which criticism is the argument most vulnerable?* Though one factor, such as technological advancements, may lead to greater labor productivity, it may not be the only such factor, or even a necessary factor, leading to great labor productivity. Therefore, the mere fact that one region's labor is more productive than another's is not sufficient to establish that the former region is more technologically advanced than the latter region is.

A The conclusion is not merely a paraphrase of the pieces of information provided in its support. Indeed, the problem with the argument is that the conclusion goes too far beyond what the premises merit.

B The premises of the argument are not inconsistent with one another.

C **Correct.** This accurately describes the flaw in the argument because the reasons given in the argument for its conclusion would be good reasons only if there were no other plausible explanations for Parland's greater labor productivity.

D The argument does not mention how long Parland has had more productive labor, or when technological improvements would have occurred.

E Neither of the premises contains anything that presupposes the conclusion to be true.

The correct answer is C.

CR08471

562. Chaco Canyon, a settlement of the ancient Anasazi culture in North America, had massive buildings. **It must have been a major Anasazi center**. Analysis of wood samples shows that some of the timber for the buildings came from the Chuska and San Mateo mountains, 50 miles from Chaco Canyon. **Only a major cultural center would have the organizational power to import timber from 50 miles away**.

In the argument given, the two portions in boldface play which of the following roles?

(A) The first is a premise used to support the argument's main conclusion; the second is the argument's main conclusion.

(B) The first is the argument's main conclusion; the second is a premise used to support that conclusion.

(C) The first is one of two premises used to support the argument's main conclusion; the second is the other of those two premises.

(D) The first is a premise used to support the argument's main conclusion; the second is a premise used to support another conclusion drawn in the argument.

(E) The first is inferred from another statement in the argument; the second is inferred from the first.

Argument Construction

Situation The ancient Anasazi settlement at Chaco Canyon had massive buildings, for which some of the timber came from mountains 50 miles away.

Reasoning *What roles do the statement that Chaco Canyon must have been a major Anasazi center and the statement that only a major center would have the organizational power to import timber from 50 miles away play in the argument?* The first and third sentences in the passage are both factual observations. Since no further support is provided for either of them, neither can be a conclusion in the argument. The fourth sentence is a speculative generalization about major cultural centers. None of the other statements gives us any reason to think this generalization is true, so it cannot be a conclusion in the argument, either. However, the third and fourth sentences together imply that Chaco Canyon was a major cultural center, and the first sentence indicates that it was Anasazi. So together, the first, third, and fourth sentences all support the claim that Chaco Canyon was a major Anasazi cultural center and thus more generally a major Anasazi center, as the second sentence asserts. Therefore, the first, third, and fourth sentences are all premises that jointly support the second sentence as a conclusion.

A As explained above, the first boldface sentence is a conclusion supported by the latter, not the other way around.

B **Correct.** As explained above, the first boldface sentence is the argument's only stated conclusion, and in that sense its main conclusion, while all the other sentences are premises used to support it.

C As explained above, the first boldface sentence is the argument's conclusion, not a premise used to support the conclusion.

D As explained above, the first boldface sentence is the argument's only stated conclusion, and there is no reason to suppose that the argument is intended to lead to any other tacit conclusion that the second boldface sentence is intended to support.

E As explained above, the first boldface sentence is inferred from the three other statements in the argument together, not from any one of them alone. The second boldface sentence is a speculative generalization that cannot be, and is not meant to be, inferred from the former or from any other statement in the argument.

The correct answer is B.

CR04364

563. The Maxilux car company's design for its new luxury model, the Max 100, included a special design for the tires that was intended to complement the model's image. The winning bid for supplying these tires was submitted by Rubco. Analysts concluded that the bid would only just cover Rubco's costs on the tires, but Rubco executives claim that winning the bid will actually make a profit for the company.

Which of the following, if true, most strongly justifies the claim made by Rubco's executives?

(A) In any Maxilux model, the spare tire is exactly the same make and model as the tires that are mounted on the wheels.

(B) Rubco holds exclusive contracts to supply Maxilux with the tires for a number of other models made by Maxilux.

(C) The production facilities for the Max 100 and those for the tires to be supplied by Rubco are located very near each other.

(D) When people who have purchased a carefully designed luxury automobile need to replace a worn part of it, they almost invariably replace it with a part of exactly the same make and type.

(E) When Maxilux awarded the tire contract to Rubco, the only criterion on which Rubco's bid was clearly ahead of its competitors' bids was price.

Argument Construction

Situation Rubco won a bid for supplying tires for the Max 100, a new luxury model by Maxilux. The bid would barely cover the cost of the tires, but Rubco executives claim that winning the bid will be profitable.

Reasoning *What would support the executives' claim?* Rubco is not expected to make a profit from supplying the tires for the new cars, so we must look for some other way that Rubco could derive a profit as a result of winning the bid. If by winning the bid Rubco created an inevitable market for itself in replacement tires—on which Rubco could earn a profit—then the executives' claim may be justified.

A We have already been told that the bid is expected to barely cover the costs of supplying the tires on the new cars, so the analysts mentioned in the passage have presumably already taken into account that there is a spare tire supplied for the Max 100.

B If winning the bid led Rubco to win more exclusive contracts with the Maxilux, that might help support the executives' claim. But this statement indicates only that Rubco already has several exclusive contracts to supply Maxilux with tires, not that winning the bid has led to, or will lead to, more such contracts, which is what would be needed.

C As in answer choice (A), this is relevant to the costs of supplying the tires for the Max 100, but presumably this was taken into account by the analysts when they concluded that the bid would barely cover Rubco's costs on the tires.

D **Correct.** This indicates that by winning the bid Rubco has created a way to profit from the contract with Maxilux, specifically, by creating a market for replacement tires.

E This is likely one of the reasons that Rubco's bid only just covers Rubco's costs on the tires; it does nothing to justify the executives' claims that the bid will lead to a profit for Rubco.

The correct answer is D.

CR05186

564. Which of the following most logically completes the passage?

Most bicycle helmets provide good protection for the top and back of the head, but little or no protection for the temple regions on the sides of the head. A study of head injuries resulting from bicycle accidents showed that a large proportion were caused by blows to the temple area. Therefore, if bicycle helmets protected this area, the risk of serious head injury in bicycle accidents would be greatly reduced, especially since _____.

(A) among the bicyclists included in the study's sample of head injuries, only a very small proportion had been wearing a helmet at the time of their accident

(B) even those bicyclists who regularly wear helmets have a poor understanding of the degree and kind of protection that helmets afford

(C) a helmet that included protection for the temples would have to be somewhat larger and heavier than current helmets

(D) the bone in the temple area is relatively thin and impacts in that area are thus very likely to cause brain injury

(E) bicyclists generally land on their arm or shoulder when they fall to the side, which reduces the likelihood of severe impacts on the side of the head

Argument Construction

Situation Bicycle helmets protect the top and back of the head, but not the sides or temples. A study found that a large proportion of head injuries caused by biking accidents were caused by blows to the temple area.

Reasoning *Why would the risk of serious head injury in bicycle accidents be greatly reduced if bicycle helmets protected the temple regions?* If for some reason a serious head injury is particularly likely when there is impact to the temple area, then bicycle helmets that protect that area would be apt to reduce the number of serious head injuries from bicycle accidents. One such reason is that the bone in the temple area is relatively thin.

A This point is irrelevant because it gives us no information about the seriousness or the likelihood of injuries due to impact to the temple area.

B Whether bicyclists who regularly wear helmets have a good understanding of what protection their helmets afford is not relevant as to whether serious head injuries are particularly likely to occur from impact to the temple area.

C This point is relevant only to what a helmet that protected the temple area would be like, not to the seriousness of injuries resulting from impact to that area. If anything, this point counts as a reason *against* the conclusion, not *for* it. If such helmets are heavier and larger, they may be used less than they otherwise would be. If fewer helmets are used, then improvements to helmet design will have less of an effect in reducing serious head injuries.

D **Correct.** This statement provides a reason why the temple area of the rider's head needs protection: impacts to this area are very likely to cause brain injuries.

E This is largely irrelevant. Even if it suggests that head injuries do not generally result from bicyclists falling to the side, it does not indicate that such injuries are rare or that there is not great risk of serious injury in those cases in which there is impact to the temple area.

The correct answer is D.

CR01867

565. Which of the following most logically completes the argument?

In a typical year, Innovair's airplanes are involved in 35 collisions while parked or being towed in airports, with a resulting yearly cost of $1,000,000 for repairs.

To reduce the frequency of ground collisions, Innovair will begin giving its ground crews additional training, at an annual cost of $500,000. Although this will cut the number of ground collisions by about half at best, the drop in repair costs can be expected to be much greater, since _____.

(A) most ground collisions happen when ground crews are rushing to minimize the time a delayed airplane spends on the ground

(B) a ground collision typically occurs when there are no passengers on the airplane

(C) the additional training will focus on helping ground crews avoid those kinds of ground collisions that cause the most costly damage

(D) the $500,000 cost figure for the additional training of ground crews includes the wages that those crews will earn during the time spent in actual training

(E) most ground collisions have been caused by the least experienced ground-crew members

Evaluation of a Plan

Situation An airline will give its ground crews additional training to reduce the frequency of the collisions its airplanes are involved in while parked or being towed in airports.

Reasoning *What premise would most logically support the conclusion that the additional training will reduce repair costs from ground collisions much more than it reduces the number of such collisions?* The key word *since* before the blank shows that the argument should be completed with a premise that supports the preceding claim that the *drop in repair costs can be expected to be much greater*. A suitable premise might provide evidence that the training will disproportionately help to prevent the ground collisions that result in the higher repair costs as opposed to the less serious collisions that result in lower repair costs.

A We are given no reason to believe that the additional training would affect how much ground crews rush to minimize delays.

B The number of passengers is not clearly relevant to the repair costs resulting from a ground collision and in any case would not be affected by additional ground crew training.

C **Correct.** If the training especially helps the ground crews avoid those kinds of collisions that cause the most costly damage, then it will probably reduce repair costs even more than it reduces the number of collisions.

D Whether the cited expense for training includes wages is irrelevant to whether the training will reduce repair costs more than it reduces the number of collisions.

E This suggests that the additional training may help reduce the number of collisions, not that it will reduce repair costs more than it reduces the number of collisions.

The correct answer is C.

CR12558

566. Many agriculturally intensive areas of the world are beginning to encounter water scarcity problems. As a result, many farmers in these areas are likely to reduce their output as the water supply they need in order to maintain production shrinks. However, one group of farmers in such a region plans to increase their production by implementing techniques for water conservation.

Which of the following, if true, would most strongly support the prediction that the group's plan will succeed?

(A) Farmers who can gain a larger share of the food market in their regions will be better positioned to control more water resources.

(B) Most agricultural practices in areas with water shortages are water-intensive.

(C) Other regions of the world not facing water shortages are likely to make up for the reduction in agricultural output.

(D) Demand for agricultural products in the group's region is not expected to decline.

(E) More than half the water used for agriculture in the farmers' region is lost to evaporation or leakage from irrigation channels.

Evaluation of a Plan

Situation Farmers in many agriculturally intensive regions will probably reduce their output because the regions' water supplies are dwindling, but one group of farmers in such a region plans to use water conservation techniques to increase their output.

Reasoning *What would provide evidence that water conservation techniques will help the farmers increase production despite their region's dwindling water supplies?* In order for the water conservation techniques to be effective, they must result in significantly more water becoming available for the farmers to use. Because overall supplies are shrinking, rather than growing, that can only happen if the farmers are currently losing or wasting a great deal of water in ways that could be prevented with water conservation techniques.

A This suggests an advantage the farmers will gain if their water conservation plan enables them to increase production, but it provides no evidence that the plan actually will enable them to increase production.

B This suggests that the plan would have to yield quite a lot of conserved water in order for the farmers to increase production, but it offers no evidence that the plan will do so. Thus, it provides some reason to question whether the plan will succeed.

C Whether regions without water shortages will increase production is not directly relevant to the question of whether a particular measure would lead to increased production in one region that does have a water shortage.

D This has some slight, indirect relevance to the question of whether the farmers' plan will succeed: it suggests that if the farmers do manage to increase production, they will continue to have a market for what they produce. However, it does not address the issue of whether they will be able to increase production. Furthermore, even if demand for agricultural products in the group's region were expected to decline, it could still remain high enough to support the farmers' increased output from their water conservation plan.

E **Correct.** This suggests that the farmers are losing a lot of water in ways that the water conservation techniques might prevent, so it provides evidence that employing some such techniques could enable the farmers to save enough water to increase their output.

The correct answer is E.

CR03367

567. Hollywood restaurant is replacing some of its standard tables with tall tables and stools. The restaurant already fills every available seat during its operating hours, and the change in seating arrangements will not result in an increase in the restaurant's seating capacity. Nonetheless, the restaurant's management expects revenue to increase as a result of the seating change without any concurrent change in menu, prices, or operating hours.

Which of the following, if true, provides the best reason for the expectation?

(A) One of the taller tables takes up less floor space than one of the standard tables.

(B) Diners seated on stools typically do not linger over dinner as long as diners seated at standard tables.

(C) Since the restaurant will replace only some of its standard tables, it can continue to accommodate customers who do not care for the taller tables.

(D) Few diners are likely to avoid the restaurant because of the new seating arrangement.

(E) The standard tables being replaced by tall tables would otherwise have to be replaced with new standard tables at a greater expense.

Argument Construction

Situation Hollywood restaurant is replacing some of its tables with taller tables and stools, and the management expects this will increase revenue, despite the fact that the restaurant already fills all of its available seats and that this change will not increase seating capacity. Furthermore, there will not be any change in menu, prices, or operating hours.

Reasoning *What would strongly support the management's expectation?* Since the new seating will not increase the restaurant's seating capacity, the management's expectations must be based on a belief that the change to taller tables and stools will somehow change diners' behavior, perhaps by leading them to order more food, or to stay at their tables for a shorter time, thereby allowing the restaurant to serve more diners during its operating hours without increasing seating capacity. If diners seated at tall tables and on tall stools spend less time lingering over their dinners, then they will leave sooner, opening up the tables for more diners. Because the restaurant, before the change, already fills every available seat during its operating hours, it is reasonable to think that it will be able to serve more diners than it currently does, thereby selling more food and thus increasing revenue.

A This would be relevant if we could infer from it that seating capacity will increase. However, the passage indicates that the new seating arrangement will not result in greater capacity.

B **Correct.** Because the restaurant will be able to serve more meals during its operating hours, the restaurant's revenue can be expected to increase.

C This may indicate that the restaurant is less likely to alienate customers who do not care for tall tables and stools, but that only supports the claim that the restaurant will not lose customers and therefore lose revenue; it does not indicate that the restaurant will see revenue increase.

D Again, this merely indicates that there will not be a loss—or much loss—of revenue, not that there will be an increase in revenue.

E Less expensive tables will decrease the restaurant's costs, but it will not increase the restaurant's revenue.

The correct answer is B.

CR07660

568. A major network news organization experienced a drop in viewership in the week following the airing of a controversial report on the economy. The network also received a very large number of complaints regarding the report. The network, however, maintains that negative reactions to the report had nothing to do with its loss of viewers.

Which of the following, if true, most strongly supports the network's position?

(A) The other major network news organizations reported similar reductions in viewership during the same week.

(B) The viewers who registered complaints with the network were regular viewers of the news organization's programs.

(C) Major network news organizations publicly attribute drops in viewership to their own reports only when they receive complaints about those reports.

(D) This was not the first time that this network news organization has aired a controversial report on the economy that has inspired viewers to complain to the network.

(E) Most network news viewers rely on network news broadcasts as their primary source of information regarding the economy.

Argument Construction

Situation A major network news organization aired a controversial report on the economy, and the following week the network's viewership declined. The network claims that the loss of viewers was not connected with negative reactions to the report.

Reasoning *Which statement most strongly supports the network's position?* If other major news network organizations had similar drops in viewership, it is implausible to think that the controversial report accounted for the other organizations' drops in viewership. On the other hand, it is not implausible to suppose that whatever did cause the drop in the viewership experienced by other network news organizations—e.g., holidays, weather, popular non-news programming—also had that effect on the organization that ran the controversial report. This would give some reason to believe that it was not the report that accounts for the organization's drop in viewership.

A **Correct.** This statement indicates that something other than the airing of the report could account for the subsequent drop in the organization's viewership.

B If anything, this statement tends to undermine the network's claim, because it suggests that the report offended people who otherwise might have continued to watch the organization's programming.

C Since the network did in fact receive complaints about the report, this statement is irrelevant.

D The fact that the network has received complaints before about controversial reports on the economy that the network's news organization has aired tells us nothing about whether this recent report caused a subsequent drop in viewership.

E The fact that viewers turn to network news broadcasts as their primary source of information about the economy tells us nothing about whether viewers might stop watching a particular network news organization's programs as a result of its airing a controversial report on the economy.

The correct answer is A.

CR04366

569. Only a reduction of 10 percent in the number of scheduled flights using Greentown's airport will allow the delays that are so common there to be avoided. Hevelia airstrip, 40 miles away, would, if upgraded and expanded, be an attractive alternative for fully 20 percent of the passengers using Greentown airport. Nevertheless, experts reject the claim that turning Hevelia into a full-service airport would end the chronic delays at Greentown.

Which of the following, if true, most helps to justify the experts' position?

(A) Turning Hevelia into a full-service airport would require not only substantial construction at the airport itself, but also the construction of new access highways.

(B) A second largely undeveloped airstrip close to Greentown airport would be a more attractive alternative than Hevelia for many passengers who now use Greentown.

(C) Hevelia airstrip lies in a relatively undeveloped area but would, if it became a full-service airport, be a magnet for commercial and residential development.

(D) If an airplane has to wait to land, the extra jet fuel required adds significantly to the airline's costs.

(E) Several airlines use Greentown as a regional hub, so that most flights landing at Greentown have many passengers who then take different flights to reach their final destinations.

Evaluation of a Plan

Situation To avoid the delays now common at Greentown's airport, the number of scheduled flights there would need to be reduced by 10 percent. If the nearby Hevelia airstrip were expanded and upgraded, it would be an attractive alternative for 20 percent of Greentown airport's passengers. Still, experts do not believe that the delays at Greentown would end even if Hevelia were turned into a full-service airport.

Reasoning *Which statement most supports the experts' position?* If the number of flights at Greentown's airport did not drop by at least 10 percent, despite the fact that 20 percent of the passengers who currently use Greentown's airport would find nearby Hevelia airstrip an attractive alternative, then the delays would not be avoided. Airlines generally use certain airports as regional hubs—an airport through which an airline routes most of its traffic—so, even if many passengers would be willing to use Hevelia airstrip, the number of flights at Greentown may not decline significantly, or at all.

A The experts' position concerns what would happen to the flight delays at Greentown airport if the Hevelia airstrip were converted into a full-service airport. So the fact that there are great costs involved in making such a conversion—possibly making such a conversion unlikely—has no bearing on the effects such a conversion would have on flight delays at Greentown if the conversion were to be carried out.

B This statement indicates that the undeveloped airstrip near Greentown might be a better way to alleviate flight delays at Greentown, but it tells us nothing about the effects that converting the Hevelia airstrip to a full-service airport would have were it to be carried out.

C This in no way explains why converting the Hevelia airstrip into a full-service airport would not alleviate the problem with flight delays at Greentown.

D This provides a reason to think that reducing the number of flights at Greentown might make the airport more efficient. But that has no bearing on the effect that converting the Hevelia airstrip to a full-service airport might have on flight delays at Greentown.

E **Correct.** This statement provides support for the experts' position because it gives a reason for thinking that the number of scheduled flights at Greentown would not be reduced, even if Hevelia airstrip became an attractive alternative for some 20 percent of Greentown's passengers.

The correct answer is E.

CR07712

570. Farmer: Worldwide, just three grain crops—rice, wheat, and corn—account for most human caloric intake. To maintain this level of caloric intake and also keep pace with global population growth, yields per acre from each of these crops will have to increase at least 1.5 percent every year, given that the supply of cultivated land is diminishing. Therefore, the government should increase funding for research into new ways to improve yields.

Which of the following is an assumption on which the farmer's argument depends?

(A) It is solely the government's responsibility to ensure that the amount of rice, wheat, and corn produced worldwide keeps pace with global population growth.

(B) Increasing government funding for research into new ways to improve the yields per acre of rice, wheat, and corn crops would help to increase total worldwide annual production of food from these crops.

(C) Increasing the yields per acre of rice, wheat, and corn is more important than increasing the yields per acre of other crops.

(D) Current levels of funding for research into ways of improving grain crop yields per acre have enabled grain crop yields per acre to increase by more than 1.5 percent per year worldwide.

(E) In coming decades, rice, wheat, and corn will become a minor part of human caloric intake, unless there is government-funded research to increase their yields per acre.

Argument Construction

Situation The farmer states that although the worldwide human population is increasing, the supply of cultivated land is decreasing. We thus need to increase yields for the food crops that account for most of human caloric intake—rice, wheat, and corn—if we are to maintain our existing caloric intake. The increase in yields, according to the farmer, would need to be at least 1.5 percent every year.

Reasoning *What must be true if we are to accept the farmer's conclusion, that the government should increase funding for research into new ways to improve crop yields, on the basis of the above statements?* The farmer uses the above statements as premises of an argument for an increase in government funding for research on crop yields. Supposing that the farmer's statements are true, we need to find in the available options the statement that, if added to the argument, may allow us to accept the farmer's conclusion, based on the argument.

A Whether or not nongovernmental entities such as NGOs (nongovernmental organizations) are responsible for helping to ensure that humans have an adequate amount of food, governments may or may not also have this responsibility.

B **Correct.** If government funding of this research does not increase crop yields, then the premises of the argument provide no support for the conclusion that the government should provide such funding. The cogency of the argument thus depends on this statement.

C Crops in addition to rice, wheat, and corn could also be very important, and perhaps essential for human existence. However, this would not diminish the importance of food crops such as rice, wheat, and corn.

D This option may suggest that current levels of funding of research into crop yields are sufficient for purposes of our obtaining the necessary crop yields.

E This option suggests that rice, wheat, and corn may be replaced by other crops, because the other crops have better yields. Because we might thus have a means for increasing crop yields that does not involve an increase in government research funding, this option may actually decrease the support for the conclusion.

The correct answer is B.

CR08770

571. The air quality board recently informed Coffee Roast, a small coffee roasting firm, of a complaint regarding the smoke from its roaster. Recently enacted air quality regulations require machines roasting more than 10 pounds of coffee to be equipped with expensive smoke-dissipating afterburners. The firm, however, roasts only 8 pounds of coffee at a time. Nevertheless, the company has decided to purchase and install an afterburner.

Which of the following, if true, most strongly supports the firm's decision?

(A) Until settling on the new air quality regulations, the board had debated whether to require afterburners for machines roasting more than 5 pounds of coffee at a time.

(B) Coffee roasted in a machine equipped with an afterburner has its flavor subtly altered.

(C) The cost to the firm of an afterburner is less than the cost of replacing its roaster with a smaller one.

(D) Fewer complaints are reported in areas that maintain strict rules regarding afterburners.

(E) The firm has reason to fear that negative publicity regarding the complaints could result in lost sales.

Evaluation of a Plan

Situation After being informed of a complaint about smoke from its coffee roaster, a firm decided to purchase and install an afterburner to reduce or eliminate emissions of smoke, even though the roaster roasts too little coffee at a time for an afterburner to be legally required.

Reasoning *What would have been a good reason for the firm to buy and install the afterburner?* The only factors mentioned that might give the firm reason to buy an afterburner are the complaint about smoke and the regulations requiring an afterburner. Since the regulations do not apply in this case, the complaint is more likely to have motivated the firm's decision. Any serious potential consequences the firm might have faced from failure to address the complaint could have provided a good reason to buy and install the afterburner.

A If this debate had still been ongoing when the firm made its decision, uncertainty about the pending regulations might have justified the decision. But the debate had already been settled before the firm decided to purchase the afterburner, and the regulations clearly did not require one.

B An unspecified alteration in flavor is not clearly a good reason to use an afterburner—the afterburner might worsen the flavor.

C The firm's roaster was already small enough that the regulations did not require it to be replaced, even without an afterburner.

D This reason relates only to rules regarding afterburners, not to Coffee Roast's purchase of an afterburner, which was not mandated by regulations. Furthermore, it could be that the air quality regulations recently enacted are among the strictest in any region, which could result in fewer complaints regardless of whether Coffee Roast installs an afterburner.

E **Correct.** Since installing an afterburner is a plausible way to address the complaint and prevent future complaints, the firm has plausible reasons to believe this strategy will help it avoid the negative publicity and lost sales it fears. These considerations could have reasonably justified its decision.

The correct answer is E.

CR03695

572. People who do regular volunteer work tend to live longer, on average, than people who do not. It has been found that "doing good," a category that certainly includes volunteer work, releases endorphins, the brain's natural opiates, which induce in people a feeling of well-being. Clearly, there is a connection: Regular releases of endorphins must in some way help to extend people's lives.

Which of the following, if true, most seriously undermines the force of the evidence given as support for the hypothesis that endorphins promote longevity?

(A) People who do regular volunteer work are only somewhat more likely than others to characterize the work they do for a living as "doing good."

(B) Although extremely high levels of endorphins could be harmful to health, such levels are never reached as a result of the natural release of endorphins.

(C) There are many people who have done some volunteer work but who do not do such work regularly.

(D) People tend not to become involved in regular volunteer work unless they are healthy and energetic to begin with.

(E) Releases of endorphins are responsible for the sense of well-being experienced by many long-distance runners while running.

Argument Evaluation

Situation People who volunteer regularly live longer on average than people who do not. Doing good work, including volunteer work, releases endorphins, which induce a feeling of well-being.

Reasoning *What additional findings would suggest that the cited evidence does not indicate that endorphins increase longevity?* The argument implicitly assumes that the reason regular volunteers tend to live longer is that volunteering lengthens their lives. It further assumes that no factor that is correlated with volunteering, other than the endorphin release, would plausibly explain how volunteering could have this effect. Findings that cast doubt on either of these assumptions would undermine the connection between the cited evidence and the conclusion that endorphins promote longevity.

A Volunteering might greatly boost volunteers' endorphin levels even if the work the volunteers do for a living is no different from other people's work.

B Even if unnaturally high endorphin levels could harm health, the levels attainable through volunteer work may promote health.

C The argument is about an observed correlation in a certain group of people (those who regularly do volunteer work). How many people are outside that group (i.e., do not regularly do volunteer work) is independent of the question of what causes the observed correlation. Even if some people volunteer only occasionally, volunteering regularly may promote longevity by causing regular releases of endorphins.

D **Correct.** This suggests that the initially better health of people who choose to volunteer could fully explain the cited correlation between volunteering and longevity.

E Unless we are also given evidence that long-distance runners tend not to live longer than other people, this does not undermine the purported evidence in the argument. Endorphins might promote longevity in both regular volunteers and long-distance runners.

The correct answer is D.

CR04140

573. A study compared a sample of Swedish people older than 75 who needed in-home assistance with a similar sample of Israeli people. The people in the two samples received both informal assistance, provided by family and friends, and formal assistance, professionally provided. Although Sweden and Israel have equally well-funded and comprehensive systems for providing formal assistance, the study found that the people in the Swedish sample received more formal assistance, on average, than those in the Israeli sample.

Which of the following, if true, does most to explain the difference that the study found?

(A) A companion study found that among children needing special in-home care, the amount of formal assistance they received was roughly the same in Sweden as in Israel.

(B) More Swedish than Israeli people older than 75 live in rural areas where formal assistance services are sparse or nonexistent.

(C) Although in both Sweden and Israel much of the funding for formal assistance ultimately comes from the central government, the local structures through which assistance is delivered are different in the two countries.

(D) In recent decades, the increase in life expectancy of someone who is 75 years old has been greater in Israel than in Sweden.

(E) In Israel, people older than 75 tend to live with their children, whereas in Sweden people of that age tend to live alone.

Argument Construction

Situation A study of elder care in Israel and Sweden found that in Sweden, of the total amount of care that people older than 75 and needing in-home assistance received, the proportion of care that was formal, i.e., provided by professional care personnel, was greater than in Israel. Both Sweden and Israel had equally good systems for providing formal care, and in both countries, the elderly also received informal care, i.e., care provided by friends and family.

Reasoning *Among the factors given, which would most contribute to explaining the difference the study found between Sweden and Israel with respect to elder care?* A good guess would be that there is a difference in some societal factor that affects the difference the study found. For example, perhaps elders in one of the countries regard maintaining independence as a higher priority than elders in the other and consequently try to rely less on friends and family? Perhaps patterns of decline in ability to remain independent are different in the two countries? Or perhaps a greater proportion of elders live alone in one of the countries than in the other?

A The difference to be explained concerns only elder care, not care of children.

B The fact that formal elder care is less available in Swedish rural areas than in Israeli rural areas might suggest that there would be a greater reliance on informal care in such areas in Sweden. But this new information throws little light on how the overall proportions of formal and informal care in each country would be affected.

C This information is not specific enough to help explain the precise difference found in the study. It is reasonable to assume that the study was conducted with sufficient rigor to take account of any relevant structural differences in the delivery of formal elder care.

D This could suggest either that the greater proportion of informal elder care in Israel contributes to greater life expectancy or that greater life expectancy signals greater fitness during old age that would make it more practical for friends and family to provide informal elder care.

E **Correct.** The prevalence in Israel of elders living in family settings—in contrast to Sweden, where elders tend to live alone—offers a plausible explanation of the difference that the study found in the patterns of elder care in Israel and Sweden. It seems reasonable to think that, all things being equal, elders living alone would use formal elder care services more often than elders living with friends or family.

The correct answer is E.

CR05077

574. Film Director: It is true that certain characters and plot twists in my newly released film *The Big Heist* are similar to characters and plot twists in *Thieves*, a movie that came out last year. Pointing to these similarities, the film studio that produced *Thieves* is now accusing me of taking ideas from that film. The accusation is clearly without merit. All production work on *The Big Heist* was actually completed months before *Thieves* was released.

Which of the following, if true, provides the strongest support for the director's position?

(A) Before *Thieves* began production, its script had been circulating for several years among various film studios, including the studio that produced *The Big Heist*.

(B) The characters and plot twists that are most similar in the two films have close parallels in many earlier films of the same genre.

(C) The film studio that produced *Thieves* seldom produces films in this genre.

(D) The director of *Thieves* worked with the director of *The Big Heist* on several earlier projects.

(E) Production work on *Thieves* began before production work on *The Big Heist* was started.

Argument Evaluation

Situation The director of the film *The Big Heist* has been accused, by the studio that produced the film *Thieves*, of taking ideas from the film. The director responds that the accusation lacks merit, since all production work on *The Big Heist* was completed before *Thieves* appeared last year in theaters.

Reasoning *Which of the five statements most strongly supports the director's position?* Crime thrillers, as a film genre, are likely to have stock characters and plot lines that reflect a long tradition. So it would be no surprise if some of the characters or plot twists in one such film would resemble, to a greater or lesser extent, the characters and plot twists in another. The studio might be correct in identifying such resemblances between *The Big Heist* and *Thieves*. But it would not necessarily be correct that characters or plot lines in *The Big Heist* were derived from *Thieves*.

A This undercuts the director's position, since it provides information that indicates an opportunity for the director to copy ideas from the script for *Thieves*.

B **Correct.** This information strengthens the support for the director's claim that the studio's accusation lacks merit. Since both *Thieves* and *The Big Heist* fall within a long tradition of crime thriller films, the characters and plot lines in both films reflect that tradition, and so any resemblances do not imply deliberate copying of the ideas in *Thieves* by the director of *The Big Heist*.

C This information seems largely irrelevant to the issue raised and does not strengthen support for the director's conclusion.

D This does little to indicate that the director's conclusion is correct. For example, the then-future director of *Thieves* might have discussed with the future director of *The Big Heist* specific ideas about character and plot for a planned crime thriller film.

E This does not support the director's claim. For example, it raises the possibility that information about *Thieves* leaked during the early stages of production—information that could have been exploited in the production of *The Big Heist*.

The correct answer is B.

CR05412

575. In Mernia commercial fossil hunters often sell important fossils they have found, not to universities or museums, but to individual collectors, who pay much better but generally do not allow researchers access to their collections. To increase the number of fossils available for research, some legislators propose requiring all fossils that are found in Mernia to be sold only to universities or museums.

Which of the following, if true, most strongly indicates that the legislators' proposal will fail to achieve its goal?

(A) Some fossil hunters in Mernia are not commercial fossil hunters, but rather are amateurs who keep the fossils that they find.

(B) Most fossils found in Mernia are common types that have little scientific interest.

(C) Commercial fossil hunters in Mernia currently sell some of the fossils they find to universities and museums.

(D) Many universities in Mernia do not engage in fossil research.

(E) Most fossils are found by commercial fossil hunters, and they would give up looking for fossils if they were no longer allowed to sell to individual collectors.

Evaluation of a Plan

Situation Fossil hunters in Mernia often sell important fossils to collectors who do not make them accessible to researchers. To increase the number of fossils available for research, some legislators propose requiring all fossils found in Mernia to be sold only to universities or museums.

Reasoning *What would most strongly suggest that requiring all fossils found in Mernia to be sold only to universities or museums would not increase the number of fossils available for research?* To increase the number of fossils available for research, the proposed requirement will have to be implemented and effectively enforced. It will presumably have to increase the total number of fossils sold to universities and museums. And those institutions will have to make more of the fossils in their collections available to researchers than the private collectors do. Evidence that any of those conditions will not be fulfilled would suggest that the legislators' proposal will fail to achieve its goal.

A Even if the legislation does not affect fossils kept by amateurs, it might still result in many more fossils being sold to universities or museums rather than to private collectors, and thus might still increase the number of fossils available for research.

B Even if few Mernian fossils are interesting to researchers, the legislation could still achieve its goal of making more fossils available for research.

C Even if commercial fossil hunters already sell a few fossils to universities and museums, the legislation could encourage them to sell many more fossils.

D The universities that do not engage in fossil research presumably will not be interested in buying fossils even if the legislation passes. But the fossil hunters can just sell their fossils to other universities and museums that do engage in fossil research.

E **Correct.** This suggests that if the legislation passes, fossils will simply be left in the ground rather than sold to private collectors. That would not increase the total number of fossils available for research.

The correct answer is E.

CR02702

576. Economist: Tropicorp, which constantly seeks profitable investment opportunities, has been buying and clearing sections of tropical forest for cattle ranching, although pastures newly created there become useless for grazing after just a few years. The company has not gone into rubber tapping, even though greater profits can be made from rubber tapping, which leaves the forest intact. Thus, some environmentalists argue that **Tropicorp's actions do not serve even its own economic interest**. However, the initial investment required for a successful rubber-tapping operation is larger than that needed for a cattle ranch; there is a shortage of workers employable in rubber-tapping operations; and taxes are higher on profits from rubber tapping than on profits from cattle ranching. Consequently, **the environmentalists' conclusion is probably wrong**.

In the economist's argument, the two **boldface** portions play which of the following roles?

(A) The first supports the conclusion of the economist's argument; the second calls that conclusion into question.

(B) The first states the conclusion of the economist's argument; the second supports that conclusion.

(C) The first supports the conclusion of the environmentalists' argument; the second states that conclusion.

(D) The first states the conclusion of the environmentalists' argument; the second states the conclusion of the economist's argument.

(E) Each supports the conclusion of the economist's argument.

Argument Construction

Situation According to an economist, the firm Tropicorp has been investing in tropical forest that it has cleared for cattle ranching. But its new pastures are useless for grazing after a few years. In contrast, rubber tapping—which would avoid cutting trees—could be more profitable. According to the economist, environmentalists consequently argue that Tropicorp's investment does not serve the firm's economic interest. However, the economist argues, investing in rubber tapping involves some potential costs and risks greater than those that investing in cattle ranching involves. Consequently, the economist argues, the environmentalists' conclusion is probably wrong.

Reasoning *What function is served by the statement that Tropicorp's actions do not serve even its own economic interest? What function is served by the statement that the environmentalists' conclusion is probably wrong?* The first statement is a conclusion that the economist attributes to environmentalists. The second statement is the conclusion of an argument presented by the economist.

A The first states the conclusion of the argument that is attributed to environmentalists; it does not support—nor is it meant to—the conclusion of the economist.

B The second statement, not the first, is the conclusion of the economist's argument.

C The first is the conclusion attributed to environmentalists and is not meant merely as support for that conclusion.

D **Correct.** The first states the conclusion of the environmentalists' argument as the economist presents it; the second is the conclusion of the economist's argument.

E Neither statement is meant as support for the economist's conclusion, nor does it offer such support.

The correct answer is D.

CR08831

577. Although the school would receive financial benefits if it had soft drink vending machines in the cafeteria, we should not allow them. Allowing soft drink machines there would not be in our students' interest. If our students start drinking more soft drinks, they will be less healthy.

The argument depends on which of the following?

(A) If the soft drink vending machines were placed in the cafeteria, students would consume more soft drinks as a result.

(B) The amount of soft drinks that most students at the school currently drink is not detrimental to their health.

(C) Students are apt to be healthier if they do not drink soft drinks at all than if they just drink small amounts occasionally.

(D) Students will not simply bring soft drinks from home if the soft drink vending machines are not placed in the cafeteria.

(E) The school's primary concern should be to promote good health among its students.

Argument Construction

Situation Allowing soft drink vending machines in a school cafeteria would financially benefit the school, but students who drink more soft drinks would become less healthy.

Reasoning *What must be true in order for the claim that students drinking more soft drinks would cause them to become less healthy to justify the conclusion that soft drink vending machines should not be allowed in the cafeteria?* The argument is that because drinking more soft drinks would be unhealthy for the students, allowing the vending machines would not be in the students' interest, so the vending machines should not be allowed. This reasoning depends on the implicit factual assumption that allowing the vending machines would result in the students drinking more soft drinks. It also depends on the implicit value judgment that receiving financial benefits should be less important to the school than preventing a situation that would make the students less healthy.

A **Correct.** If the cafeteria vending machines would not result in students consuming more soft drinks, then allowing the machines would not harm the students' health in the way the argument assumes.

B Even if the amount of soft drinks the students currently drink were unhealthy, enabling the students to drink more could make them even less healthy.

C Even if drinking small amounts of soft drinks occasionally would not harm the students, vending machines in the cafeteria could lead the students to drink excessive amounts.

D Even if students who cannot buy soft drinks in the cafeteria sometimes bring them from home instead, adding vending machines in the cafeteria could increase the students' overall soft drink consumption.

E A concern does not have to be the primary one in order to be valid and important. It could be held that promoting students' good health should not be the schools' primary concern but should still be a more important concern than the financial benefits from the vending machines.

The correct answer is A.

CR01112

578. Many athletes inhale pure oxygen after exercise in an attempt to increase muscular reabsorption of oxygen. Measured continuously after exercise, however, the blood lactate levels of athletes who inhale pure oxygen are practically identical, on average, to those of athletes who breathe normal air. The lower the blood lactate level is, the higher the muscular reabsorption of oxygen is.

If the statements above are all true, they most strongly support which of the following conclusions?

(A) Athletes' muscular reabsorption of oxygen is not increased when they inhale pure oxygen instead of normal air.

(B) High blood lactate levels cannot be reduced.

(C) Blood lactate levels are a poor measure of oxygen reabsorption by muscles.

(D) The amount of oxygen reabsorbed by an athlete's muscles always remains constant.

(E) The inhaling of pure oxygen has no legitimate role in athletics.

Argument Construction

Situation Blood lactate levels after exercise are practically identical in athletes who breathe normal air and in those who inhale pure oxygen after exercise. The lower the blood lactate level, the higher the muscular reabsorption of oxygen.

Reasoning *What conclusion do the stated facts most strongly support?* We are told that lower blood lactate levels correspond consistently to higher muscular reabsorption of oxygen. Since athletes who breathe pure oxygen after exercise have blood lactate levels practically identical to those in athletes who breathe normal air, probably muscular reabsorption of oxygen does not differ significantly between athletes who breathe pure oxygen and those who breathe pure air.

A **Correct.** As explained above, the stated facts suggest that muscular reabsorption of oxygen does not differ significantly between athletes who breathe pure oxygen and those who breathe pure air. So breathing pure oxygen instead of normal air after exercise probably does not increase athletes' muscular reabsorption of oxygen.

B None of the statements indicates that blood lactate levels cannot be reduced by means other than inhaling pure oxygen.

C We are told that blood lactate levels are negatively correlated with muscular reabsorption of oxygen. This negative correlation might allow muscular reabsorption of oxygen to be precisely determined by measuring blood lactate levels.

D Muscular reabsorption of oxygen might vary for reasons unrelated to whether an athlete has been inhaling pure oxygen.

E Inhaling pure oxygen might have some legitimate role unrelated to muscular reabsorption of oxygen.

The correct answer is A.

CR02143

579. Boreal owls range over a much larger area than do other owls of similar size. Scientists have hypothesized that **it is scarcity of prey that leads the owls to range so widely**. This hypothesis would be hard to confirm directly, since it is not possible to produce a sufficiently accurate count of the populations of small mammals inhabiting the forests where boreal owls live. Careful study of owl behavior has, however, shown that **boreal owls do range over larger areas when they live in regions where food of the sort eaten by small mammals is comparatively sparse**. This indicates that the scientists' hypothesis is not sheer speculation.

In the argument given, the two **boldfaced** portions play which of the following roles?

(A) The first presents an explanatory hypothesis; the second states the main conclusion of the argument.

(B) The first presents an explanatory hypothesis; the second presents evidence tending to support this hypothesis.

(C) The first presents an explanatory hypothesis; the second presents evidence to support an alternative explanation.

(D) The first describes a position that the argument opposes; the second presents evidence to undermine the support for the position being opposed.

(E) The first describes a position that the argument opposes; the second states the main conclusion of the argument.

Argument Construction

Situation The boreal owl range over a much larger area than owls of similar size. Scientists hypothesize they do so because of prey scarcity. Counting the owls' prey—small mammals—in the boreal owls' habitat is inherently difficult. This makes the scientists' hypothesis hard to confirm directly. However, it has been found that boreal owls range widely when they inhabit regions with relatively little food for the small mammals they prey on.

Reasoning *What function is served by the statement that it is scarcity of prey that leads the owls to range so widely? What function is served by the statement that boreal owls range widely if food for their small-mammal prey is relatively sparse in the region they inhabit?* The first statement expresses a hypothesis that seeks to explain the comparatively wide range of boreal owls. The second statement serves to provide some indirect evidence for the scientists' hypothesis.

A The main conclusion of the argument is that the scientists' hypothesis is not sheer speculation, i.e., that the scientists have based their hypothesis on some evidence that they have discovered. The first statement presents the scientists' hypothesis. The second statement cites some evidence for the hypothesis and is not the main conclusion of the argument.

B **Correct.** As stated, the first presents an explanatory hypothesis, while the second cites some indirect evidence for the hypothesis.

C The second statement cites some indirect evidence for the scientists' hypothesis, not for some other hypothesis.

D The argument does not oppose the scientists' hypothesis, presented in the first statement; the second statement cites evidence for the hypothesis, and does not cite evidence for a position the argument opposes.

E The second statement does not present the argument's main conclusion. The main conclusion is that the scientists' hypothesis is not mere speculation.

The correct answer is B.

CR02888

580. Last year a record number of new manufacturing jobs were created. Will this year bring another record? Well, any new manufacturing job is created either within an existing company or by the start-up of a new company. **Within existing firms, new jobs have been created this year at well below last year's record pace**. At the same time, there is considerable evidence that the number of new companies starting up will be no higher this year than it was last year and **there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups**. So clearly, the number of new jobs created this year will fall short of last year's record.

In the argument given, the two portions in **boldface** play which of the following roles?

(A) The first is a claim that the argument challenges; the second is an explicit assumption on which that challenge is based.

(B) The first is a claim that the argument challenges; the second is a judgment advanced in support of the main conclusion of the argument.

(C) The first provides evidence in support of the main conclusion of the argument; the second is an objection that has been raised against that main conclusion.

(D) The first provides evidence in support of the main conclusion of the argument; the second is a judgment advanced in support of that main conclusion.

(E) The first and the second are each claims that have been advanced in support of a position that the argument opposes.

Argument Construction

Situation Manufacturing jobs are created either within existing companies or in start-ups. Manufacturing jobs are being created at a much slower rate this year than last year. It seems likely that the number of new start-ups will not exceed last year's number and that the average number of manufacturing jobs per start-up will not exceed last year's number. So fewer manufacturing jobs are likely to be created this year than last year.

Reasoning *What function is served by the statement that within existing firms, new jobs have been created this year at well below last year's record pace? What function is served by the statement that there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups?* The first statement is one of the statements used as support for the argument's main conclusion (the prediction about this year's job creation). The second statement gives another premise used as support for that prediction.

A The argument does not challenge the claim made by the first statement; it uses the first and the second statement as support for the argument's main conclusion, the prediction about this year's job creation.

B The argument does not challenge the claim made by the first statement, but uses the first and second statements as support for the argument's main conclusion.

C The first provides evidence in support of the main conclusion of the argument; the second is not an objection that has been raised against the main conclusion.

D **Correct.** The first provides evidence in support of the main conclusion of the argument; the second also provides support for the main conclusion.

E Neither the first nor the second is meant to support a position that the argument opposes; rather, they are both meant to support the argument's main conclusion.

The correct answer is D.

CR07809

581. A study of ticket sales at a summer theater festival found that people who bought tickets to individual plays had a no-show rate of less than 1 percent, while those who paid in advance for all ten plays being performed that summer had a no-show rate of nearly 30 percent. This may be at least in part because the greater the awareness customers retain about the cost of an item, the more likely they are to use it.

Which of the following would, if true, best serve as an alternative explanation of the results of the study?

(A) The price per ticket was slightly cheaper for those who bought all ten tickets in advance.

(B) Many people who attended the theater festival believed strongly that they should support it financially.

(C) Those who attended all ten plays became eligible for a partial refund.

(D) Usually, people who bought tickets to individual plays did so immediately prior to each performance that they attended.

(E) People who arrived just before the performance began could not be assured of obtaining seats in a preferred location.

Argument Construction

Situation People who bought tickets to individual plays at a theater festival had a much lower no-show rate than did people who paid in advance for all ten plays.

Reasoning *What factor other than greater awareness of the ticket costs could explain why people who bought tickets individually were more likely to attend the plays?* The passage suggests that people who bought tickets individually were more likely to attend the plays because they were more vividly aware of what they had paid for each ticket. But there are other possible explanations—perhaps the people who bought the tickets individually were more eager to attend each play for its own sake, or had other characteristics or incentives that made them more likely to attend the plays.

A A slight price difference would not plausibly explain why the no-show rate was thirty times greater among those who bought all the tickets in advance than among those who bought them individually.

B This could be true of many people who bought their tickets individually as well as many who bought them in advance.

C This would provide an added incentive for those who bought tickets in advance to attend all the plays.

D **Correct.** If people who bought individual tickets usually did so right before each performance, they would have much less time after buying the tickets to change their minds about whether to attend than would people who bought all the tickets in advance.

E If anything, this might present an additional difficulty for those who bought individual tickets without advance planning, so it would not help to explain the lower no-show rate among buyers of individual tickets.

The correct answer is D.

CR12019

582. Although there is no record of poet Edmund Spenser's parentage, we do know that as a youth Spenser attended the Merchant Tailors' School in London for a period between 1560 and 1570. Records from this time indicate that the Merchant Tailors' Guild then had only three members named Spenser: Robert Spenser, listed as a gentleman; Nicholas Spenser, elected the Guild's Warden in 1568; and John Spenser, listed as a "journeyman cloth-maker." Of these, the last was likely the least affluent of the three—and most likely Edmund's father, since school accounting records list Edmund as a scholar who attended the school at a reduced fee.

Which of the following is an assumption on which the argument depends?

(A) Anybody in sixteenth century London who made clothing professionally would have had to be a member of the Merchant Tailors' Guild.

(B) The fact that Edmund Spenser attended the Merchant Tailors' School did not necessarily mean that he planned to become a tailor.

(C) No member of the Guild could become Guild warden in sixteenth century London unless he was a gentleman.

(D) Most of those whose fathers were members of the Merchant Tailors' Guild were students at the Merchant Tailors' School.

(E) The Merchant Tailors' School did not reduce its fees for the children of the more affluent Guild members.

Argument Construction

Situation Records indicate that the poet Edmund Spenser attended the Merchant Tailors' School for a reduced fee as a youth. There is no record of his parentage, but at the time the Merchant Tailors' Guild had only three members named Spenser, of whom the least affluent was probably John Spenser.

Reasoning *What must be true in order for the cited facts to support the conclusion that John Spenser was probably Edmund Spenser's father?* The implicit reasoning is that since Edmund Spenser attended the Merchant Tailors' School at a reduced fee, his father must have been poor. And since John Spenser was probably the poorest of the three men named Spenser in the Merchant Tailors' Guild, he was probably Edmund Spenser's father. This reasoning assumes that only the children of poor parents had reduced fees at the Merchant Tailors' School, that the children at the school generally had fathers in the Merchant Tailors' Guild, that children in that time and place generally shared their fathers' surnames, and that the two other Spensers in the Merchant Tailors' Guild were not poor enough for their children to qualify for reduced fees.

A John Spenser, as a tailor and member of the guild, could have been Edmund Spenser's father even if some other professional tailors did not belong to the guild and did not have children at the school.

B Although Edmund Spenser became a poet as an adult, he and all his classmates might have attended the school as children because they planned to become tailors.

C The argument assumes that a Guild's Warden probably would have been wealthier than a journeyman cloth-maker, but that might have been probable even if the Guild's Warden were not a "gentleman."

D Even if most children of fathers in the guild did not attend the school, all the children who did attend the school might have had fathers in the guild.

E **Correct.** If the school reduced its fees for children of wealthier guild members, then the fact that Edmund Spenser's fees were reduced would not provide evidence that his father was the poorest of the three Spensers in the guild, as the argument requires.

The correct answer is E.

CR03749

583. Rainwater contains hydrogen of a heavy form called deuterium. The deuterium content of wood reflects the deuterium content of rainwater available to trees during their growth. Wood from trees that grew between 16,000 and 24,000 years ago in North America contains significantly more deuterium than wood from trees growing today. But water trapped in several North American caves that formed during that same early period contains significantly less deuterium than rainwater in North America contains today.

Which of the following, if true, most helps to reconcile the two findings?

(A) There is little deuterium in the North American caves other than the deuterium in the water trapped there.

(B) Exposure to water after a tree has died does not change the deuterium content of the wood.

(C) Industrialization in North America over the past 100 years has altered the deuterium content of rain.

(D) Trees draw on shallow groundwater from rain that falls during their growth, whereas water trapped in caves may have fallen as rainwater thousands of years before the caves formed.

(E) Wood with a high deuterium content is no more likely to remain preserved for long periods than is wood with a low deuterium content.

Argument Construction

Situation In North America, wood from trees that grew 16,000 to 24,000 years ago contains more deuterium than wood from trees growing today. But water in caves that formed during that same period contains less deuterium than rainwater contains today.

Reasoning *What could explain the puzzling discrepancy between the observed deuterium levels in wood and in caves?* Since the deuterium content of wood from trees reflects the deuterium content of rainwater available to the trees while they grew, the deuterium levels observed in wood suggests that North American rainwater contained more deuterium 16,000 to 24,000 years ago than it contains today. But this conclusion seems at odds with the low deuterium levels in water in caves that formed 16,000 to 24,000 years ago. Several factors might explain the discrepancy: the water in those caves might not be rainwater from the period when the caves formed; or some natural process might have altered the deuterium levels in the cave water or the wood; or the wood or caves in which deuterium levels were measured might be statistically abnormal somehow.

A If the caves had absorbed deuterium out of the rainwater trapped in them, there would probably be deuterium in the cave walls. So the observation that there is little deuterium in the caves apart from that in the water eliminates one possible explanation for the oddly low deuterium levels in the cave water.

B This suggests that the deuterium levels in the wood accurately reflect higher deuterium levels in rainwater that fell 16,000 to 24,000 years ago, but it does not explain why the deuterium levels are so low in water in the caves that formed then.

C This could explain why deuterium levels in rainwater have changed, but it does not help explain the discrepancy between the high deuterium levels in the wood and the low deuterium levels in the cave water.

D **Correct.** If the water in the caves fell as rainwater thousands of years before the caves formed, it may date from a period when rainwater contained much less deuterium than during the period 16,000 to 24,000 years ago, and much less than today.

E If wood with high deuterium content were more likely to be preserved, then wood from 16,000 to 24,000 years ago might have a high deuterium content even if the rainwater then had a low deuterium content. So the observation that wood with more deuterium is not more likely to be preserved eliminates one possible explanation for the discrepancy.

The correct answer is D.

CR04925

584. Enforcement of local speed limits through police monitoring has proven unsuccessful in the town of Ardane. In many nearby towns, speed humps (raised areas of pavement placed across residential streets, about 300 feet apart) have reduced traffic speeds on residential streets by 20 to 25 percent. In order to reduce traffic speed and thereby enhance safety in residential neighborhoods, Ardane's transportation commission plans to install multiple speed humps in those neighborhoods.

Which of the following, if true, identifies a potentially serious drawback to the plan for installing speed humps in Ardane?

(A) On residential streets without speed humps, many vehicles travel at speeds more than 25 percent above the posted speed limit.

(B) Because of their high weight, emergency vehicles such as fire trucks and ambulances must slow almost to a stop at speed humps.

(C) The residential speed limit in Ardane is higher than that of the nearby towns where speed humps were installed.

(D) Motorists who are not familiar with the streets in Ardane's residential districts would be likely to encounter the speed humps unawares unless warned by signs and painted indicators.

(E) Bicyclists generally prefer that speed humps be constructed so as to leave a space on the side of the road where bicycles can travel without going over the humps.

Evaluation of a Plan

Situation Ardane's difficulty in getting compliance with speed limits has led it to propose the installation of speed humps to slow traffic. In nearby towns, speed humps have reduced speeds in residential areas by up to 25 percent.

Reasoning *Which one of the statements presented identifies a major disadvantage of the proposed installation of speed humps?* Is it possible that they might slow traffic too much? Clearly, there is a general need for traffic to flow smoothly. Would speed humps affect all types of traffic equally? Perhaps not. For example, certain emergency vehicles must sometimes need to travel quickly through residential neighborhoods. A problem with speed humps is that some heavier vehicles must go very slowly over speed humps.

A This indicates a drawback of not installing speed humps.

B **Correct.** This information indicates a significant drawback—possibly leading to loss of life and property—of the plan to install the speed humps.

C This suggests that installing speed humps might lower speeds significantly below the current speed limits. If speeds became very low, the result could be traffic gridlock that would have unforeseen consequences. However, we have insufficient information to evaluate such possibilities.

D This is unlikely to be a drawback, since such warning signs are typically put in place whenever speed humps are installed.

E This information provides no evidence of a drawback in Ardane's plan for speed humps, since the design of Ardane's planned speed humps is not indicated.

The correct answer is B.

CR00748

585. Which of the following most logically completes the argument below?

NowNews, although still the most popular magazine covering cultural events in Kalopolis, has recently suffered a significant drop in advertising revenue because of falling circulation. Many readers have begun buying a competing magazine that, at 50 cents per copy, costs less than *NowNews* at $1.50 per copy. In order to boost circulation and thus increase advertising revenue, *NowNews's* publisher has proposed making it available at no charge, but this proposal has a serious drawback, since _____.

(A) Those Kalopolis residents with the greatest interest in cultural events are regular readers of both magazines.

(B) One reason *NowNews's* circulation fell was that its competitor's reporting on cultural events was superior.

(C) The newsstands and stores that currently sell *NowNews* will no longer carry it if it is being given away for free.

(D) At present, 10 percent of the total number of copies of each issue of *NowNews* are distributed free to students on college campuses in the Kalopolis area.

(E) *NowNews's* competitor would begin to lose large amounts of money if it were forced to lower its cover price.

Argument Construction

Situation *NowNews* is suffering declines in circulation and advertising revenue due to competition from a lower-priced magazine. The publisher proposes offering *NowNews* for free to reverse these declines.

Reasoning *What would suggest that the publisher's proposal will fail to increase circulation and advertising revenue?* The proposal's intended effect is simply to increase advertising revenue by increasing circulation. Any evidence that offering the magazine for free will not result in more copies being circulated or will not attract advertisers would therefore be evidence of a drawback in the proposal. So a statement offering such evidence would logically complete the argument.

A The fact that certain highly motivated Kalopolis residents still read *NowNews* even at a cost of $1.50 per issue leaves open the possibility that providing the magazine free might still boost readership.

B This suggests that improving its cultural reporting might help *NowNews* increase its circulation, not that the publisher's proposal will fail to do so.

C **Correct.** If the proposal leads newsstands and stores to stop carrying *NowNews*, circulation and advertising revenue would probably decline as a result.

D Even if 10 percent of the copies of *NowNews* are already distributed for free, distributing the remaining 90 percent for free could still increase circulation and advertising revenue as the publisher intends.

E Forcing a competing magazine to lower its cover price and lose lots of money would be an advantage rather than a drawback of the proposal, as far as the publisher of *NowNews* was concerned.

The correct answer is C.

CR07304

586. Archaeologist: Researchers excavating a burial site in Cyprus found a feline skeleton lying near a human skeleton. Both skeletons were in the same sediment at the same depth and equally well-preserved, suggesting that the feline and human were buried together about 9,500 years ago. This shows that felines were domesticated around the time farming began, when they would have been useful in protecting stores of grain from mice.

Which of the following, if true, would most seriously weaken the archaeologist's argument?

(A) Archaeologists have not found any remains of stores of grain in the immediate vicinity of the burial site.

(B) The burial site in Cyprus is substantially older than any other known burial site in which a feline skeleton and a human skeleton appear to have been buried together.

(C) Paintings found near the burial site seem to show people keeping felines as domestic companions, but do not show felines hunting mice.

(D) In Cyprus, there are many burial sites dating from around 9,500 years ago in which the remains of wild animals appear to have been buried alongside human remains.

(E) Before felines were domesticated, early farmers had no effective way to protect stores of grain from mice.

Argument Evaluation

Situation A human skeleton and a feline skeleton were apparently buried together in Cyprus about 9,500 years ago.

Reasoning *What would most strongly suggest that the skeletons do not show that felines were domesticated around the time farming began?* The argument implicitly assumes that farming in Cyprus began around 9,500 years ago, so evidence against that assumption would weaken the argument. The argument could also be weakened by evidence that felines were domesticated much earlier, that the feline skeleton was not from a domesticated cat, or that the two skeletons were not actually buried together around 9,500 years ago.

A Even if archaeologists searched for evidence of a grain store, the fact that no such evidence was found near the burial site is at best only weak evidence that no grain store existed there or slightly farther away.

B The lack of corroborating evidence from other burial sites would weaken the argument slightly but would still be compatible with the hypothesis that this site revealed one of the very first burials of a domesticated cat.

C This would cast doubt on the hypothesis that cats were domesticated mainly to protect stores of grain, but not on the argument's conclusion that cats were domesticated around the time farming began.

D **Correct.** If many wild animals were buried alongside humans in Cyprus around 9,500 years ago, then the feline skeleton is just as likely to be that of a wild animal than that of a domesticated cat.

E Since this would provide an additional reason why early farmers might have domesticated the local cats, it would strengthen rather than weaken the argument.

The correct answer is D.

CR09117

587. The heavy traffic in Masana is a growing drain on the city's economy—the clogging of the streets of the central business district alone cost the economy more than $1.2 billion over the past year. In order to address this problem, officials plan to introduce congestion pricing, by which drivers would pay to enter the city's most heavily trafficked areas during the busiest times of the day.

Which of the following, if true, would most strongly indicate that the plan will be a success?

(A) Approximately one-fifth of the vehicles in the central business district are in transit from one side of the city to the other.

(B) Planners expect that, without congestion pricing, traffic in Masana is likely to grow by 6 percent in the next five years.

(C) In other urban areas, congestion pricing has strongly encouraged carpooling (sharing of rides by private commuters).

(D) Several studies have shown that a reduction in traffic of 15 percent in Masana could result in 5,500 or more new jobs.

(E) Over 30 percent of the vehicles in the city's center are occupied by more than one person.

Evaluation of a Plan

Situation Traffic congestion in Masana has been harming the city's economy. To address the problem, officials plan to make drivers pay to enter the city's most heavily trafficked areas during the busiest times of day.

Reasoning *What would most strongly suggest that the plan will reduce the harm to Masana's economy from traffic congestion?* In order to succeed, the plan will have to be implemented and effectively enforced. Furthermore, the prices drivers pay will have to be high enough to significantly change their behavior in ways that reduce the amount of traffic congestion in the city. Finally, the economic benefits from the reduced traffic congestion will have to substantially outweigh any economically damaging side effects of the congestion pricing. Any evidence that any of these conditions will hold would provide at least some support for the prediction that the plan will succeed.

A This provides no evidence that the congestion pricing would affect the behavior of either the one-fifth of drivers whose vehicles traverse the city or of the other four-fifths of drivers, nor does it give any evidence that the plan would produce overriding economic benefits.

B This indicates that the traffic problem will grow worse if the plan is not implemented, but it does not provide any evidence that the plan will help address the problem.

C **Correct.** This indicates that similar plans have successfully changed drivers' behavior in other cities in a way likely to reduce the number of cars on the road in heavily trafficked areas at busy times of day without producing harmful economic side effects. Thus, it provides evidence that the strategy could also be successful in Masana.

D Although this suggests that reducing traffic congestion would be economically beneficial, it doesn't provide any evidence that the plan will succeed in reducing traffic congestion.

E This suggests that many drivers in the city center are already carpooling, which, if anything, indicates that the plan will be less able to further affect those drivers' behavior and thus could be less effective than it might otherwise be.

The correct answer is C.

CR09151

588. Economist: The most economically efficient way to reduce emissions of air pollutants is to tax them in proportion to the damage they are likely to cause. But in Country Y, many serious pollutants are untaxed and unregulated, and policy makers strongly oppose new taxes. Therefore, the best way to achieve a reduction in air pollutant emissions in Country Y would be to institute fixed upper limits on them.

Which of the following is an assumption of the economist's argument?

(A) Policy makers in Country Y oppose all new taxes equally strongly, regardless of any benefits they may provide.

(B) Country Y's air pollutant emissions would not fall significantly if they were taxed in proportion to the damage they are likely to cause.

(C) Policy makers in Country Y strongly favor reductions in air pollutant emissions.

(D) Country Y's policy makers believe that air pollutant emissions should be reduced with maximum economic efficiency.

(E) Policy makers in Country Y do not oppose setting fixed upper limits on air pollutant emissions as strongly as they oppose new taxes.

Argument Construction

Situation Although taxing air pollution emissions in proportion to the damage they cause is the most economically efficient way to reduce those emissions, many serious pollutants in Nation Y are untaxed and unregulated, and the nation's policy makers strongly oppose new taxes. Therefore, fixed upper limits on such emissions would more effectively reach this goal.

Reasoning *What must be true in order for the factors the economist cites to support the claim that fixing upper limits on air pollutant emissions in Nation Y would be the best way to reduce those emissions?* Political opposition to taxation in Nation Y is the only factor the economist cites to support the argument's conclusion that it would be best to institute fixed upper limits on air pollutants. In order for the premise to support the conclusion, there must be less political opposition in Nation Y to instituting such limits than there would be to the proportional taxation approach the economist prefers.

A Even if the policy makers oppose some new taxes less than others, they could still oppose the proportional taxation approach strongly enough for it to be utterly infeasible.

B Even if the proportional taxation scheme would significantly reduce emissions, it still might not be the best approach for Nation Y if it would generate too much political opposition to be viable there.

C Even if policy makers in Nation Y do not strongly favor reducing emissions, fixing upper limits on emissions might still be a better and more politically feasible way to reduce emissions than any alternative is.

D Since fixing upper emissions limits would be no more economically efficient than the proportional taxation scheme, the policy makers' support for economic efficiency would not make the former approach any more politically feasible than the latter.

E **Correct.** If the policy makers opposed fixing upper emissions limits as strongly as they oppose new taxes, then their opposition to new taxes would no longer support the conclusion that fixing the emissions limits is a better way to reduce emissions.

The correct answer is E.

CR04986

589. Humans get Lyme disease from infected ticks. Ticks get infected by feeding on animals with Lyme disease, but the ease of transmission from host animal to tick varies. With most species of host animal, transmission of Lyme disease to ticks is extremely rare, but white-footed mice are an exception, readily passing Lyme disease to ticks. And white-footed mouse populations greatly expand, becoming the main food source for ticks, in areas where biodiversity is in decline.

The information in the passage most strongly supports which of the following?

(A) In areas where many humans are infected with Lyme disease, the proportion of ticks infected with Lyme disease is especially high.

(B) Very few animals that live in areas where there are no white-footed mice are infected with Lyme disease.

(C) Humans are less at risk of contracting Lyme disease in areas where biodiversity is high.

(D) Ticks feed on white-footed mice only when other host species are not available to them.

(E) The greater the biodiversity of an area, the more likely any given host animal in that area is to pass Lyme disease to ticks.

Argument Construction

Situation White-footed mice readily pass Lyme disease to ticks, which pass it to humans. White-footed mouse populations expand where biodiversity is declining.

Reasoning *What conclusion do the stated facts support?* Since declining biodiversity causes white-footed mouse populations to increase, and white-footed mice are especially likely to pass Lyme disease to ticks, and ticks pass it to humans, declining biodiversity could reasonably be expected to increase the incidence of Lyme disease in both ticks and humans.

A In areas where many humans are infected with Lyme disease, the total number of ticks may be unusually high, so even if the number of infected ticks is unusually high, the proportion of infected ticks may not be unusually high

B Most animals with Lyme disease may get it from sources other than ticks that have fed on infected mice.

C **Correct.** If biodiversity is high, then any biodiversity decline that has already begun has likely not yet reached a point where white-footed mouse populations have greatly expanded, so the risk of people contracting Lyme disease is still relatively less than in areas where biodiversity is low and where significant decline in biodiversity has likely already occurred.

D Even if ticks feed on white-footed mice when few other species are available for them to feed on, they may also sometimes feed on white-footed mice when there are many other species for them to feed on.

E The passage suggests that the overall incidence of Lyme disease is probably lower in more biodiverse areas, so any given host animal in those areas would probably be less likely to pass Lyme disease to a tick.

The correct answer is C.

CR04935

590. Many industrialized nations are trying to reduce atmospheric concentrations of carbon dioxide, a gas released by the burning of fossil fuels. One proposal is to replace conventional cement, which is made with calcium carbonate, by a new "eco-cement." This new cement, made with magnesium carbonate, absorbs large amounts of carbon dioxide when exposed to the atmosphere. Therefore, using eco-cement for new concrete building projects will significantly help reduce atmospheric concentrations of carbon dioxide.

Which of the following, if true, most strengthens the argument?

(A) The cost of magnesium carbonate, currently greater than the cost of calcium carbonate, probably will fall as more magnesium carbonate is used in cement manufacture.

(B) Eco-cement is strengthened when absorbed carbon dioxide reacts with the cement.

(C) Before the development of eco-cement, magnesium-based cement was considered too susceptible to water erosion to be of practical use.

(D) The manufacture of eco-cement uses considerably less fossil fuel per unit of cement than the manufacture of conventional cement does.

(E) Most building-industry groups are unaware of the development or availability of eco-cement.

Argument Evaluation

Situation Many nations are trying to reduce atmospheric concentrations of carbon dioxide. One proposed method is to use a new type of "eco-cement" that absorbs carbon dioxide from air.

Reasoning *What evidence, combined with the cited facts, would most support the prediction that using eco-cement will significantly help reduce atmospheric concentrations of carbon dioxide?* The prediction assumes that the use of eco-cement would be an effective way to reduce carbon dioxide levels. Any evidence supporting this assumption will support the prediction.

A Since eco-cement uses magnesium carbonate, the prediction that magnesium carbonate prices will fall suggests that a potential financial barrier to widespread eco-cement use will diminish. However, those prices may not fall enough to make eco-cement cost-competitive with regular cement.

B Even if absorbed carbon dioxide strengthens eco-cement, the strengthened eco-cement might still be much weaker than regular cement and thus might never become widely used, in which case it will not significantly help reduce atmospheric concentrations of carbon dioxide.

C Even if eco-cement is less susceptible to water erosion than earlier forms of magnesium-based cement were, it might still be much more susceptible to water erosion than regular cement is, and thus might never become widely used.

D **Correct.** This suggests that manufacturing eco-cement produces much less carbon dioxide than manufacturing regular cement does, so it supports the claim that widespread use of eco-cement would be an effective way to reduce carbon dioxide levels.

E If anything, this lack of awareness makes it less likely that eco-cement will become widely used, which in turn makes it less likely that eco-cement will significantly help reduce atmospheric concentrations of carbon dioxide.

The correct answer is D.

CR00895

591. Advertisement: When your car's engine is running at its normal operating temperature, any major brand of motor oil will protect it about as well as Tuff does. When the engine is cold, it is a different story: Tuff motor oil flows better at lower temperatures than its major competitors do. So, if you want your car's engine to have maximum protection, you should use Tuff.

Which of the following, if true, most strengthens the argument in the advertisement?

(A) Tuff motor oil provides above-average protection for engines that happen to overheat.

(B) Tuff motor oil is periodically supplied free of charge to automobile manufacturers to use in factory-new cars.

(C) Tuff motor oil's share of the engine oil market peaked three years ago.

(D) Tuff motor oil, like any motor oil, is thicker and flows less freely at cold temperatures than at hot temperatures.

(E) Tuff motor oil is manufactured at only one refinery and shipped from there to all markets.

Argument Evaluation

Situation An advertisement argues that since Tuff motor oil flows better than its major competitors at low temperatures and works about as well as they do at normal temperatures, it provides *maximum protection* for car engines.

Reasoning *What additional evidence would suggest that Tuff motor oil provides the best available protection for car engines?* The argument requires the assumptions that no type of motor oil other than the "major brands" provides superior protection, that flowing better at lower temperatures ensures superior protection at those temperatures, and that Tuff protects car engines at least as well as its competitors do at above-normal temperatures. Any evidence supporting any of these assumptions would strengthen the argument.

A **Correct.** If Tuff provides above-average protection when engines overheat, in addition to the solid protection it provides at normal and low temperatures, it may well provide the best available protection overall.

B The company that makes Tuff might give automobile manufacturers free motor oil as a promotional gimmick even if Tuff is an inferior product.

C Tuff's sales might have declined over the past three years because consumers have realized that Tuff is an inferior product.

D The similar responses of Tuff and other motor oils to temperature changes do not suggest that Tuff provides better protection overall than those other motor oils do.

E Even if Tuff is manufactured at only one refinery, it may still be an inferior product.

The correct answer is A.

CR13108

592. *The Testament of William Thorpe* was published around 1530 as an appendix to Thorpe's longer Examination. Many scholars, however, doubt the attribution of the *Testament* to Thorpe because, whereas the *Examination* is dated 1406, the *Testament* is dated 1460. One scholar has recently argued that the 1460 date be amended to 1409, based on the observation that when these numbers are expressed as Roman numerals, MCCCCLX and MCCCCIX, it becomes easy to see how the dates might have become confused through scribal error.

Which of the following, if true, would most support the scholar's hypothesis concerning the date of the *Testament*?

(A) The sole evidence that historians have had that William Thorpe died no earlier than 1460 was the presumed date of publication of the *Testament*.

(B) In the preface to the 1530 publication, the editor attributes both works to William Thorpe.

(C) Few writers in fifteenth-century England marked dates in their works using only Roman numerals.

(D) The *Testament* alludes to a date, "Friday, September 20," as apparently contemporaneous with the writing of the *Testament*, and September 20 fell on a Friday in 1409 but not in 1460.

(E) The *Testament* contains few references to historical events that occurred later than 1406.

Argument Construction

Situation *The Testament of William Thorpe*, dated 1460, was published around 1530 as an appendix to Thorpe's *Examination*, dated 1406. But when expressed in Roman numerals, 1460 could easily be confused with 1409.

Reasoning *Given the facts cited, what would provide additional evidence that Thorpe's* Testament *dates from 1409 rather than 1460?* The scholar's hypothesis that the work dates from 1409 is based on the observation that in Roman numerals, 1409 might easily have been improperly transcribed as 1460. This hypothesis could be supported by evidence that the manuscripts were dated in Roman numerals, or by any independent evidence that 1409 is a more likely date for the *Testament* than 1460.

A This suggests that scholars have no biographical evidence that the *Testament* was published in 1460, but they could still have abundant evidence of other types to support that date, such as the text's cultural allusions or references to other works.

B The editor of the 1530 publication could easily have been mistaken about the authorship of one or both works. And even if the editor were correct, Thorpe might have lived long enough to write one work in 1406 and the other in 1460.

C This would cast doubt on the scholar's argument by providing evidence that the original manuscripts were not dated only in Roman numerals.

D **Correct.** This provides strong evidence directly supporting the hypothesis that the *Testament* dates from 1409 specifically.

E Even if the *Testament* contained only one reference to a historical event that occurred later than 1406 (for example, one event in 1459), that reference alone could provide strong evidence that the work dates from 1460 rather than 1409.

The correct answer is D.

CR00777

593. A prominent investor who holds a large stake in the Burton Tool Company has recently claimed that the company is mismanaged, citing as evidence the company's failure to slow down production in response to a recent rise in its inventory of finished products. It is doubtful whether an investor's sniping at management can ever be anything other than counterproductive, but in this case, it is clearly not justified. It is true that **an increased inventory of finished products often indicates that production is outstripping demand**, but in Burton's case it indicates no such thing. Rather, **the increase in inventory is entirely attributable to products that have already been assigned to orders received from customers**.

In the argument given, the two **boldfaced** portions play which of the following roles?

(A) The first states a generalization that underlies the position that the argument as a whole opposes; the second provides evidence to show that the generalization does not apply in the case at issue.

(B) The first states a generalization that underlies the position that the argument as a whole opposes; the second clarifies the meaning of a specific phrase as it is used in that generalization.

(C) The first provides evidence to support the conclusion of the argument as a whole; the second is evidence that has been used to support the position that the argument as a whole opposes.

(D) The first provides evidence to support the conclusion of the argument as a whole; the second states that conclusion.

(E) The first and the second each provide evidence against the position that the argument as a whole opposes.

Argument Construction

Situation An investor has criticized a company, based on the company's recent increase in inventory and on its not decreasing production as a result of this increase.

Reasoning *What roles do the two boldfaced statements play in the argument?* The argument suggests that the investor's criticism is based on a principle that increased inventory of finished products often indicates that production is faster than it should be, given the existing demand for a company's products. However, the argument then states that the increase in inventory at the company in question is "entirely attributable" to existing orders of products. The argument thus suggests that the investor's criticism is misplaced, based on a suggestion as to (1) a principle that the investor could be using to support her argument and (2) an explanation as to why the principle does not apply to the company. The two boldfaced portions state these respective elements.

A **Correct.** The first boldfaced portion states the principle that may provide the basis of the investor's criticism, which the argument as a whole opposes. The second boldfaced portion is a statement that, if true, the generalization would not apply to the company in question.

B This option correctly describes the first of the boldfaced portions. However, rather than clarifying an aspect of the meaning of the first generalization, the second boldfaced portion indicates why the first generalization may not apply to the company.

C This option incorrectly describes both of the boldfaced portions. The first boldfaced portion states a general principle that could support the position that the argument *opposes*. The second boldfaced portion then criticizes the application of the principle.

D Because the second boldfaced portion describes a fundamental premise rather than the conclusion, the description in this option of the second boldfaced portion is incorrect.

E If we think of an argument as a set of statements that are meant to support, or provide evidence for, a conclusion, then, because the boldfaced statements are indeed part of the argument, they may be seen as providing evidence for the position the argument opposes. However, a description of the roles of the boldfaced statements in this argument would need to provide more detail, such as what option A provides.

The correct answer is A.

CR10028

594. To reduce productivity losses from employees calling in sick, Corporation X implemented a new policy requiring employees to come into work unless they were so sick that they had to go to a doctor. But a year after the policy was implemented, a study found that Corporation X's overall productivity losses due to reported employee illnesses had increased.

Which of the following, if true, would best explain why the policy produced the reverse of its intended effect?

(A) After the policy was implemented, employees more frequently went to the doctor when they felt sick.

(B) Before the policy was implemented, employees who were not sick at all often called in sick.

(C) Employees coming into work when sick often infect many of their coworkers.

(D) Unusually few employees became genuinely sick during the year after the policy was implemented.

(E) There are many other factors besides employee illness that can adversely affect productivity.

Evaluation of a Plan

Situation After a company started requiring employees to come to work unless they were sick enough to have to go to a doctor, the company's productivity losses from reported employee illness increased.

Reasoning *What would explain why the policy increased productivity losses from reported employee illness?* Any factors that could have plausibly caused the policy to increase employee absenteeism from reported illness or to reduce the employees' productivity at work as a result of reported illness could explain why the policy increased productivity losses from reported illness.

A Even though the policy required sick employees to consult a doctor, there is no reason to think that employees' doing so would have made them less productive than they would otherwise have been when absent from work.

B This suggests that the policy made it more difficult for employees to falsely claim illness as an excuse for a work absence. Reduction in absences should result in productivity gains rather than losses.

C **Correct.** This could have been a result of the policy and would have led to productivity losses possibly greater than those seen before the policy was introduced.

D This would help to explain lower productivity losses from reported illness after the policy was implemented, not higher productivity losses.

E The question is what could explain how the policy increased productivity losses from reported employee illness specifically, not productivity losses from any other factors.

The correct answer is C.

CR08443

595. Advertising by mail has become much less effective, with fewer consumers responding. Because consumers are increasingly overwhelmed by the sheer amount of junk mail they receive, most discard almost all offers without considering them. Thus, an effective way for corporations to improve response rates would be to more carefully target the individuals to whom they mail advertising, thereby cutting down on the amount of junk mail each consumer receives.

Which of the following, if true, would most support the recommendation above?

(A) There are cost-effective means by which corporations that currently advertise by mail could improve response rates.

(B) Many successful corporations are already carefully targeting the individuals to whom they mail advertising.

(C) Any consumer who, immediately after receiving an advertisement by mail, merely glances at it is very likely to discard it.

(D) Improvements in the quality of the advertising materials used in mail that is carefully targeted to individuals can improve the response rate for such mail.

(E) Response rates to carefully targeted advertisements by mail are considerably higher, on average, than response rates to most other forms of advertising.

Evaluation of a Plan

Situation Advertising by mail has become less effective because consumers overwhelmed with the amount of junk mail they receive discard almost all of it without considering it.

Reasoning *What would most help to support the claim that making mail advertising more carefully targeted would improve response rates?* The passage recommends targeted advertising, reasoning that since targeted advertising would reduce the total amount of junk mail consumers receive, it would generate higher response rates. Any additional evidence for the claim that carefully targeted advertising would improve response rates would support this recommendation.

A Even if targeted advertising and every other means of improving response rates were too expensive to be cost-effective, targeted advertising could still be effective for any corporation willing to pay the expense.

B If many corporations already mail targeted advertising, and mail advertising is nonetheless yielding declining response rates, that suggests that targeted mail is an ineffective way to increase response rates.

C This could be equally true for targeted and untargeted mail advertising, so it does not suggest that the former is more effective.

D The question under consideration is whether more carefully targeted mail advertising would in itself increase response rates, not whether higher quality advertising would do so.

E **Correct.** This provides some evidence that carefully targeted mail advertising is associated with higher response rates than untargeted mail advertising is, and therefore that targeting mail advertising more carefully would improve response rates.

The correct answer is E.

CR01905

596. Petrochemical industry officials have said that the extreme pressure exerted on plant managers during the last five years to improve profits by cutting costs has done nothing to impair the industry's ability to operate safely. However, environmentalists contend that the recent rash of serious oil spills and accidents at petrochemical plants is traceable to cost-cutting measures.

Which of the following, if true, would provide the strongest support for the position held by industry officials?

(A) The petrochemical industry benefits if accidents do not occur, since accidents involve risk of employee injury as well as loss of equipment and product.

(B) Petrochemical industry unions recently demanded that additional money be spent on safety and environmental protection measures, but the unions readily abandoned those demands in exchange for job security.

(C) Despite major cutbacks in most other areas of operation, the petrochemical industry has devoted more of its resources to environmental and safety measures in the last five years than in the preceding five years.

(D) There is evidence that the most damaging of the recent oil spills would have been prevented had cost-cutting measures not been instituted.

(E) Both the large fines and the adverse publicity generated by the most recent oil spill have prompted the petrochemical industry to increase the resources devoted to oil-spill prevention.

Argument Evaluation

Situation Petrochemical industry officials claim that pressure on plant managers to cut costs over the past five years has not made the industry's operations any less safe. Environmentalists claim that recent oil spills and accidents show otherwise.

Reasoning *What evidence would most strongly suggest that the cost-cutting pressure was not responsible for the recent rash of oil spills and accidents?* Evidence that the plant managers did not cut costs in any specific ways likely to have increased the likelihood of oil spills and accidents would support the industry officials' position that the cost-cutting pressure has not made petrochemical operations any less safe.

A Even if the petrochemical industry has good reasons to try to prevent accidents, the recent rash of serious accidents suggests that it is failing to do so and that the cost-cutting pressure might be responsible.

B This suggests that the unions, whose members could directly observe the cost-cutting pressure's effects, share the environmentalists' belief that this pressure contributed to the oil spills and accidents. Because the unions abandoned their demands, their concerns probably have not been addressed.

C **Correct.** This suggests that, as the industry officials claim, the cost-cutting pressure has not in itself reduced the industry's effectiveness at preventing oil spills and accidents. Thus, it suggests that other factors are probably responsible for the recent problems.

D This clearly suggests that the cost-cutting measures have indeed caused the industry to operate less safely, as the environmentalists claim.

E Although this suggests that the industry is now trying to address the recent problems, the cost-cutting measures might nonetheless have caused all those problems.

The correct answer is C.

CR01368

597. A company has developed a new sensing device that, according to the company's claims, detects weak, ultralow-frequency electromagnetic signals associated with a beating heart. These signals, which pass through almost any physical obstruction, are purportedly detected by the device even at significant distances. Therefore, if the company's claims are true, their device will radically improve emergency teams' ability to locate quickly people who are trapped within the wreckage of collapsed buildings.

Which of the following, if true, most strengthens the argument?

(A) People trapped within the wreckage of collapsed buildings usually have serious injuries that require prompt medical treatment.

(B) The device gives a distinctive reading when the signals it detects come from human beings rather than from any other living beings.

(C) Most people who have survived after being trapped in collapsed buildings were rescued within two hours of the building's collapse.

(D) Ultralow-frequency signals are not the only electromagnetic signals that can pass through almost any physical obstruction.

(E) Extensive training is required in order to operate the device effectively.

Argument Evaluation

Situation A new sensing device can detect—at significant distances and even behind obstructions such as walls—weak, ultralow-frequency electromagnetic signals that are characteristic of heartbeats. It is predicted, based on this information, that the new device will shorten the time it currently takes to locate people buried under collapsed buildings but still alive.

Reasoning *What new information, if accurate, would provide further evidence that would support the prediction?* The existing evidence fails to tell us whether the new device can distinguish between human heartbeats and heartbeats from other species. If the device does not quickly provide signals characteristic of human heartbeats, then the prediction might not be correct; even if the prediction eventually turns out to be correct, the evidence given for it is insufficient. Any new information that implies the signals provided by the device can discern a human heartbeat from those of nonhuman species will strengthen support for the prediction.

A This implies that prompt rescue of people trapped under collapsed buildings is vitally important. The prediction is that the new device will speed rescue of such people, but the new information here does nothing to indicate that the prediction is accurate.

B **Correct.** This information fills an important gap (already discussed) in the evidence for the prediction.

C Even if this is true, shortening the time for locating and rescuing people from collapsed buildings would clearly be beneficial. However, the new information given here does not make it more likely that the prediction is correct.

D If this is correct, then if anything, it somewhat undermines the evidence given for the prediction, since it raises the possibility that the detection ability of the device might be impeded by "noise" from irrelevant electromagnetic signals near the collapsed building.

E This could lead to practical obstacles when using the device even in emergency situations, with the result that the device might never actually be used by competent personnel to "improve emergency teams' ability" because the "extensive training" would cost too much.

The correct answer is B.

CR11639

598. Economist: The price of tap water in our region should be raised drastically. **Supplies in local freshwater reservoirs have been declining for years** because water is being used faster than it can be replenished. Since the price of tap water has been low, **few users have bothered to adopt even easy conservation measures**.

The two sections in boldface play which of the following roles in the economist's argument?

(A) The first is a conclusion for which support is provided, and which in turn supports the main conclusion; the second is the main conclusion.

(B) The first is an observation for which the second provides an explanation; the second is the main conclusion but not the only conclusion.

(C) The first is a premise supporting the argument's main conclusion; so is the second.

(D) The first is the only conclusion; the second provides an explanation for the first.

(E) The first is the main conclusion; the second is a conclusion for which support is provided, and which in turn supports the first.

Argument Construction

Situation Local water supplies have been declining for years because of excessive water use and low prices. Few users have adopted even easy conservation measures.

Reasoning *What roles do the two boldface statements play in the argument?* Both are factual observations. Since no further evidence or support is provided for either, neither can be a conclusion in the argument. However, interconnected causal explanations, signaled by *because* and *since,* are provided for both. The observation in the first boldface statement is causally explained by the further observation that water is being used faster than it can be replenished, which in turn is causally explained by the entire final sentence. The observation in the second boldface statement is causally explained by the observation that the price of tap water has been low. The only remaining portion of the argument is the initial sentence, a recommendation supported by these four observations together, and by the causal claims in which they are embedded. Thus, the four observations (including the two boldface statements) and the causal claims containing them are all premises, and the initial statement is the argument's only conclusion.

A As explained above, the two boldface statements are premises of the argument. Although causal explanations are provided for both, no support or evidence is provided for either.

B As explained above, the second boldface statement does provide part of the causal explanation for the observation in the first boldface statement. But no support is provided for either statement, so neither is a conclusion.

C **Correct.** As explained above, each of the statements is a premise that serves along with other claims to support the recommendation in the initial sentence, which is the argument's only conclusion, and in that sense its main conclusion.

D As explained above, the second boldface statement does provide part of the causal explanation for the observation in the first boldface statement. But no support is provided for either statement, so neither is a conclusion in the argument.

E As explained above, the two boldface statements are premises of the argument. Although causal explanations are provided for both, no support or evidence is provided for either.

The correct answer is C.

CR13127

599. Politician: Hybrid cars use significantly less fuel per kilometer than nonhybrids. And fuel produces air pollution, which contributes to a number of environmental problems. Motorists can save money by driving cars that are more fuel efficient, and they will be encouraged to drive hybrid cars if we make them aware of that fact. Therefore, we can help reduce the total amount of pollution emitted by cars in this country by highlighting this advantage of hybrid cars.

Which of the following, if true, would most indicate a vulnerability of the politician's argument?

(A) People with more fuel-efficient cars typically drive more than do those with less fuel-efficient cars.

(B) Not all air pollution originates from automobiles.

(C) Hybrid cars have already begun to gain popularity.

(D) Fuel-efficient alternatives to hybrid cars will likely become available in the future.

(E) The future cost of gasoline and other fuel cannot be predicted with absolute precision or certainty.

Argument Evaluation

Situation According to a politician, hybrid cars use less fuel per kilometer than nonhybrids, and fuel produces air pollution. Motorists can save money by driving fuel-efficient cars, and will be encouraged to do so if made aware of the fact. The politician concludes that highlighting this fact will result in a reduction in air pollution.

Reasoning *What would suggest that telling motorists they can save money by driving fuel-efficient cars would not reduce automotive air pollution, despite the facts cited by the politician?* The politician's implicit reasoning is that since hybrid cars use less fuel per kilometer, and fuel produces air pollution, motorists who drive hybrid cars must produce less air pollution than those who drive nonhybrids. The politician concludes that encouraging motorists to drive hybrid cars by telling them they would save money on fuel will therefore reduce automotive air pollution. Evidence that motorists who drive hybrid cars produce just as much automotive air pollution as those who drive nonhybrids would undermine this argument.

A **Correct.** If drivers of hybrid cars tend to drive more kilometers than drivers of nonhybrids, then they may consume just as much fuel and produce just as much air pollution as the nonhybrid car drivers do, despite their lower fuel use per kilometer.

B The politician's argument is only about air pollution from cars specifically, not air pollution from all sources.

C Even if hybrid cars are beginning to gain popularity, informing motorists of the cost savings from fuel efficiency could help these cars become more popular than they would otherwise be.

D Encouraging motorists to switch to hybrid cars now could reduce fuel use and automotive air pollution in the near future even if other, more fuel-efficient vehicles will become available further in the future.

E Even if the future cost of fuel cannot be predicted accurately, encouraging motorists to switch to hybrid cars could reduce air pollution as the politician argues.

The correct answer is A.

CR03862

600. Which of the following most logically completes the passage?

Pecan growers get a high price for their crop when pecans are comparatively scarce, but the price drops sharply when pecans are abundant. Thus, in high-yield years, growers often hold back part of their crop in refrigerated warehouses for one or two years, hoping for higher prices in the future. This year's pecan crop was the smallest in five years. It is nonetheless quite possible that a portion of this year's crop will be held back, since _____.

(A) each of the last two years produced record-breaking pecan yields

(B) the quality of this year's pecan crop is no worse than the quality of the pecan crops of the previous five years

(C) pecan prices have not been subject to sharp fluctuations in recent years

(D) for some pecan growers, this year's crop was no smaller than last year's

(E) the practice of holding back part of one year's crop had not yet become widespread the last time the pecan crop was as small as it was this year

Argument Construction

Situation The price of pecans tends to drop sharply in years when pecans are abundant. So in high-yield years, growers often hold back part of the harvest in refrigerated warehouses. This year's harvest was the smallest in five years.

Reasoning *What would provide the best completion of the argument?* The argument's conclusion is that some of this year's crop might be held back. The blank to be completed should provide a reason in support of that conclusion. What would lead us to believe that some of this year's crop might go into cold storage even though the crop was unusually small? Only in high-yield years does this usually happen. But suppose there is *already* a large quantity of pecans in cold storage from previous harvests. Given this information, it would make perfect sense to expect that the pecans already in cold storage would be marketed first, while some of the latest crop would be stored. This would avoid the market oversupply and lower producer prices that might result if both all of this year's crop and all of the already stored pecans were marketed this year.

A **Correct.** This answer choice provides information that makes it more probable that the conclusion is true.

B The argument provides no information whatsoever that would suggest the decision to store or not to store pecans is based on evaluation of the crop's quality.

C This information is of little or no relevance. It is reasonable to think that predictions about pecan prices this year would affect the decision to store or not to store. But the information in this answer choice sheds little or no light on what this year's pecan prices might be, given that, as the passage tells us, this year's crop is exceptionally small.

D It is not surprising that some growers had crops this year that were as big as their crops the year before. But what matters, what affects the price of pecans, is the overall size of total pecan production and the abundance or scarcity of pecans at the time.

E This piece of history about marketing and storage practices explains why pecans were not placed in storage in previous small-yield years, but it provides no reason to believe that some of the new pecan crop will be stored this year.

The correct answer is A.

CR09534

601. Coffee shop owner: A large number of customers will pay at least the fair market value for a cup of coffee, even if there is no formal charge. Some will pay more than this out of appreciation of the trust that is placed in them. And our total number of customers is likely to increase. We could therefore improve our net cash flow by implementing an honor system in which customers pay what they wish for coffee by depositing money in a can.

Manager: We're likely to lose money on this plan. Many customers would cheat the system, paying a very small sum or nothing at all.

Which of the following, if true, would best support the owner's plan, in light of the manager's concern?

(A) The new system, if implemented, would increase the number of customers.

(B) By roasting its own coffee, the shop has managed to reduce the difficulties (and cost) of maintaining an inventory of freshly roasted coffee.

(C) Many customers stay in the cafe for long stretches of time.

(D) The shop makes a substantial profit from pastries and other food bought by the coffee drinkers.

(E) No other coffee shop in the area has such a system.

Evaluation of a Plan

Situation The owner and the manager of a coffee shop disagree about whether allowing customers to pay for coffee on an honor system would increase or decrease profits.

Reasoning *What would be the best evidence that the honor-system plan would increase profits even if many customers cheated the system?* The owner argues that profits would increase because many customers will choose to pay as much or more than before and the total number of customers will likely increase. But the manager points out that many customers would also choose to pay little or nothing. Assuming that the manager is correct about that, what further support could the owner present for the claim that the plan would still be profitable?

A Since the owner has already basically asserted this, asserting it again would not provide any significant additional support for the plan.

B This suggests that the shop is already profitable, not that the honor-system plan would make it more profitable.

C Customers who stay in the cafe for long stretches would not necessarily pay any more per cup on the honor-system plan than other customers would.

D **Correct.** If the customer base increases (as both the owner and the manager seem to agree), more customers will likely purchase highly profitable pastries and other foods, thus boosting profits.

E The reason no other coffee shop in the area has an honor system may be that their owners and managers have determined that it would not be profitable.

The correct answer is D.

CR06749

602. Which of the following most logically completes the argument?

By competing with rodents for seeds, black ants help control rodent populations that pose a public health risk. However, a very aggressive species of black ant, the Loma ant, which has recently invaded a certain region, has a venomous sting that is often fatal to humans. Therefore, the planned introduction into that region of ant flies, which prey on Loma ants, would benefit public health, since _____.

(A) ant flies do not attack black ants other than Loma ants

(B) Loma ants are less effective than many bird species in competing with rodents for seeds

(C) certain other species of black ants are more effective than Loma ants in competing with rodents for seeds

(D) the sting of Loma ants can also be fatal to rodents

(E) some pesticides that could be used to control Loma ants are harmful to the environment

Argument Construction

Situation Black ants help to control populations of rodents by competing with them for seeds. But a very aggressive species of black ant, the Loma ant, has a sting that can be fatal to humans. Ant flies prey on Loma ants and their presence can thereby benefit public health.

Reasoning *Which of the possible completions of the passage provides the most support for the conclusion?* The argument's conclusion is that introducing ant flies into the region where Loma ants have recently invaded would benefit public health. We know from the passage that black ants, generally, benefit public health by keeping down rodent populations. However, the sting of Loma ants, a species of black ant, can be fatal to humans. Ant flies prey on Loma ants. To that extent their introduction in the region would tend to benefit public health by making fatal Loma stinging of humans less likely. But if these ant flies also prey on black ants other than the Loma ants, then to that extent they would undermine another public health benefit associated with controlling rodents. Thus, the information that ant flies do not prey on black ants other than Loma ants would provide strong logical support for the conclusion.

A **Correct.** This most logically completes the argument because it addresses a potential downside of introducing the ant flies into the region. The potential downside is that it might reduce the desirable effect that other species of black ants have in keeping down the rodent populations.

B We have no idea whether the bird species that are more effective than Loma ants at competing with rodents for seeds are even present in the region in question.

C This does not help the conclusion very much because we do not know *from the passage* whether ant flies prey on other species of black ants besides Loma ants.

D If anything, this is a reason *not to introduce ant flies* into the region. This answer choice at least suggests that Loma ants might have some positive effect on public health because they might keep down rodent populations by reducing their survival chances.

E This provides very little support for the conclusion. It does not exclude the possibility that there are pesticides—perhaps several—that would control Loma ants effectively without harming the environment. So it is not a strong reason for introducing ant flies.

The correct answer is A.

CR01900

603. The city of Workney, in raising bus fares from $1.00 to $1.25, proposed that 18 fare tokens be sold for $20.00 to alleviate the extra burden of the fare increase on the city's poor people. Critics suggested alternatively that 9 fare tokens be sold for $10.00, because a $20.00 outlay would be prohibitive for poor riders.

The alternative proposal depends on which of the following assumptions?

(A) Poor residents of Workney will continue to ride the buses in the same numbers despite the fare increase.

(B) Riders who are poor would be more likely to take advantage of the savings afforded by the 9-token offer than would riders who are not poor.

(C) The outlay of $10.00 for the purchase of 9 fare tokens would not be prohibitive for bus riders who are poor.

(D) The proposed fare increase is needed for the purchase of new buses for the city's bus system.

(E) Fewer riders would regularly purchase 18 fare tokens at once than would purchase only 9 fare tokens at once.

Evaluation of a Plan

Situation The city of Workney raised bus fares from $1.00 to $1.25. To help poor people, the city proposed allowing people to block-purchase 18 tickets for $20.00. But critics argued that poor people could ill-afford to pay $20.00 at one time. Their alternative proposal was: allow block-purchase of 9 tickets for $10.00.

Reasoning *What did the critics assume in making their alternative proposal?* The city's block-purchase proposal would result in a saving of $4.50 over the cost of purchasing 18 tickets separately, and the critics' alternative block-purchase proposal would result in a saving of $2.25 over the cost of purchasing 9 tickets separately. The savings per ticket on either proposal would be identical. But the critics' proposal assumes that people who would not be able to block-purchase 18 tickets for $20.00 would be able to block-purchase 9 tickets for $10.00.

A The critics assume something incompatible with this. Their alternative was proposed because they believed that some poor residents would otherwise, at least sometimes, be unable to purchase bus tickets.

B Nothing in the critics' proposal allows only poor people to benefit from the proposal. No information is given about the relative likelihoods of poor or nonpoor riders availing of the block-purchase option perhaps for convenience rather than for savings.

C **Correct.** The critics' proposal depends on the assumption that poor bus riders would generally be able to block-purchase 9 tickets for $10.00.

D The alternative proposal does not depend on assuming this. The proposal could still make sense even if the additional revenue from the fare increases was needed to pay higher wages to bus drivers.

E The critics' proposal does not depend on assuming this. How many riders would block-purchase 18 tickets? How many would block-purchase 9 tickets? These are open questions, as far as the critics' proposal is concerned. The 9-ticket option might or might not turn out to be more frequently purchased than the 18-ticket option. The critics' concern is to make the 9-ticket option available to poor riders, not to prevent nonpoor riders from availing of it.

The correct answer is C.

CR03272

604. Birds have been said to be descended from certain birdlike dinosaur species with which they share distinctive structural features. The fossil record, however, shows that this cannot be so, since there are bird fossils that are much older than the earliest birdlike dinosaur fossils that have been found.

Which of the following is an assumption on which the argument relies?

(A) The birdlike dinosaurs have no living descendants.

(B) There are no flightless dinosaur species that have the distinctive structural features shared by birds and birdlike dinosaurs.

(C) There are no birdlike dinosaur fossils that are older than the bird fossils but have not yet been unearthed.

(D) It could not have been the case that some birds were descended from one of the birdlike dinosaur species and other birds from another.

(E) Birds cannot have been descended from dinosaur species with which the birds do not share the distinctive structural features.

Argument Construction

Situation Although birds have been said to be descended from birdlike dinosaurs, some bird fossils predate the earliest known birdlike dinosaur fossils.

Reasoning *What must be true in order for the premise that some bird fossils predate the earliest known birdlike dinosaur fossils to support the conclusion that birds are not descended from birdlike dinosaurs?* The argument implicitly reasons that since the cited bird fossils predate the earliest known birdlike dinosaur fossils, they must be from birds that lived before the earliest birdlike dinosaurs, and which therefore could not have been descended from birdlike dinosaurs. This reasoning assumes that any birdlike dinosaurs that lived before the first birds would have left fossils that still exist. It also assumes that no undiscovered birdlike dinosaur fossils predate the cited bird fossils.

A The argument is only about whether birds are descended from birdlike dinosaurs. Whether birdlike dinosaurs have any living descendants other than birds is irrelevant.

B The argument is only about birds and birdlike dinosaurs. It is not about other types of dinosaurs that were not birdlike.

C **Correct.** If any undiscovered birdlike dinosaur fossils predate the cited bird fossils, then the latter fossils' age does not support the conclusion that birds are not descended from birdlike dinosaurs.

D The argument purports to establish that the relative ages of bird fossils and birdlike dinosaur fossils show that birds cannot be descended from any of the known birdlike dinosaur species. In doing this, it acknowledges multiple birdlike dinosaur species and leaves open the question of whether some birds may be descended from one such species and other birds from another such species.

E The argument does not claim that the known fossil record shows that birds cannot be descended from dinosaurs. It only claims that the record shows that they cannot be descended from the birdlike dinosaurs that shared their distinctive structural features.

The correct answer is C.

CR08239

605. City council member: Demand for electricity has been increasing by 1.5 percent a year, and there simply is no more space to build additional power plants to meet future demand increases. We must therefore begin to curtail usage, which is why I propose passing ordinances requiring energy-conservation measures in all city departments.

The city council member's proposal assumes which of the following?

(A) Existing power plants do not have the capacity to handle all of the projected increase in demand for electricity.

(B) No city departments have implemented energy-conservation measures voluntarily.

(C) Passing ordinances designed to curtail electricity usage will not have negative economic consequences for the city.

(D) Residential consumers are not responsible for the recent increases in demand for electricity.

(E) City departments that successfully conserve energy will set a good example for residential and industrial consumers of electricity.

Argument Construction

Situation A city council member proposes energy-conservation measures for all city government departments because there is no room to build new power plants to meet future increases in the demand for electricity.

Reasoning *What must be true in order for the factors the city council member cites to help justify the proposal?* The city council member says electricity usage must be curtailed on account of an increasing demand for electricity and a lack of space for new power plants that could meet future demand increases. In order for this reasoning to help justify the proposal, the cited factors must actually establish a need to curtail electricity usage.

A **Correct.** If current power plants could satisfy the projected increased demand for electricity, then the increasing demand and the lack of room to build new plants would not establish a need to curtail electricity usage.

B The proposed ordinances could still be necessary even if one city department had voluntarily implemented energy-conservation measures.

C Passing the ordinances could still be necessary even if they would have some negative economic effects.

D No matter who is responsible for the recent increases in demand, curtailing the city government's electricity usage could still help to reduce demand.

E Ordinances to curtail the city government's energy usage could be economically necessary regardless of whether or not departments that obey the ordinances set a good example.

The correct answer is A.

CR00862

606. Which of the following most logically completes the argument below?

Using broad-spectrum weed killers on weeds that are competing with crops for sunlight, water, and nutrients presents a difficulty: how to keep the crop from being killed along with the weeds. For at least some food crops, specially treated seed that produces plants resistant to weed killers is under development. This resistance wears off as the plants mature. Therefore, the special seed treatment will be especially useful for plants that _____.

(A) produce their crop over an extended period of time, as summer squash does

(B) produce large seeds that are easy to treat individually, as corn and beans do

(C) provide, as they approach maturity, shade dense enough to keep weeds from growing

(D) are typically grown in large tracts devoted to a single crop

(E) are cultivated specifically for the seed they produce rather than for their leaves or roots

Argument Construction

Situation A difficulty in using broad-spectrum weed killers is keeping them from killing the food crops along with the weeds. Specially treated seed is being developed that will protect certain food crop plants in their earlier stages of growth.

Reasoning *Which is the best completion for the conclusion?* The conclusion is incompletely stated as "Therefore, the special seed treatment will be especially useful for plants that _____." The question is what sorts of plants does the passage suggest the seed treatment would be especially useful for. We have been told that this treatment makes the plants resistant to weed killer, but that this resistance wears off when the plant matures. So the treatment will be most useful with plants that are not harmed by weed killer and that suffer no significant disadvantage when the resistance wears off as the plant matures. Choice (C) is the correct answer choice because it describes a sort of plant that can combat weeds and requires no weed killer once the plant matures.

A Given that the seed treatment wears off as the plant matures, it would not be especially useful for plants that produce their crops over an extended period.

B We have not been told whether small seeds are more difficult to treat, and so we have no basis to conclude that the special seed treatment would be *especially useful* for plants that have large seeds that are easy to treat individually. We have also been given no reason to think that it is better to treat seeds individually.

C **Correct.** Plants that, as they approach maturity, produce shade dense enough to keep weeds from growing, would benefit from resistance to weed killer when young and would not need weed killer when they have matured and lost their resistance.

D We have been given no reason to think that the seed treatment would be especially useful for plants grown in a large tract devoted to a single crop. For example, why would it be less useful for small tracts with a variety of crops?

E A plant harvested for its roots, fruits, or leaves, rather than for its seeds, would derive no less an advantage from resistance to weed killers in earlier stages of growth.

The correct answer is C.

CR00713

607. Previously, Autoco designed all of its cars itself and then contracted with specialized parts suppliers to build parts according to its specifications. Now it plans to include its suppliers in designing the parts they are to build. Since many parts suppliers have more designers with specialized experience than Autoco has, Autoco expects this shift to reduce the overall time and cost of the design of its next new car.

Which of the following, if true, most strongly supports Autoco's expectation?

(A) When suppliers provide their own designs, Autoco often needs to modify its overall design.

(B) In order to provide designs for Autoco, several of the parts suppliers will have to add to their existing staffs of designers.

(C) Parts and services provided by outside suppliers account for more than 50 percent of Autoco's total costs.

(D) When suppliers built parts according to specifications provided by Autoco, the suppliers competed to win contracts.

(E) Most of Autoco's suppliers have on hand a wide range of previously prepared parts designs that can readily be modified for a new car.

Evaluation of a Plan

Situation A car manufacturer plans to have its parts suppliers start helping to design the parts they build for the manufacturer. Many parts suppliers have more designers with specialized experience than the manufacturer has.

Reasoning *What would make it more likely that having the parts suppliers help design the parts will reduce the time and cost of designing the manufacturer's next new car?* In order for the change to reduce the time and cost, the parts suppliers involved in designing the next car will probably have to do their portion of the design process faster and cheaper than the manufacturer would have, and the design collaboration process will have to avoid producing substantial new inefficiencies.

A The additional need to modify the overall design would probably make the design process slower and more expensive, not faster and cheaper.

B The additional need to hire more designers would probably increase design costs, not reduce them.

C Although this suggests that the change is likely to substantially affect the design's expense, it does not indicate whether the expense will increase or decrease.

D If anything, this competition probably made Autoco's previous design process cheaper. It does not suggest that the new design process, which may involve less competition, will be faster or cheaper than the previous one.

E **Correct.** Modifying the previously prepared parts designs will probably be faster and cheaper than creating new designs from scratch.

The correct answer is E.

CR02830

608. In response to viral infection, the immune systems of mice typically produce antibodies that destroy the virus by binding to proteins on its surface. Mice infected with the herpesvirus generally develop keratitis, a degenerative disease affecting part of the eye. Since proteins on the surface of cells in this part of the eye closely resemble those on the herpesvirus surface, scientists hypothesize that these cases of keratitis are caused by antibodies to the herpesvirus.

Which of the following, if true, most helps to support the scientists' reasoning?

(A) Other types of virus have surface proteins that closely resemble proteins found in various organs of mice.

(B) Mice that are infected with the herpesvirus but do not develop keratitis produce as many antibodies as infected mice that do develop keratitis.

(C) Mice infected with a new strain of the herpesvirus that has different surface proteins did not develop keratitis.

(D) Mice that have never been infected with the herpesvirus can sometimes develop keratitis.

(E) There are mice that are unable to form antibodies in response to herpes infections, and these mice contract herpes at roughly the same rate as other mice.

Argument Evaluation

Situation Mice infected with the herpesvirus tend to develop keratitis, an eye disease. The surface of the eye cells have proteins that resemble those on the herpesvirus surface. Based on this finding, scientists have hypothesized that keratitis develops in mice because antibodies that attack herpesvirus surface proteins can also attack eyes.

Reasoning *What other information, if correct, would provide the strongest support for the scientists' hypothesis?* The clue that led the scientists to form their hypothesis was the close resemblance of the proteins on the mouse eye surface to those on the herpesvirus surface. The resemblance could cause antibodies to bind to both types of proteins, in one case eliminating the herpesvirus and in the other case causing keratitis.

A Even if this is correct, we lack information as to whether the antibodies to those other types of virus can damage the organs that display the closely resembling proteins. If such a damage process were confirmed, it could count as evidence—even if not sufficient—to confirm the scientists' hypothesis.

B If anything, this would, absent further information, raise doubts about the correctness of the scientists' proposed explanation.

C **Correct.** This provides strong confirmation of the scientists' hypothesis. The proteins on the new strain of the herpesvirus no longer sufficiently resemble the proteins on the eye surface to cause the antibodies to attack those proteins and cause keratitis.

D For all we know, keratitis may have multiple independent causes and may sometimes be caused by processes other than the protein misidentification hypothesized by the scientists. This information neither confirms nor refutes the scientists' hypothesis.

E The rates at which mice contract herpes is not discussed. We lack any information as to whether mice that lack antibodies to the herpesvirus sometimes contract keratitis along with herpes infection.

The correct answer is C.

CR02885

609. Last year a record number of new manufacturing jobs were created. Will this year bring another record? Well, **any new manufacturing job is created either within an existing company or by the start-up of a new company**. Within existing firms, new jobs have been created this year at well below last year's record pace. At the same time, there is considerable evidence that the number of new companies starting up this year will be no higher than it was last year and **there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups**. So clearly, the number of new jobs created this year will fall short of last year's record.

In the argument given, the two portions in **boldface** play which of the following roles?

(A) The first provides evidence in support of the main conclusion of the argument; the second is a claim that the argument challenges.

(B) The first is a generalization that the argument seeks to establish; the second is a conclusion that the argument draws in order to support that generalization.

(C) The first is a generalization that the argument seeks to establish; the second is a judgment that has been advanced in order to challenge that generalization.

(D) The first is presented as an obvious truth on which the argument is based; the second is a claim that has been advanced in support of a position that the argument opposes.

(E) The first is presented as an obvious truth on which the argument is based; the second is a judgment advanced in support of the main conclusion of the argument.

Argument Construction

Situation Manufacturing jobs are created either within existing companies or in start-ups. Manufacturing jobs are being created at a much slower rate this year than last year. It seems likely that the number of new start-ups will not exceed last year's number and that the average number of manufacturing jobs per start-up will not exceed last year's number. So fewer manufacturing jobs are likely to be created this year than last year.

Reasoning *What function is served by the statement that any new manufacturing job is created either within an existing company or by the start-up of a new company? What function is served by the statement that there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups?* The first statement makes explicit a general background assumption that there are just two ways in which manufacturing jobs are created. This assumption is used, along with other information, to support the argument's main conclusion, i.e., the prediction about job creation this year. The second statement gives a premise meant to help support the prediction about this year's manufacturing-job creation.

A The first is a general statement making explicit an assumption on which the argument's reasoning depends, but the second is a statement affirmed as part of the argument and does not express a claim that the argument challenges.

B The first is a generalization that is simply stated, without any support being offered. The second is not a conclusion and is not offered in support of the first.

C The second is not presented as a challenge to the generalization that is given in the first statement. The argument does not seek to establish the first statement, but merely asserts it.

D The second is information offered in support of the argument's main conclusion rather than a statement offered in support of a position opposed by the argument.

E **Correct.** The first, stating a truism, is merely asserted and requires no support in the argument, for which it provides a foundation; the second is a piece of information meant to support the prediction that is the argument's main conclusion.

The correct answer is E.

CR02886

610. Last year a record number of new manufacturing jobs were created. Will this year bring another record? Well, **any new manufacturing job is created either within an existing company or by the start-up of a new company**. Within existing firms, new jobs have been created this year at well below last year's record pace. At the same time, there is considerable evidence that the number of new companies starting up will be no higher this year than it was last year and there is no reason to think that the new companies starting up this year will create more jobs per company than did last year's start-ups. So clearly, **the number of new jobs created this year will fall short of last year's record**.

In the argument given, the two portions in boldface play which of the following roles?

(A) The first is presented as an obvious truth on which the argument is based; the second is the main conclusion of the argument.

(B) The first is presented as an obvious truth on which the argument is based; the second is a conclusion drawn in order to support the main conclusion of the argument.

(C) The first and the second each provide evidence in support of the main conclusion of the argument.

(D) The first is a generalization that the argument seeks to establish; the second is the main conclusion of the argument.

(E) The first is a generalization that the argument seeks to establish; the second is a conclusion that has been drawn in order to challenge that generalization.

Argument Construction

Situation Manufacturing jobs are created either within existing companies or in start-ups. Manufacturing jobs are being created at a much slower rate this year than last year. It seems likely that the number of new start-ups will not exceed last year's number and that the average number of new manufacturing jobs per start-up will not exceed last year's number. So fewer manufacturing jobs are likely to be created this year than last year.

Reasoning *What function is served by the statement that any new manufacturing job is created either within an existing company or by the start-up of a new company? What function is served by the statement that the number of new jobs created this year will fall short of last year's record number?* The first statement makes explicit a general background assumption that manufacturing jobs are created in just two ways. This assumption is used, along with other information, to support the argument's main conclusion. The second statement gives the argument's main conclusion, a prediction about how this year's manufacturing-job creation will compare with last year's.

A **Correct.** The first statement states a truism that is meant to provide support for the second statement; the second statement is the argument's main conclusion.

B The second statement is the argument's main conclusion, not an intermediate conclusion used to support the argument's main conclusion.

C The second statement is the main conclusion of the argument, not a statement used as support for the main conclusion.

D The argument merely asserts, and does not "seek to establish," the first statement. The first statement is a truism that does not need to be supported with evidence.

E The second statement is the argument's main conclusion and is not meant to present a challenge to the first statement. The first statement serves to provide partial support for the argument's main conclusion.

The correct answer is A.

CR00827

611. In Stenland, many workers have been complaining that they cannot survive on minimum wage, the lowest wage an employer is permitted to pay. The government is proposing to raise the minimum wage. Many employers who pay their workers the current minimum wage argue that if it is raised, unemployment will increase because they will no longer be able to afford to employ as many workers.

Which of the following, if true in Stenland, most strongly supports the claim that raising the minimum wage there will not have the effects that the employers predict?

(A) For any position with wages below a living wage, the difficulty of finding and retaining employees adds as much to employment costs as would raising wages.

(B) Raising the minimum wage does not also increase the amount employers have to contribute in employee benefits.

(C) When inflation is taken into account, the proposed new minimum wage is not as high as the current one was when it was introduced.

(D) Many employees currently being paid wages at the level of the proposed new minimum wage will demand significant wage increases.

(E) Many employers who pay some workers only the minimum wage also pay other workers wages that are much higher than the minimum.

Argument Evaluation

Situation Stenland's government proposes to raise the minimum wage because many workers have complained they cannot survive on it. But many employers claim that raising the minimum wage will increase unemployment.

Reasoning *What evidence would most strongly suggest that raising the minimum wage will not increase unemployment?* The employers with minimum-wage workers implicitly reason that because raising the minimum wage will increase the wages they have to pay each worker, it will reduce the number of workers they can afford to employ, and thus will increase unemployment. Evidence that the increased wage would not actually increase the employers' expenses per employee would cast doubt on their prediction, as would evidence that reducing the number of minimum-wage workers would not increase the nation's overall unemployment rate.

A **Correct.** This suggests that raising the minimum wage would make it easier for employers to find and retain minimum-wage employees, and that the savings would fully offset the cost of paying the higher wages. If there were such offsetting savings, the employers should still be able to afford to employ as many workers as they currently do.

B Even if raising the minimum wage does not increase employers' costs for employee benefits, paying the higher wage might still in itself substantially increase employers' overall costs per employee.

C For all we know, the current minimum wage might have substantially increased unemployment when it was introduced.

D These additional demands would probably raise employers' overall costs per employee, making it more likely that increasing the minimum wage would increase overall unemployment.

E Even if some workers receive more than the minimum wage, raising that wage could still raise employers' expenses for employing low-wage workers, making it too expensive for the employers to employ as many workers overall.

The correct answer is A.

CR07810

612. Biologists with a predilection for theory have tried—and largely failed—to define what it is that makes something a living thing. Organisms take in energy-providing materials and excrete waste products, but so do automobiles. Living things replicate and take part in evolution, but so do some computer programs. We must be open to the possibility that there are living things on other planets. Therefore, we will not be successful in defining what it is that makes something a living thing merely by examining living things on Earth—the only ones we know. Trying to do so is analogous to trying to specify _____.

Which of the following most logically completes the passage?

(A) the laws of physics by using pure mathematics
(B) what a fish is by listing its chemical components
(C) what an animal is by examining a plant
(D) what a machine is by examining a sketch of it
(E) what a mammal is by examining a zebra

Argument Construction

Situation Some biologists have tried, unsuccessfully, to find a theoretically defensible account of what it means for something to be a living thing. Some of the suggested definitions are too broad, because they include things that we would not regard as living. To find life on other planets, we must not narrow our conception of life by basing it simply on the kinds of life encountered on Earth.

Reasoning *Which of the five choices would be the logically most appropriate completion of the argument?* The argument points out that life-forms elsewhere in the universe may be very different from any of the life-forms on Earth. Both life-forms on Earth and life-forms discovered elsewhere would all qualify as members of a very large class, the class of all life-forms. Taking life-forms on Earth, a mere subset of the class of all life-forms, as representative of all life-forms would be a logical mistake and would not lead to success in defining what it means for something to be a living thing.

A This would not involve a logical mistake like the one already identified.
B This would not involve a logical mistake closely resembling the one already identified.
C Plants are not a subclass of the class of animals, so this does not involve the logical mistake of taking a subclass as representative of a larger class.
D This does not involve a logical mistake closely resembling the one already discussed.
E **Correct.** This involves the logical mistake of taking the class of zebras, a subclass of the class of mammals, as representative of the class of mammals. Logically, it resembles taking the class of life-forms on Earth as representative of the class of all life-forms.

The correct answer is E.

CR01145

613. When trying to identify new technologies that promise to transform the marketplace, market researchers survey the managers of those companies that are developing new technologies. Such managers have an enormous stake in succeeding, so they invariably overstate the potential of their new technologies. Surprisingly, however, market researchers typically do not survey a new technology's potential buyers, even though it is the buyers—not the producers—who will ultimately determine a technology's commercial success.

Which of the following, if true, best accounts for the typical survey practices among market researchers?

(A) If a new technology succeeds, the commercial benefits accrue largely to the producers, not to the buyers, of that technology.

(B) People who promote the virtues of a new technology typically fail to consider that the old technology that is currently in use continues to be improved, often substantially.

(C) Investors are unlikely to invest substantial amounts of capital in a company whose own managers are skeptical about the commercial prospects of a new technology they are developing.

(D) The potential buyers for not-yet-available technologies can seldom be reliably identified.

(E) The developers of a new technology are generally no better positioned than its potential buyers to gauge how rapidly the new technology can be efficiently mass-produced.

Argument Construction

Situation Market researchers seeking to identify new technologies that have the potential to transform the marketplace survey managers of companies developing new technologies, but typically not the potential buyers of new technologies, even though managers tend to overstate the potential of their new technologies and it is the buyers who determine the products' commercial success.

Reasoning *What best explains why it is managers, not buyers, that the market researchers survey?* Why, despite the information in the passage, are managers of technology companies surveyed while potential buyers are typically not? A partial explanation would be that it is difficult to reliably determine who the potential buyers of new technologies will be. If market researchers cannot identify who the potential buyers of as-yet unavailable technologies will be, that explains why they are not typically surveyed—and why the next best alternative may be to survey managers.

A This answer choice tells us who would benefit from commercial success of new technologies. But it says nothing about whose opinion would be most valuable in predicting the commercial success of new technologies.

B At most, this could help explain why managers overstate the potential of their new technologies. But it does not explain the motives of market researchers in relying on the managers' rather than buyers' opinions about new technologies.

C Given that managers of technology companies will want to attract investors, this helps to explain why the managers would tend to overstate the potential of their new technologies. But it does not help to explain the survey practices.

D **Correct.** This accounts for why potential buyers of new technologies are not typically sought out in surveys by market researchers: It is difficult to determine in advance who they are.

E This, like answer choice (C), tends to make the practices of market researchers more difficult rather than easier to understand. If developers of new technologies are no better at gauging how rapidly a new technology can be mass-produced (a factor affecting commercial success), then all the more reason to survey potential buyers rather than the managers.

The correct answer is D.

CR02829

614. Sammy: For my arthritis, I am going to try my aunt's diet: large amounts of wheat germ and garlic. She was able to move more easily right after she started that diet.

Pat: When my brother began that diet, his arthritis got worse. But he has been doing much better since he stopped eating vegetables in the nightshade family, such as tomatoes and peppers.

Which of the following, if true, would provide a basis for explaining the fact that Sammy's aunt and Pat's brother had contrasting experiences with the same diet?

(A) A change in diet, regardless of the nature of the change, frequently brings temporary relief from arthritis symptoms.

(B) The compounds in garlic that can lessen the symptoms of arthritis are also present in tomatoes and peppers.

(C) Arthritis is a chronic condition whose symptoms improve and worsen from time to time without regard to diet.

(D) In general, men are more likely to have their arthritis symptoms alleviated by avoiding vegetables in the nightshade family than are women.

(E) People who are closely related are more likely to experience the same result from adopting a particular diet than are people who are unrelated.

Argument Construction

Situation Sammy's aunt's arthritis apparently improved after she consumed large amounts of wheat germ and garlic. Pat's brother's arthritis deteriorated after he followed the same diet. Since he stopped eating vegetables in the nightshade family, such as tomatoes and peppers, his arthritis has improved.

Reasoning *What could account for the fact that Sammy's aunt's arthritis improved and Pat's brother's arthritis got worse after they both followed the wheat germ and garlic diet?* The fact that a person has a health improvement following a diet is, by itself, very weak evidence for the claim that the diet caused the improvement. More generally, the fact that one event follows another is seldom, by itself, evidence that the earlier event caused the later. This applies to both the experience of Sammy's aunt and that of Pat's brother with the wheat germ and garlic diet.

A In theory, this could be somewhat relevant to Sammy's aunt's experience but not to Pat's brother's experience. It is, however, insufficient to explain either.

B Even if this is true, it might be the case that a large quantity of the compounds in question must be consumed in concentrated form to benefit arthritis. No evidence is given to indicate whether this is so. Regardless, the puzzle as to why the wheat germ and garlic diet was followed by arthritis improvement in one case and not in the other remains.

C **Correct.** If we know there are typically fluctuations in the severity of arthritis symptoms and these can occur independent of diet, then the divergent experiences of the two people can be attributed to such fluctuations—even if it is conceded that some diets can affect arthritis symptoms in some manner. The wheat germ and garlic diet may, or may not, be such a diet.

D This could throw light on Pat's brother's experience but not on Sammy's aunt's experience.

E If this is correct, it is still far too general to provide a basis for explaining why the experiences of the two people were different. Does it apply to arthritis? We're not told. Nor are we told that it applies to the wheat germ and garlic diet. Is Pat's brother closely related to Sammy's aunt? We don't know.

The correct answer is C.

CR05756

615. Infotek, a computer manufacturer in Katrovia, has just introduced a new personal computer model that sells for significantly less than any other model. Market research shows, however, that very few Katrovian households without personal computers would buy a computer, regardless of its price. Therefore, introducing the new model is unlikely to increase the number of computers in Katrovian homes.

Which of the following is an assumption on which the argument depends?

(A) Infotek achieved the lower price of the new model by using components of lower quality than those used by other manufacturers.

(B) The main reason cited by consumers in Katrovia for replacing a personal computer is the desire to have an improved model.

(C) Katrovians in households that already have computers are unlikely to purchase the new Infotek model as an additional computer for home use.

(D) The price of other personal computers in Katrovia is unlikely to drop below the price of Infotek's new model in the near future.

(E) Most personal computers purchased in Katrovia are intended for home use.

Argument Construction

Situation In Katrovia, a new personal computer model costs less than any other model. But market research shows that very few Katrovian households without personal computers would buy even cheap ones.

Reasoning *What must be true in order for the stated facts to support the conclusion that introducing the new computer model is unlikely to increase the overall number of computers in Katrovian homes?* The market research supports the conclusion that no new computer model is likely to significantly increase the number of computers in Katrovian homes that currently lack computers. But the overall number of computers in Katrovian homes will still increase if Katrovian homes that already have computers buy additional computers while keeping their existing ones. So the argument has to assume that the new computer model will not increase the number of additional computers purchased for Katrovian homes that already have computers.

A Even if Infotek used high-quality components in the new computer model, Katrovians might still refuse to buy it.

B Replacing a personal computer does not change the overall number of personal computers in homes, so Katrovians' motives for replacing their computers are irrelevant to the argument.

C **Correct.** As explained above, unless computers of the new model are purchased as additional computers for Katrovian homes that already have computers, the new model's introduction is unlikely to increase the overall number of computers in Katrovian homes.

D The assumption that other personal computer prices would stay relatively high does not help establish the link between its premises and its conclusion. If answer choice D were false, the argument would be no weaker than it is without any consideration of other computers' potential prices.

E If most personal computers purchased in Katrovia were not intended for home use, then the new model's introduction would be even less likely to increase the number of personal computers in Katrovian homes. So the argument does not depend on assuming that most of the computers purchased are for home use.

The correct answer is C.

CR05501

616. Fast-food restaurants make up 45 percent of all restaurants in Canatria. Customers at these restaurants tend to be young; in fact, studies have shown that the older people get, the less likely they are to eat in fast-food restaurants. Since the average age of the Canatrian population is gradually rising and will continue to do so, the number of fast-food restaurants is likely to decrease.

Which of the following, if true, most seriously weakens the argument?

(A) Fast-food restaurants in Canatria are getting bigger, so each one can serve more customers.

(B) Some older people eat at fast-food restaurants more frequently than the average young person.

(C) Many people who rarely eat in fast-food restaurants nevertheless eat regularly in restaurants.

(D) The overall population of Canatria is growing steadily.

(E) As the population of Canatria gets older, more people are eating at home.

Argument Evaluation

Situation In Canatria, the older people get, the less likely they are to eat in fast-food restaurants. The average age of Canatrians is increasing.

Reasoning *What evidence would most weaken the support provided by the cited facts for the prediction that the number of fast-food restaurants in Canatria is likely to decrease?* The argument implicitly reasons that since studies have shown that Canatrians tend to eat in fast-food restaurants less as they get older, and since Canatrians are getting older on average, the proportion of Canatrians eating in fast-food restaurants will decline. The argument assumes that this means the overall number of fast-food restaurant customers will decline and that demand will decrease enough to reduce the number of fast-food restaurants that can sustain profitability. Consequently, fewer new fast-food restaurants will open or more old ones will close, or both. Thus, the number of fast-food restaurants in Canatria will fall. Any evidence casting doubt on any inference in this chain of implicit reasoning will weaken the argument.

A This strengthens the argument by providing additional evidence that the total number of fast-food restaurants will decrease. If the average number of customers per fast-food restaurant is increasing, then fewer fast-food restaurants will be needed to serve the same—or a lesser—number of customers.

B Even if a few individuals do not follow the general trends described, those trends could still reduce the overall demand for and number of fast-food restaurants.

C The argument is only about fast-food restaurants, not restaurants of other types.

D **Correct.** This suggests that even if the proportion of Canatrians eating at fast-food restaurants declines, the total number doing so may not decline. Thus, the total demand for and profitability of fast-food restaurants may not decline either, so the total number of fast-food restaurants in Canatria may not decrease.

E If anything, this strengthens the argument by pointing out an additional trend likely to reduce the demand for, and thus the number of, fast-food restaurants in Canatria.

The correct answer is D.

CR04805

617. Last year a chain of fast-food restaurants, whose menu had always centered on hamburgers, added its first vegetarian sandwich, much lower in fat than the chain's other offerings. Despite heavy marketing, the new sandwich accounts for a very small proportion of the chain's sales. The sandwich's sales would have to quadruple to cover the costs associated with including it on the menu. Since such an increase is unlikely, the chain would be more profitable if it dropped the sandwich.

Which of the following, if true, most seriously weakens the argument?

(A) Although many of the chain's customers have never tried the vegetarian sandwich, in a market research survey most of those who had tried it reported that they were very satisfied with it.

(B) Many of the people who eat at the chain's restaurants also eat at the restaurants of competing chains and report no strong preference among the competitors.

(C) Among fast-food chains in general, there has been little or no growth in hamburger sales over the past several years as the range of competing offerings at other restaurants has grown.

(D) When even one member of a group of diners is a vegetarian or has a preference for low-fat food, the group tends to avoid restaurants that lack vegetarian or low-fat menu options.

(E) An attempt by the chain to introduce a lower-fat hamburger failed several years ago, since it attracted few new customers and most of the chain's regular customers greatly preferred the taste of the regular hamburgers.

Argument Evaluation

Situation Last year a fast-food restaurant chain specializing in hamburgers started offering a low-fat vegetarian sandwich and marketed it heavily. The new sandwich's sales are far too low to cover the costs associated with including it on the menu.

Reasoning *What evidence would most weaken the support provided by the cited facts for the prediction that it would be more profitable for the chain to drop the sandwich?* The implicit argument is that since the new sandwich's sales are too low to cover the costs associated with including it on the menu, offering the sandwich diminishes the chain's profitability and will continue to do so if the sandwich continues to be offered. This reasoning assumes that the sandwich provides the chain no substantial indirect financial benefits except through its direct sales. It also assumes that the sandwich's sales will not increase sufficiently to make the sandwich a viable product. Any evidence casting doubt on either of these assumptions will weaken the argument.

A This gives information only about the respondents to the survey who had tried the sandwich (possibly very few), who were probably already more open to liking a vegetarian sandwich than any of the chain's other customers. So their responses are probably unrepresentative of the chain's customers in general and do not suggest that the sandwich has enough market potential.

B Although the issue of competition with other restaurants is not raised in the information provided, this new information, if anything, strengthens the argument, by suggesting that the introduction of the new sandwich has not significantly enhanced customer preference for eating at the restaurants that offer the new sandwich.

C This suggests that the cause of stagnation in fast-food restaurants' hamburger sales has been competition from non-fast-food restaurants, but not that the non-fast-food restaurants competed by offering vegetarian options.

D **Correct.** This suggests that even if the sandwich's sales are low, it may indirectly increase the chain's overall profits by encouraging large groups to eat at the chain.

E This strengthens the argument by suggesting that the chain's customers are generally not interested in low-fat menu options such as the new sandwich.

The correct answer is D.

CR03727

618. Transportation expenses accounted for a large portion of the total dollar amount spent on trips for pleasure by residents of the United States in 1997, and about half of the total dollar amount spent on transportation was for airfare. However, the large majority of United States residents who took trips for pleasure in 1997 did not travel by airplane but used other means of transportation.

If the statements above are true, which of the following must also be true about United States residents who took trips for pleasure in 1997?

(A) Most of those who traveled by airplane did so because the airfare to their destination was lower than the cost of other available means of transportation.

(B) Most of those who traveled by airplane did so because other means of transportation to their destination were unavailable.

(C) Per mile traveled, those who traveled by airplane tended to spend more on transportation to their destination than did those who used other means of transportation.

(D) Overall, people who did not travel by airplane had lower average transportation expenses than people who did.

(E) Those who traveled by airplane spent about as much, on average, on other means of transportation as they did on airfare.

Argument Construction

Situation In 1997, about half of total transportation spending by U.S. residents taking trips for pleasure was for airfare. But the large majority of U.S. residents who took trips for pleasure in 1997 did not travel by airplane.

Reasoning *What can be deduced from the stated facts?* The information provided indicates that among U.S. residents who took trips for pleasure in 1997, those who traveled by airplane were a small minority. Yet this small minority's spending for airfare accounted for half of all transportation spending among residents taking trips for pleasure. It follows that on average, those who traveled by airplane must have spent far more per person on transportation than those who did not travel by airplane.

A This does not follow logically from the information given. Most of those who traveled by airplane may have done so even if flying was more expensive than other modes of transportation—for example, because flying was faster or more comfortable.

B This does not follow from the information given. Most of those who traveled by airplane may have done so even if many other modes of transportation were available—the other modes may all have been less desirable.

C This does not follow from the information given. Those who traveled by airplane may have traveled much farther on average than those who used other means of transportation, so their transportation spending per mile traveled need not have been greater.

D **Correct.** As explained above, those who traveled by airplane must have spent more per person on transportation than those who did not travel by airplane, on average. In other words, those who did not travel by airplane must have had lower average transportation expenses than those who did.

E This does not follow from the information given. Although half the total dollar spending on transportation was for airfare, much of the transportation spending that was not for airfare was by the large majority of U.S. residents who did not travel by airplane.

The correct answer is D.

CR12051

619. Voters commonly condemn politicians for being insincere, but politicians often must disguise their true feelings when they make public statements. If they expressed their honest views—about, say, their party's policies—then achieving politically necessary compromises would be much more difficult. Clearly, the very insincerity that people decry shows that our government is functioning well.

Which of the following, if true, most seriously undermines this reasoning?

(A) Achieving political compromises is not all that is necessary for the proper functioning of a government.

(B) Some political compromises are not in the best long-term interest of the government.

(C) Voters often judge politicians by criteria other than the sincerity with which they express their views.

(D) A political party's policies could turn out to be detrimental to the functioning of a government.

(E) Some of the public statements made by politicians about their party's policies could in fact be sincere.

Argument Evaluation

Situation Politicians must often make insincere public statements because expressing their true feelings would make it harder for them to achieve politically necessary compromises.

Reasoning *What would suggest that the argument's premises do not establish that politicians' insincerity shows our government is functioning well?* The implicit reasoning is that insincerity helps politicians achieve politically necessary compromises, and these compromises help our government to function well, so insincerity must show that our government is functioning well. Evidence that these necessary compromises do not ensure that our government functions well would undermine the argument's reasoning, as would evidence that politicians' insincerity has other substantial effects that hinder the government's functioning.

A **Correct.** If governments may function poorly even when insincerity allows necessary political compromises to be made, then the argument's premises do not establish that politicians' insincerity shows our government is functioning well.

B The argument does not require that all political compromises help government to function well, only that politically necessary compromises do.

C Even if voters often judge politicians by criteria other than their sincerity, they may also often decry politicians' insincerity, not realizing or caring that such insincerity helps the government function well.

D Even if a political party's policies impair the government's functioning, politically necessary compromises by politicians in that party could improve the government's functioning.

E Even if politicians sometimes speak sincerely about their party's policies, their general willingness to be insincere as needed to achieve politically necessary compromises could be a sign that the government is functioning well.

The correct answer is A.

CR06728

620. To reduce waste of raw materials, the government of Sperland is considering requiring household appliances to be broken down for salvage when discarded. To cover the cost of salvage, the government is planning to charge a fee, which would be imposed when the appliance is first sold. Imposing the fee at the time of salvage would reduce waste more effectively, however, because consumers tend to keep old appliances longer if they are faced with a fee for discarding them.

Which of the following, if true, most seriously weakens the argument?

(A) Increasing the cost of disposing of an appliance properly increases the incentive to dispose of it improperly.

(B) The fee provides manufacturers with no incentive to produce appliances that are more durable.

(C) For people who have bought new appliances recently, the salvage fee would not need to be paid for a number of years.

(D) People who sell their used, working appliances to others would not need to pay the salvage fee.

(E) Many nonfunctioning appliances that are currently discarded could be repaired at relatively little expense.

Evaluation of a Plan

Situation A government is considering requiring household appliances to be broken down for salvage when discarded. To cover the salvage costs, the government plans to charge a fee on appliance sales.

Reasoning *What would suggest that charging the fee at the time of salvage would less effectively reduce waste than charging the fee at the time of sale would?* The argument is that charging the fee at the time of salvage would reduce waste of raw materials because it would encourage consumers to keep their appliances longer before salvaging them. This argument could be weakened by pointing out other factors that might increase waste if the fee is charged at the time of salvage or reduce waste if the fee is charged at the time of sale.

A **Correct.** This suggests that charging the fee at the time of salvage rather than the time of sale would encourage consumers to discard their appliances illegally, thereby increasing waste of raw materials by reducing the proportion of discarded appliances that are salvaged.

B This factor would remain the same regardless of whether the fee was charged at the time of sale or the time of salvage.

C This might be a reason for consumers to prefer the fee be charged at the time of salvage rather than the time of sale, but it does not suggest that charging the fee at the time of salvage would reduce waste less effectively.

D This provides an additional reason to expect that charging the fee at the time of salvage would help reduce waste, so it strengthens rather than weakens the argument.

E This would give consumers an additional reason to keep using their old appliances and postpone paying a fee at the time of salvage, so it strengthens rather than weakens the argument.

The correct answer is A.

CR02866

621. When there is less rainfall than normal, the water level of Australian rivers falls and the rivers flow more slowly. Because algae whose habitat is river water grow best in slow-moving water, the amount of algae per unit of water generally increases when there has been little rain. By contrast, however, following a period of extreme drought, algae levels are low even in very slow-moving river water.

Which of the following, if true, does most to explain the contrast described above?

(A) During periods of extreme drought, the populations of some of the species that feed on algae tend to fall.

(B) The more slowly water moves, the more conducive its temperature is to the growth of algae.

(C) When algae populations reach very high levels, conditions within the river can become toxic for some of the other species that normally live there.

(D) Australian rivers dry up completely for short intervals in periods of extreme drought.

(E) Except during periods of extreme drought, algae levels tend to be higher in rivers in which the flow has been controlled by damming than in rivers that flow freely.

Argument Construction

Situation When Australian rivers flow slowly due to little rain, algae populations in those rivers increase. But after periods of extreme drought, algae levels are low even in water moving at speeds that would normally show population increases.

Reasoning *What would explain the contrast between algae levels in slow-moving water resulting from little rain and slow-moving water after a drought?* There must be some difference between what happens during periods in which there is simply less rainfall than normal and periods in which there is extreme drought, a difference that affects the algae population.

A This indicates one of the consequences of drought, and slightly suggests that this might be due to a lower algae level. But it does nothing to explain why algae levels might be lower after a drought.

B This could explain why some rivers that are slow-moving and have little water might have a high algae level—but not why the algae level is low in such rivers after a period of drought.

C This explains why levels of other species might be low when algae populations are high, not why algae populations are high when there is little rain, but low following a period of extreme drought.

D **Correct.** This statement properly identifies something that helps explain the contrast. According to the information given, the habitat of the algae under discussion is river water. If the river dries up, the algae will probably not survive. Then after the drought, algae population levels would likely take a while to rise again.

E This emphasizes that there is a contrast between what happens to algae during periods of extreme drought and what happens to them at other times, but it does not help explain that contrast.

The correct answer is D.

CR04924

622. Increased use of incineration is sometimes advocated as a safe way to dispose of chemical waste. But opponents of incineration point to the 40 incidents involving unexpected releases of dangerous chemical agents that were reported just last year at two existing incinerators commissioned to destroy a quantity of chemical waste material. Since designs for proposed new incinerators include no additional means of preventing such releases, leaks will only become more prevalent if use of incineration increases.

Which of the following, if true, most seriously weakens the argument?

(A) At the two incinerators at which leaks were reported, staff had had only cursory training on the proper procedures for incinerating chemical waste.

(B) Other means of disposing of chemical waste, such as chemical neutralization processes, have not been proven safer than incineration.

(C) The capacity of existing incinerators is sufficient to allow for increased incineration of chemical waste without any need for new incinerators.

(D) The frequency of reports of unexpected releases of chemical agents at newly built incinerators is about the same as the frequency at older incinerators.

(E) In only three of the reported incidents of unexpected chemical leaks did the releases extend outside the property on which the incinerators were located.

Argument Evaluation

Situation Last year, at two chemical waste incinerators, there were forty reported incidents involving unexpected releases of dangerous chemicals. Designs for proposed new incinerators include no additional safeguards against such releases. Therefore, increased use of incineration will likely make such releases more prevalent.

Reasoning *What would undermine the support provided for the conclusion that leaks will become more prevalent if more chemical waste is disposed of through incineration?* The argument draws a general conclusion about chemical waste incineration from evidence about only two particular incinerators. This reasoning would be undermined by any evidence that the leaks at those two incinerators were the result of something other than insufficient safeguards against such releases.

A **Correct.** If the staff training at the two incinerators was cursory, then the leaks may have been the results of staff not knowing how to use safeguards with which the incinerators are equipped that, if properly used, would have prevented the release of dangerous chemicals. Therefore, if staff at newer incinerators will be better trained, leaks might not become more prevalent even if chemical waste incineration becomes more common.

B Other chemical waste disposal methods may be safer than incineration even if no one has proven so; and even if they're not safer overall, they may involve fewer leaks.

C Continuing to use existing incinerators might well produce just as many leaks as switching to new incinerators would.

D This suggests that new incinerators produce as many leaks as older incinerators do, a finding that provides additional evidence that increased incineration even with proposed new incinerators would lead to more leaks.

E The argument is not about how far the releases from leaks extend, only about how many of them are likely to occur.

The correct answer is A.

CR10049

623. Public health expert: **Increasing the urgency of a public health message may be counterproductive**. In addition to irritating the majority who already behave responsibly, **it may undermine all government pronouncements on health by convincing people that such messages are overly cautious**. And there is no reason to believe that those who ignore measured voices will listen to shouting.

The two sections in boldface play which of the following roles in the public health expert's argument?

(A) The first is a conclusion for which support is provided, but is not the argument's main conclusion; the second is an unsupported premise supporting the argument's main conclusion.

(B) The first is a premise supporting the only explicit conclusion; so is the second.

(C) The first is the argument's main conclusion; the second supports that conclusion and is itself a conclusion for which support is provided.

(D) The first is a premise supporting the argument's only conclusion; the second is that conclusion.

(E) The first is the argument's only explicit conclusion; the second is a premise supporting that conclusion.

Argument Construction

Situation A public health expert argues against increasing the urgency of public health messages by pointing out negative effects that may arise from such an increase, as well as by questioning its efficacy.

Reasoning *What roles are played in the argument by the two claims in boldface?* The first claim in boldface states that increasing the urgency of public health messages may be counterproductive. After making this claim, the public health expert mentions two specific reasons this could be so: it could irritate people who already behave responsibly, and it could convince people that all public health messages are too cautious. (The latter reason in the second claim in boldface). The phrase [*i*]*n addition to* indicates that neither claim in the second sentence is intended to support or explain the other. However, since each claim in the second sentence gives a reason to believe the claim in the first sentence, each independently supports the first sentence as a conclusion. The word [*a*]*nd* beginning the third sentence reveals that its intended role in the argument is the same as that of the two claims in the second sentence.

A Everything stated after the first sentence is intended to help support it, so the first sentence is the argument's main conclusion.

B Everything stated after the first sentence is intended to help support it, so the first sentence is a conclusion, not a premise.

C Each of the three claims in the second and third sentences is presented as an independent reason to accept the general claim in the first sentence. Therefore, nothing in the passage is intended to support the second statement in boldface as a conclusion.

D Everything stated after the first sentence is intended to help support it, so the first sentence is a conclusion, not a premise.

E **Correct.** Each of the three claims in the second and third sentences is presented as an independent reason to accept the general claim in the first sentence. Thus, each of those claims is a premise supporting the claim in the first sentence as the argument's only conclusion.

The correct answer is E.

CR01163

624. Several industries have recently switched at least partly from older technologies powered by fossil fuels to new technologies powered by electricity. It is thus evident that less fossil fuel is being used as a result of the operations of these industries than would have been used if these industries had retained their older technologies.

Which of the following, if true, most strengthens the argument above?

(A) Many of the industries that have switched at least partly to the new technologies have increased their output.

(B) Less fossil fuel was used to manufacture the machinery employed in the new technologies than was originally used to manufacture the machinery employed in the older technologies.

(C) More electricity is used by those industries that have switched at least partly to the new technologies than by those industries that have not switched.

(D) Some of the industries that have switched at least partly to the new technologies still use primarily technologies that are powered by fossil fuels.

(E) The amount of fossil fuel used to generate the electricity needed to power the new technologies is less than the amount that would have been used to power the older technologies.

Argument Evaluation

Situation Several industries have now switched, at least partly, to technologies using electricity rather than fossil fuels. Thus, less fossil fuel will be consumed as a result of the operation of these industries than otherwise would have been.

Reasoning *Which option most strengthens the argument?* One way to strengthen an argument is to eliminate or minimize one of its flaws or weaknesses. Because the conclusion is stated in terms of "fossil fuel consumed *as a result* of the operation of these industries," the claim would encompass even any fossil fuel that might be used to generate the electricity that the newer technologies use. Yet the premise of the argument does not address this issue. So the argument is strengthened if it turns out that less fossil fuel was used to produce the electricity than would have been used to power the older technologies.

A In an indirect way, this option slightly weakens rather than strengthens the argument. For *if* fossil fuels are used to produce the electricity now used by the industries and *if* it is because of these newer technologies that output has increased, the argument's conclusion is *less* likely.

B It does not matter how much fossil fuel was used to manufacture the older technologies *originally*. That has no bearing on whether more fossil fuel would have been expended as a result of the *continued operation* of the industries if the partial switch to newer technologies had not occurred.

C This is what we would expect, but it in no way strengthens the argument.

D This may seem to weaken the argument by indicating that the switch from older technologies will have less of an impact on fossil fuel consumption by these industries than we might have assumed. But since the conclusion makes no claim about how much consumption has been reduced, it is not clear that this option has any bearing on the strength of the argument one way or the other.

E **Correct.** This is the option that most strengthens the argument.

The correct answer is E.

CR00792

625. The difference in average annual income in favor of employees who have college degrees, compared with those who do not have such degrees, doubled between 1980 and 1990. Some analysts have hypothesized that increased competition between employers for employees with college degrees drove up income for such employees.

Which of the following, if true, most seriously undermines the explanation described above?

(A) During the 1980s a growing percentage of college graduates, unable to find jobs requiring a college degree, took unskilled jobs.

(B) The average age of all employees increased slightly during the 1980s.

(C) The unemployment rate changed very little throughout most of the 1980s.

(D) From 1980 to 1990 the difference in average income between employees with advanced degrees and those with bachelor's degrees also increased.

(E) During the 1980s there were some employees with no college degree who earned incomes comparable to the top incomes earned by employees with a college degree.

Argument Evaluation

Situation The amount by which average annual income for employees with college degrees exceeds that for employees without such degrees doubled between 1980 and 1990.

Reasoning *What evidence would most strongly suggest that increased competition among employers for employees with college degrees does not explain the relative increase in those employees' incomes?* Such increased competition could not explain the relative increase in income for employees with college degrees if the competition did not actually increase, or if such competition occurred but did not result in employers paying higher wages or salaries, or if the increase in competition to hire employees without college degrees was even greater. So evidence that any of those conditions existed would undermine the analysts' explanation.

A **Correct.** This suggests that the supply of college graduates grew relative to employers' demand for them, and hence that employers' competition for college-educated employees did not actually increase.

B The average age might have increased equally for employees with college degrees and for those without them, so the increase is not clearly relevant to explaining why the difference between these two groups' average incomes grew.

C Even if the overall unemployment rate did not change, competition for college-educated employees could have increased while competition for other employees decreased.

D This statement gives information comparing income trends among two groups of those with college degrees, and is irrelevant to the comparison of income trends for those with college degrees and those without college degrees.

E Even if there was strong competition and high pay for certain unusual types of employees without college degrees, increasing competition for employees with college degrees might have explained the overall growing difference in average pay between employees with college degrees and those without.

The correct answer is A.

CR01239

626. Which of the following most logically completes the passage?

According to the last pre-election poll in Whippleton, most voters believe that the three problems government needs to address, in order of importance, are pollution, crime, and unemployment. Yet in the election, candidates from parties perceived as strongly against pollution were defeated, while those elected were all from parties with a history of opposing legislation designed to reduce pollution. These results should not be taken to indicate that the poll was inaccurate, however, since _____.

(A) some voters in Whippleton do not believe that pollution needs to be reduced

(B) every candidate who was defeated had a strong antipollution record

(C) there were no issues other than crime, unemployment, and pollution on which the candidates had significant differences of opinion

(D) all the candidates who were elected were perceived as being stronger against both crime and unemployment than the candidates who were defeated

(E) many of the people who voted in the election refused to participate in the poll

Argument Construction

Situation A pre-election poll indicated that most voters believed the three problems government needs to address, in order of importance, are pollution, crime, and unemployment. But in the election, candidates from parties with a history of opposing anti-pollution legislation beat candidates from parties perceived as more strongly against pollution.

Reasoning *What would most help explain how the poll might have been accurate despite the election results?* Since the poll indicated that voters were most concerned about pollution, it suggested that candidates from anti-pollution parties would be more likely to be elected, other things being equal—and yet those candidates were not elected. There are many possible explanations for this outcome that are compatible with the poll having been accurate. For example, voters might have been swayed by the candidates' personalities, qualifications, or advertising more than by their positions on the issues. Or some candidates might have convinced voters that their personal positions on the issues were different from those of their parties. Or voters might have chosen candidates based on their positions on crime and unemployment, considering those issues together more important than pollution alone. Any statement suggesting that any such factors explained the election results would logically complete the passage by providing a reason to believe that the poll could have been accurate despite those results.

A If the number of voters who did not believe that pollution needed to be reduced was large enough to explain the election results, then the poll was probably inaccurate. So this does not explain how the poll might have been accurate despite those results.

B This eliminates the possibility that candidates were defeated for having weak antipollution records conflicting with their parties' antipollution stances, so it eliminates one explanation of how the poll might have been accurate despite the election results. Thus, it slightly weakens the conclusion of the argument instead of providing a premise to support it.

C This eliminates the possibility that differences of opinion among the candidates on these other issues might explain the election results, but it does not explain how the poll could have been accurate despite the election results.

D **Correct.** The poll indicated that voters believed that the government needs to address crime and unemployment as well as pollution. So if the poll was accurate, the election outcome might have resulted from voters considering candidates' positions on crime and unemployment to be jointly more important than their positions on pollution.

E If anything, this provides a reason to doubt that the poll accurately reflected voters' opinions. It does not explain how the poll might have accurately reflected those opinions despite the election results.

The correct answer is D.

CR01153

627. Manufacturing plants in Arundia have recently been acquired in substantial numbers by investors from abroad. Arundian politicians are proposing legislative action to stop such investment, justifying the proposal by arguing that foreign investors, opportunistically exploiting a recent fall in the value of the Arundian currency, were able to buy Arundian assets at less than their true value.

Which of the following, if true, casts the most serious doubt on the adequacy of the Arundian politicians' justification for the proposed legislation?

(A) The Arundian government originally welcomed the fall in the value of the Arundian currency because the fall made Arundian exports more competitive on international markets.

(B) Foreign investors who acquired Arundian manufacturing plants generally did so with no intention of keeping and running those plants over the long term.

(C) Without the recent fall in the value of the Arundian currency, many of the Arundian assets bought by foreign investors would have been beyond the financial reach of those investors.

(D) In Concordia, a country broadly similar to Arundia, the share of manufacturing assets that is foreign-controlled is 60 percent higher than it is in Arundia.

(E) The true value of an investment is determined by the value of the profits from it, and the low value of the Arundian currency has depressed the value of any profits earned by foreign investors from Arundian assets.

Argument Evaluation

Situation After a recent fall in the value of Arundian currency, foreign investors have been acquiring many Arundian manufacturing plants. Arundian politicians are proposing legislation to stop such investment.

Reasoning *What would most undermine the Arundian politicians' justification for the proposed legislation?* The politicians are justifying their proposal by claiming that foreign investors have been exploiting the fall in the currency's value by buying Arundian assets at less than their *true value* (whatever that means). Any evidence that their claim is false or meaningless would undermine their justification for the proposal, as would any evidence that the claim, even if true, does not provide a good reason to stop the foreign investments.

A This suggests that the foreign investors got a good deal on the manufacturing plants, since it provides evidence that those plants will now be more competitive and profitable. So, if anything, it supports the politicians' justification for their proposal rather than undermining it.

B This suggests that the foreign investors generally believe the manufacturing plants are undervalued, and intend to sell them at a profit as soon as the currency rises enough. So it supports the politicians' justification for their proposal rather than undermining it.

C This suggests that the recent fall in the currency's value made Arundian assets cost less than usual for foreign investors, thus arguably allowing the investors to buy the assets at less than their *true value*. So, if anything, it supports the politicians' justification for their proposal rather than undermining it.

D The Arundian politicians might consider the example of Concordia to be a warning of the disaster that could befall Arundia unless the legislation is enacted. So the situation in Concordia might be cited as support for the politicians' justification of their proposal.

E **Correct.** This implies that the fall in the Arundian currency's value has reduced the *true value* of Arundian manufacturing plants and any profits they may make, so it undermines the politicians' claim that the foreign investors exploited the fall in the currency's value to acquire the plants for less than their *true value*.

The correct answer is E.

CR04964

628. Proposed new safety rules for the Beach City airport would lengthen considerably the minimum time between takeoffs from the airport. In consequence, the airport would be able to accommodate 10 percent fewer flights than currently use the airport daily. The city's operating budget depends heavily on taxes generated by tourist spending, and most of the tourists come by plane. Therefore, the proposed new safety rules, if adopted, will reduce the revenue available for the operating budget.

The argument depends on assuming which of the following?

(A) There are no periods of the day during which the interval between flights taking off from the airport is significantly greater than the currently allowed minimum.

(B) Few, if any, of the tourists who use the Beach City airport do so when their main destination is a neighboring community and not Beach City itself.

(C) If the proposed safety rules are adopted, the reduction in tourist numbers will not result mainly from a reduction in the number of tourists who spend relatively little in Beach City.

(D) Increasing the minimum time between takeoffs is the only way to achieve necessary safety improvements without a large expenditure by the city government on airport enhancements.

(E) The response to the adoption of the new safety rules would not include an increase in the number of passengers per flight.

Argument Construction

Situation Proposed safety rules for a city airport would reduce the number of daily flights the airport can accommodate. The city's operating budget depends heavily on taxes generated by tourists, who mostly come by plane. Therefore, adopting the safety rules will result in lower revenue available for the operating budget.

Reasoning *What must be true in order for the cited facts to support the conclusion that the proposed rules would reduce the revenue for the operating budget?* The implicit reasoning is that since the rules would reduce the number of flights that can be accommodated, they would thereby reduce the number of tourists arriving by plane, which in turn would reduce the tax revenue that tourist spending generates for the operating budget. This assumes that the actual number of daily flights would fall along with the number that the airport can accommodate; that fewer daily flights would mean fewer people flying into the airport; that fewer people flying into the airport would mean fewer tourists flying into the airport; that fewer tourists flying into the airport would mean fewer tourists visiting the city; that fewer tourists visiting the city would mean less taxable spending by tourists; and that less taxable spending by tourists would mean less revenue overall for the operating budget.

A Even if flights depart the airport less frequently during some periods of the day, increasing the minimum time between flights at busy times of day could reduce the total number of daily flights from the airport.

B Even if half the tourists flying into the airport were bound for other nearby towns, the other half could still spend enough in town to generate lots of revenue for the operating budget.

C It is possible that most tourists spend relatively little in the city, but a few spend a lot. In that case, even if a reduction in tourist numbers resulted mainly from a declining number of tourists who spend relatively little, it could also greatly reduce the already small number of tourists who spend a lot.

D This suggests that the proposed rules might be financially better for the city than any alternative way to improve safety, whereas the argument's conclusion is that the proposed rules are financially disadvantageous.

E **Correct.** If adopting the proposed rules would result in a large increase in the number of passengers per flight, fewer daily flights would not necessarily mean fewer passengers or fewer tourists overall.

The correct answer is E.

CR01096

629. The introduction of new drugs into the market is frequently prevented by a shortage of human subjects for the clinical trials needed to show that the drugs are safe and effective. Since the lives and health of people in future generations may depend on treatments that are currently experimental, practicing physicians are morally in the wrong when, in the absence of any treatment proven to be effective, they fail to encourage suitable patients to volunteer for clinical trials.

Which of the following, if true, casts most doubt on the conclusion of the argument?

(A) Many drugs undergoing clinical trials are intended for the treatment of conditions for which there is currently no effective treatment.

(B) Patients do not share the physician's professional concern for public health, but everyone has a moral obligation to alleviate suffering when able to do so.

(C) Usually, half the patients in a clinical trial serve as a control group and receive a nonactive drug in place of the drug being tested.

(D) An experimental drug cannot legally be made available to patients unless those patients are subjects in clinical trials of the drug.

(E) Physicians have an overriding moral and legal duty to care for the health and safety of their current patients.

Argument Evaluation

Situation A shortage of human subjects for clinical trials needed to show that new drugs are safe and effective often prevents those drugs from being introduced into the market. The lives and health of future generations may depend on treatments that are now experimental.

Reasoning *What would cast doubt on the judgment that doctors are morally obligated to encourage their patients to volunteer for clinical trials?* Note that the argument's conclusion, unlike its premises, is a moral judgment. This judgment could be cast into doubt by a moral principle that would be likely to conflict with it under the conditions described. For example, a principle suggesting that it is sometimes morally unacceptable for doctors to encourage their patients to volunteer for clinical trials would also suggest that they are not morally obligated to encourage their patients to volunteer for clinical trials, since anything morally obligatory must also be morally acceptable.

A If anything, this highlights how important it is to ensure that these drugs undergo clinical trials to benefit future generations, so it supports rather than casts doubt on the argument's conclusion.

B This suggests that patients are morally obligated to volunteer for clinical trials to help prevent suffering in future generations. If anything, this supports the claim that doctors are morally obligated to encourage their patients to volunteer.

C The clinical trial will probably not harm any patients in the control group, yet their participation will benefit future generations. So, if anything, this supports the claim that doctors should encourage their patients to volunteer.

D This legal barrier makes it even more essential for the drugs to undergo clinical trials in order to benefit patients, so it supports rather than casts doubt on the argument's conclusion.

E **Correct.** Since the experimental drugs' safety is being tested during the trials, the drugs may prove unsafe for subjects in the trials. If doctors have an overriding moral duty to keep their current patients safe, then it may be morally unacceptable for them to encourage those patients to volunteer for the trials.

The correct answer is E.

CR01285

630. As a construction material, bamboo is as strong as steel and sturdier than concrete. Moreover, in tropical areas bamboo is a much less expensive construction material than either steel or concrete and is always readily available. In tropical areas, therefore, building with bamboo makes better economic sense than building with steel or concrete, except where land values are high.

Which of the following, if true, most helps to explain the exception noted above?

(A) Buildings constructed of bamboo are less likely to suffer earthquake damage than are steel and concrete buildings.

(B) Bamboo is unsuitable as a building material for multistory buildings.

(C) In order to protect it from being damaged by termites and beetles, bamboo must be soaked, at some expense, in a preservative.

(D) In some tropical areas, bamboo is used to make the scaffolding that is used during large construction projects.

(E) Bamboo growing in an area where land values are increasing is often cleared to make way for construction.

Argument Construction

Situation Bamboo is as strong as steel and sturdier than concrete when used as a construction material. In tropical areas, bamboo is much less expensive and is always readily available.

Reasoning *What explains the exception specified in the conclusion?* The argument's conclusion is that in tropical areas bamboo is a more economical building material than steel or concrete, *except where land values are high*. The information in the passage makes clear why bamboo is a more economical building material in tropical areas than are concrete or steel. So the question is: Why must an exception be made for areas where land values are high? Multistory buildings are particularly desirable in areas where land values are high, but bamboo may not be suitable for such buildings.

A This explains why bamboo would be preferable to steel or concrete in tropical areas especially prone to earthquakes. However, there is no clear connection to be made between areas where land values are high and areas especially prone to earthquakes.

B **Correct.** Multistory buildings provide a greater area of floor space for a given site area, and in that sense are more economical. A single-story building with the same floor space will occupy a much bigger site, so the higher the land values, the more likely it is that a multistory building will be built on that land. Thus, given this information, bamboo is less suitable for areas where land values are high.

C This undermines, to some extent, the claim that bamboo is an economical building material. But it does nothing to explain why it would be less economical specifically in areas where land values are high.

D This is irrelevant. Bamboo is used to build scaffolding for construction projects and as a building material for permanent structures. There is no way to infer from this that bamboo is less economical specifically in areas where land values are high.

E The fact that bamboo is cleared from an area to make room for construction in no way implies that bamboo would not be a suitable and economical building material for the area once it has been cleared.

The correct answer is B.

CR00788

631. Newspaper editors should not allow reporters to write the headlines for their own stories. The reason for this is that, while the headlines that reporters themselves write are often clever, what typically makes them clever is that they allude to little-known information that is familiar to the reporter but that never appears explicitly in the story itself.

Which of the following, if true, most strengthens the argument?

(A) The reporter who writes a story is usually better placed than the reporter's editor is to judge what the story's most newsworthy features are.

(B) To write a headline that is clever, a person must have sufficient understanding of the story that the headline accompanies.

(C) Most reporters rarely bother to find out how other reporters have written stories and headlines about the same events that they themselves have covered.

(D) For virtually any story that a reporter writes, there are at least a few people who know more about the story's subject matter than does the reporter.

(E) The kind of headlines that newspaper editors want are those that anyone who has read a reporter's story in its entirety will recognize as clever.

Argument Evaluation

Situation The headlines newspaper reporters write for their own stories are often clever only because they allude to little-known information that never appears explicitly in the stories themselves.

Reasoning *What would most help the argument support the conclusion that newspaper editors should not allow reporters to write headlines for their own stories?* The argument's only explicit premise is that the headlines newspaper reporters write for their own stories are often clever only because they allude to little-known information that never appears explicitly in the stories themselves. In order for this premise to support the conclusion that newspaper editors should not allow reporters to write their own headlines, it would be helpful to be given a reason why editors should avoid headlines alluding to such little-known information.

A This suggests that reporters are likely to write better headlines for their stories than editors are, so it weakens the argument that editors should not allow reporters to write their own headlines.

B Since a reporter who wrote a story is likely to understand that story well, this does not provide a reason why editors should not allow reporters to write their own headlines.

C If most reporters did what is suggested, they could perhaps hone their headline-writing skills—unless almost all reporters are weak in such skills, as suggested in the given information. The fact that they do not bother to do so may help explain why reporters' headline-writing skills are weak. An explanation of why this is so does not provide additional support for the argument's conclusion.

D The people who know more about a story's subject matter than the reporter writing the story might be just as likely to see the cleverness of allusions to little-known information as the reporters are. So, to the extent that this is relevant at all, it slightly weakens the argument by suggesting that obscurely clever headlines sometimes function as intended.

E **Correct.** The argument's explicit premise suggests that typically a reporter's headline for his or her own story cannot be recognized as clever by a reader who has read the whole story. So if editors want headlines that anyone who has read the accompanying stories would recognize as clever, they have a reason not to let reporters write the headlines.

The correct answer is E.

CR03251

632. Scientists have modified feed corn genetically, increasing its resistance to insect pests. Farmers who tried out the genetically modified corn last season applied less insecticide to their corn fields and still got yields comparable to those they would have gotten with ordinary corn. Ordinary corn seed, however, costs less, and what these farmers saved on insecticide rarely exceeded their extra costs for seed. Therefore, for most feed-corn farmers, switching to genetically modified seed would be unlikely to increase profits.

Which of the following would it be most useful to know in order to evaluate the argument?

(A) Whether there are insect pests that sometimes reduce feed-corn yields, but against which commonly used insecticides and the genetic modification are equally ineffective

(B) Whether the price that farmers receive for feed corn has remained steady over the past few years

(C) Whether the insecticides typically used on feed corn tend to be more expensive than insecticides typically used on other crops

(D) Whether most of the farmers who tried the genetically modified corn last season applied more insecticide than was actually necessary

(E) Whether, for most farmers who plant feed corn, it is their most profitable crop

Argument Evaluation

Situation Farmers who grew feed corn genetically engineered to be pest resistant got yields comparable to those of farmers growing ordinary feed corn, but did so while using less pesticide. Since the amount saved on pesticide was rarely in excess of the extra costs for the genetically modified corn, most farmers will probably not increase profits by choosing the genetically engineered variety.

Reasoning *Which would be most useful to know in evaluating the argument?* To answer a question such as this, one should look for information that would strengthen or weaken the argument. If one had information that the farmers growing the genetically modified corn could have increased their yields last year at lower cost, this would be helpful in evaluating the argument, because this would show that the argument is weak.

A It does not matter to the argument whether there are pests against which pesticides and genetic resistance are equally ineffective, because that is compatible with there being pests against which they are not equally effective.

B Whether prices of feed corn go up or down affects the comparison groups equally.

C The relative cost of insecticides for other crops has no bearing on the argument because the argument is concerned with only feed corn.

D **Correct.** This option provides the information that it would be most useful to know in evaluating the argument. It shows that farmers growing genetically modified corn last year could have attained higher profits than they in fact did.

E The argument concerns only the relative profitability of growing one variety of feed corn versus another.

The correct answer is D.

CR07318

633. Debater: The average amount of overtime per month worked by an employee in the manufacturing division of the Haglut Corporation is 14 hours. Most employees of the Haglut Corporation work in the manufacturing division. Furthermore, the average amount of overtime per month worked by any employee in the company generally does not fluctuate much from month to month. Therefore, each month, most employees of the Haglut Corporation almost certainly work at least some overtime.

The debater's argument is most vulnerable to criticism on which of these grounds?

(A) It takes for granted that the manufacturing division is a typical division of the corporation with regard to the average amount of overtime its employees work each month.

(B) It takes for granted that if a certain average amount of overtime is worked each month by each employee of the Haglut Corporation, then approximately the same amount of overtime must be worked each month by each employee of the manufacturing division.

(C) It confuses a claim from which the argument's conclusion about the Haglut Corporation would necessarily follow with a claim that would follow from the argument's conclusion only with a high degree of probability.

(D) It overlooks the possibility that even if, on average, a certain amount of overtime is worked by the members of some group, many members of that group may work no overtime at all.

(E) It overlooks the possibility that even if most employees of the corporation work some overtime each month, any one corporate employee may, in some months, work no overtime.

Argument Evaluation

Situation Most of the employees of the Haglut Corporation work in the manufacturing division, where employees average 14 hours per month in overtime. The average amount of overtime per month for employees at Haglut does not fluctuate much from month to month.

Reasoning *What is the argument's greatest weakness?* The argument's conclusion is that almost certainly each month most of the employees of Haglut work at least some overtime. Answer choice (D) identifies the argument's greatest weakness because it points out how the conclusion of the argument could be false even if all of the supporting information were true. For example, it could be that less than half of the employees work any overtime at all, but those that do work overtime work much more than 14 hours per month.

A The argument leaves open the possibility that in some divisions of the corporation, the average monthly overtime of its employees is quite different from 14 hours, even if (as the argument states) that average does not change much from month to month.

B The argument does not assume that there is a monthly amount of overtime worked by each employee of the manufacturing division equivalent to the company-wide average monthly overtime per employee.

C This does not identify a weakness that can be detected in the argument. Since the claims mentioned here are not specified, the passage provides no evidence that clearly indicates that this type of confusion is playing a role in the argument.

D **Correct.** The argument ignores the possibility that most of the employees of Haglut work no overtime at all in a particular month—which is quite consistent with the argument's assertion that the average number of monthly overtime hours per employee within the manufacturing division is 14.

E The possibility described by this is not overlooked by the argument, because this possibility is consistent with the conclusion. It could easily be that most employees of the corporation work some overtime each month—as the conclusion envisions—but that there are always some employees who do not work any overtime.

The correct answer is D.

CR05446

634. Proponents of the recently introduced tax on sales of new luxury boats had argued that a tax of this sort would be an equitable way to increase government revenue because the admittedly heavy tax burden would fall only on wealthy people and neither they nor anyone else would suffer any economic hardship. In fact, however, 20 percent of the workers employed by manufacturers of luxury boats have lost their jobs as a direct result of this tax.

The information given, if true, most strongly supports which of the following?

(A) The market for luxury boats would have collapsed even if the new tax on luxury boats had been lower.

(B) The new tax would produce a net gain in tax revenue for the government only if the yearly total revenue that it generates exceeds the total of any yearly tax-revenue decrease resulting from the workers' loss of jobs.

(C) Because many people never buy luxury items, imposing a sales tax on luxury items is the kind of legislative action that does not cost incumbent legislators much popular support.

(D) Before the tax was instituted, luxury boats were largely bought by people who were not wealthy.

(E) Taxes can be equitable only if their burden is evenly distributed over the entire population.

Argument Construction

Situation Proponents of a recently introduced tax on sales of new luxury boats argued that it would be an equitable way to increase government revenue because the tax would fall only on the wealthy and cause no economic hardship. But because of the tax, 20 percent of luxury-boat manufacturing workers have lost their jobs.

Reasoning *What conclusion do the statements about the proponents' argument and the tax's effects support?* Since the tax caused many workers to lose their jobs, apparently the proponents were incorrect in asserting that it would cause no one to suffer any economic hardship. Thus, their justification for concluding that the tax is an equitable way to increase government revenue is factually inaccurate, casting doubt on that conclusion.

A The passage indicates that the tax directly caused a significant decrease (though not necessarily a collapse) in the market for luxury boats. But the passage contains no evidence about whether such a decrease might not have occurred if the new tax had been somewhat lower.

B **Correct.** Since the tax caused the workers to lose their jobs, it might have made the government lose revenue from payroll taxes that the laid-off workers would have paid if they had kept their jobs. So if the yearly total revenue generated directly and indirectly by the tax were less than those total yearly payroll taxes and any other tax revenue that was lost as a result of the tax, the tax would have caused a net loss in tax revenue.

C The passage contains no information about what types of legislative actions cost, or do not cost, incumbent legislators popular support.

D Although the passage suggests that some of the tax proponents' assumptions were wrong, it contains no information suggesting that those proponents were wrong in thinking that luxury boats are purchased mainly by wealthy people.

E The passage does not provide any basis for determining what makes a tax equitable or about whether the luxury boat tax is equitable. The tax's proponents evidently felt that a tax whose burden falls only on the wealthy rather than evenly on the entire population can be equitable.

The correct answer is B.

CR05191

635. In Wareland last year, 16 percent of licensed drivers under 21 and 11 percent of drivers ages 21–24 were in serious accidents. By contrast, only 3 percent of licensed drivers 65 and older were involved in serious accidents. These figures clearly show that the greater experience and developed habits of caution possessed by drivers in the 65-and-older group make them far safer behind the wheel than the younger drivers are.

Which of the following is an assumption on which the argument depends?

(A) Drivers 65 and older do not, on average, drive very many fewer miles per year than drivers 24 and younger.

(B) Drivers 65 and older do not constitute a significantly larger percentage of licensed drivers in Wareland than drivers ages 18–24 do.

(C) Drivers 65 and older are less likely than are drivers 24 and younger to drive during weather conditions that greatly increase the risk of accidents.

(D) The difference between the accident rate of drivers under 21 and of those ages 21–24 is attributable to the greater driving experience of those in the older group.

(E) There is no age bracket for which the accident rate is lower than it is for licensed drivers 65 and older.

Argument Evaluation

Situation Last year in Wareland, a much higher percentage of drivers 24 and under than of drivers 65 and older were in serious accidents.

Reasoning *What must be true for the observation about the accident rates to support the conclusion that the greater experience and caution of drivers 65 and older make them safer behind the wheel than the younger drivers?* Several factors other than greater experience and caution could explain the lower accident rate among the older drivers. For example, the older drivers might simply drive much less than the younger ones, but still get in just as many accidents per mile driven. Or perhaps because the older drivers are more often retired, their schedules less often lead them to drive at times of day when accident rates are greater for everyone. Or they might be more likely to live in rural areas with less traffic and lower accident rates. The argument depends on assuming that none of these factors fully explains the difference in accident rates.

A **Correct.** Although we are given no information about the possible extent of any difference in average miles driven, the (somewhat vague) information that drivers 65 and older drive *very many fewer miles per year*, on average, than drivers 24 and younger would cast serious doubt on the statistical argument given. The argument assumes that the difference in miles driven is not sufficiently substantial to undermine the argument.

B The argument is only about the discrepancy between the percentages of the drivers in two specific age groups who were in serious accidents last year. The percentages of licensed drivers who fall in these age groups are irrelevant.

C Even if drivers 65 and older are just as likely as younger drivers to drive in inclement weather, they may do so far more carefully than the younger drivers, so the older drivers' greater experience and caution could still explain their lower accident rates.

D Even if greater experience does not explain the difference between the accident rates of the two younger groups of drivers, it might still explain the differences between the accident rate of those two younger groups taken together and that of drivers aged 65 and older.

E The accident rate could be lower for drivers in late middle age than for those 65 and older because drivers in late middle age are also cautious and experienced, but their reflexes and vision tend to be less impaired. Even if that were true, the experience and caution of the drivers 65 and older might still make them safer than drivers 24 and under.

The correct answer is A.

CR05614

636. In the past the country of Malvernia has relied heavily on imported oil. Malvernia recently implemented a program to convert heating systems from oil to natural gas. Malvernia currently produces more natural gas each year than it uses, and oil production in Malvernian oil fields is increasing at a steady pace. If these trends in fuel production and usage continue, therefore, Malvernian reliance on foreign sources for fuel is likely to decline soon.

Which of the following would it be most useful to establish in evaluating the argument?

(A) When, if ever, will production of oil in Malvernia outstrip production of natural gas?

(B) Is Malvernia among the countries that rely most on imported oil?

(C) What proportion of Malvernia's total energy needs is met by hydroelectric, solar, and nuclear power?

(D) Is the amount of oil used each year in Malvernia for generating electricity and fuel for transportation increasing?

(E) Have any existing oil-burning heating systems in Malvernia already been converted to natural-gas-burning heating systems?

Argument Evaluation

Situation Malvernia has relied heavily on imported oil, but recently began a program to convert heating systems from oil to natural gas. Malvernia produces more natural gas than it uses, so it will probably reduce its reliance on imported oils if these trends continue.

Reasoning *Which option provides the information that it would be most useful to know in evaluating the argument?* In other words, we are looking for the option which—depending on whether it was answered yes or no—would either most weaken or most strengthen the argument. The argument indicates that Malvernia will be using less oil for heating and will be producing more oil domestically. But the conclusion that Malvernia's reliance on foreign oil will decline, assuming the current trends mentioned continue, would be seriously undermined if there was something in the works that was bound to offset these trends, for instance, if it turned out that the country's need for oil was going to rise drastically in the coming years.

A Since both counteract the need for imported oil, it makes little difference to the argument whether domestic oil production exceeds domestic natural gas.

B Whether there are many countries that rely more on foreign oil than Malvernia would have little impact on whether Malvernia's need for foreign oil can be expected to decline.

C Since there is no information in the argument about whether Malvernia can expect an increase or decrease from these other energy sources, it does not matter how much they now provide.

D **Correct.** This option provides the information that it would be most useful to know in evaluating the argument.

E The argument tells us that a program has begun *recently* to convert heating systems from oil to gas. So, even if no such conversions have been completed, the argument still indicates that they can be expected to occur.

The correct answer is D.

CR03618

637. Exposure to certain chemicals commonly used in elementary schools as cleaners or pesticides causes allergic reactions in some children. Elementary school nurses in Renston report that the proportion of schoolchildren sent to them for treatment of allergic reactions to those chemicals has increased significantly over the past ten years. Therefore, either Renston's schoolchildren have been exposed to greater quantities of the chemicals, or they are more sensitive to them than schoolchildren were ten years ago.

Which of the following is an assumption on which the argument depends?

(A) The number of school nurses employed by Renston's elementary schools has not decreased over the past ten years.

(B) Children who are allergic to the chemicals are no more likely than other children to have allergies to other substances.

(C) Children who have allergic reactions to the chemicals are not more likely to be sent to a school nurse now than they were ten years ago.

(D) The chemicals are not commonly used as cleaners or pesticides in houses and apartment buildings in Renston.

(E) Children attending elementary school do not make up a larger proportion of Renston's population now than they did ten years ago.

Argument Construction

Situation Some children have allergic reactions to some of the chemicals commonly used in elementary schools as cleaners and pesticides. The number of children sent to elementary school nurses in Renston for allergic reactions to such chemicals has risen significantly over the past ten years.

Reasoning *What must the argument assume?* The argument's conclusion presents just two alternatives: either the children are exposed to more of the chemicals than children in earlier years *or* they are more sensitive. But there is a third possible explanation for the significant increase in school-nurse visits that the school nurses have reported: that children are just more inclined to go to the school nurse when they experience an allergic reaction than were children several years ago. For the conclusion to follow from its premises, the argument must assume that this is not the correct explanation.

A If the number of elementary school nurses in Renston elementary schools had decreased over the past ten years, that would in no way explain the rise in the proportion of children reporting to school nurses for allergic reactions.

B Only school-nurse visits for allergic reactions to the cleaners and pesticides used in elementary schools are in question in the argument. Of course there could be school-nurse visits for allergic reactions to other things, but that issue does not arise in the argument.

C **Correct.** This can be seen by considering whether the argument would work if we assume that this were false, i.e., that a school-nurse visit *is* more likely in such cases. As noted above, this provides an alternative to the two explanations that the conclusion claims are the sole possibilities.

D This does not need to be assumed by the argument. The argument's conclusion suggests that children may in recent years have had greater exposure to the chemicals, not that this exposure has occurred exclusively in the schools. The argument does not rely on this latter assumption.

E The argument does not need to make this assumption. The argument is framed in terms of proportions of children having school-nurse visits for certain allergic reactions. *How many* children there are or what proportion such children are of Renston's total population is not directly relevant to the argument.

The correct answer is C.

CR01854

638. Normally, the pineal gland governs a person's sleep-wake cycle by secreting melatonin in response to the daily cycle of light and darkness as detected by the eye. Nonetheless, many people who are totally blind due to lesions in the visual cortex of the brain easily maintain a 24-hour sleep-wake cycle. So the neural pathway by which the pineal gland receives information from the eye probably does not pass through the visual cortex.

For purposes of evaluating the argument it would be most useful to establish which of the following?

(A) Whether melatonin supplements help people who have difficulty maintaining a 24-hour sleep cycle to establish such a pattern

(B) Whether the melatonin levels of most totally blind people who successfully maintain a 24-hour sleep-wake cycle change in response to changes in exposure to light and darkness

(C) Whether melatonin is the only substance secreted by the pineal gland

(D) Whether most people who do not have a 24-hour sleep-wake cycle nevertheless have a cycle of consistent duration

(E) Whether there are any people with normal vision whose melatonin levels respond abnormally to periods of light and darkness

Argument Evaluation

Situation Normally, a person's sleep-wake cycle is governed by the pineal gland secreting melatonin in response to the daily cycle of light and darkness as detected by the eye. Yet many people who are totally blind due to lesions of the visual cortex easily maintain a 24-hour sleep-wake cycle.

Reasoning *What additional information would be most helpful in evaluating the argument?* The argument's conclusion is that the neural pathway by which the pineal gland receives information probably does not pass through the visual cortex. This is suggested by the fact that people without a well-functioning visual cortex (e.g., people with a certain type of blindness) can nonetheless maintain a 24-hour sleep-wake cycle. Is it by the pineal gland's secretion of melatonin that they do so? The argument tells us that *normally* (i.e., in sighted people), this is the mechanism for sleep regulation. But the argument depends on assuming that a similar mechanism is operating in people who are blind but have well-regulated sleep cycles. The best choice will be the one that helps us decide whether that assumption is correct.

A This question would not give us an answer that would help in evaluating the argument. A "no" answer would not clarify whether the pineal gland-melatonin mechanism operates in people who are blind. A "yes" answer would do no better. The question refers only to people who have sleep dysfunctions (which the argument does not address).

B **Correct.** Answering this question would provide the most useful information for evaluating the argument. A "yes" answer would help confirm a key assumption of the argument: that blind people rely on the pineal gland-melatonin mechanism for sleep regulation. A "no" answer would help disconfirm that assumption.

C Whether or not there are other substances secreted by the pineal gland makes no difference to the reasoning. The argument relies on the premise that the pineal gland governs the sleep cycle *by secreting melatonin*. For example, if the pineal gland sometimes secreted adrenaline, that would still have no bearing on the argument.

D The consistency or inconsistency of the duration of some people's sleep patterns has no relevance to the reasoning. Their sleep patterns could be due to any of a number of factors.

E This does not help, for there could be sighted people whose melatonin levels respond abnormally simply because of a pineal-gland abnormality.

The correct answer is B.

CR00942

639. **In countries where automobile insurance includes compensation for whiplash injuries sustained in automobile accidents, reports of having suffered such injuries are twice as frequent as they are in countries where whiplash is not covered.** Presently, no objective test for whiplash exists, so it is true that spurious reports of whiplash injuries cannot be readily identified. Nevertheless, these facts do not warrant the conclusion drawn by some commentators that in the countries with the higher rates of reported whiplash injuries, half of the reported cases are spurious. Clearly, **in countries where automobile insurance does not include compensation for whiplash, people often have little incentive to report whiplash injuries that they actually have suffered.**

In the argument given, the two boldfaced portions play which of the following roles?

(A) The first is a claim that the argument disputes; the second is a conclusion that has been based on that claim.

(B) The first is a claim that has been used to support a conclusion that the argument accepts; the second is that conclusion.

(C) The first is evidence that has been used to support a conclusion for which the argument provides further evidence; the second is the main conclusion of the argument.

(D) The first is a finding whose implications are at issue in the argument; the second is a claim presented in order to argue against deriving certain implications from that finding.

(E) The first is a finding whose accuracy is evaluated in the argument; the second is evidence presented to establish that the finding is accurate.

Argument Evaluation

Situation Reported whiplash injuries are twice as common in countries where car insurance companies pay compensation for such injuries as they are in countries where insurance companies do not. Although there is no objective test for whiplash, this does not mean, as some suggest, that half of the reports of such injuries are fake. It could simply be that where insurance will not pay for such injuries, people are less inclined to report them.

Reasoning *What roles do the two boldfaced portions play in the argument?* The first portion tells us about the correlation between reported cases of whiplash in countries and the willingness of insurance companies in those countries to compensate for whiplash injuries. The argument next states that whiplash is difficult to objectively verify. The argument then asserts that *although* this last fact, taken together with the first boldfaced portion, has led some to infer that over half of the reported cases in countries with the highest whiplash rates are spurious, such an inference is unwarranted. The second boldfaced portion then helps to explain why such an inference is not necessarily warranted by offering an alternative explanation.

A The claim made in the first boldfaced portion is never disputed in the argument; at dispute is how to account for the fact that this claim is true. The second is not the argument's conclusion.

B In a manner of speaking, perhaps, the argument uses the first portion to support its conclusion; but there is no indication that it has been used elsewhere to do so. In any case, the second boldfaced portion is not the argument's conclusion.

C The first has been used to support a conclusion that the argument *rejects*; the second boldfaced portion is not the argument's conclusion.

D **Correct.** This option correctly identifies the roles played in the argument by the boldfaced portions.

E The accuracy of the first boldfaced portion is never questioned in the argument; nor is the second intended to somehow help show that the first is accurate. Rather, the argument assumes that the first portion is accurate.

The correct answer is D.

CR03859

640. Last year Comfort Airlines had twice as many delayed flights as the year before, but the number of complaints from passengers about delayed flights went up three times. It is unlikely that this disproportionate increase in complaints was rooted in an increase in overall dissatisfaction with the service Comfort Airlines provides, since the airline made a special effort to improve other aspects of its service last year.

Which of the following, if true, most helps to explain the disproportionate increase in customer complaints?

(A) Comfort Airlines had more flights last year than the year before.

(B) Last year a single period of unusually bad weather caused a large number of flights to be delayed.

(C) Some of the improvements that Comfort Airlines made in its service were required by new government regulations.

(D) The average length of a flight delay was greater last year than it was the year before.

(E) The average number of passengers per flight was no higher last year than the year before.

Argument Construction

Situation Last year Comfort Airlines had twice as many delayed flights as it did the year before, but three times as many passenger complaints about delayed flights. The airline made a special effort to improve other aspects of its service last year.

Reasoning *What could explain why the number of complaints about delayed flights increased disproportionately to the number of delayed flights last year?* In other words, why did the average number of passenger complaints per delayed flight go up last year? One obvious possibility is that the average number of passengers per delayed flight was greater last year than it had been the year before. Another is that the flight delays tended to cause worse problems for passengers last year than they had the year before, so that on average each delay was more upsetting for the passengers.

A This helps explain why the airline had more delayed flights last year, but not why the increase in complaints about delayed flights was disproportionate to the increase in delayed flights.

B This helps explain why the airline had more delayed flights last year. But, if anything, the situation should have reduced the number of passenger complaints per delayed flight, since many passengers should have realized that the unusually bad weather was not the airline's fault.

C If any of the improvements concerned handling of flight delays, for example, and passengers were aware that government regulations addressed this, then passengers might have complained more than previously. But the information we are given here is too general and too vague to explain the disproportionate increase in complaints.

D **Correct.** Longer flight delays would have more severely inconvenienced passengers and thus would probably have generated more passenger complaints per delay.

E This rules out the possibility that an increased number of passengers per delayed flight could have caused the disproportionate increase in the number of complaints about delayed flights. But no alternative explanation is offered.

The correct answer is D

CR01337

641. Last year a global disturbance of weather patterns disrupted harvests in many of the world's important agricultural areas. Worldwide production of soybeans, an important source of protein for people and livestock alike, was not adversely affected, however. Indeed, last year's soybean crop was actually slightly larger than average. Nevertheless, the weather phenomenon is probably responsible for a recent increase in the world price of soybeans.

Which of the following, if true, provides the strongest justification for the attribution of the increase in soybean prices to the weather phenomenon?

(A) Last year's harvest of anchovies, which provide an important protein source for livestock, was disrupted by the effects of the weather phenomenon.

(B) Most countries that produce soybeans for export had above-average harvests of a number of food crops other than soybeans last year.

(C) The world price of soybeans also rose several years ago, immediately after an earlier occurrence of a similar global weather disturbance.

(D) Heavy rains attributable to the weather phenomenon improved grazing pastures last year, allowing farmers in many parts of the world to reduce their dependence on supplemental feed.

(E) Prior to last year, soybean prices had been falling for several years.

Argument Construction

Situation A weather disturbance last year disrupted harvests worldwide but did not reduce production of soybeans, a protein source for both people and livestock. Soybean prices increased nonetheless, likely a result of the weather.

Reasoning *What evidence would suggest that the weather disturbance caused the increase in soybean prices even though it did not reduce soybean production?* Prices tend to increase when the supply of a product falls relative to the demand for the product. But the production of soybeans did not fall. Evidence that the weather disturbance either hindered the global distribution of soybeans or increased global demand for soybeans could support the claim that the weather disturbance caused the increase in soybean prices.

A **Correct.** If the weather disturbance reduced the anchovy harvest, and anchovies provide protein for livestock just as soybeans do, then more soybeans for livestock feed would be needed to compensate for the lack of anchovies. The resulting increase in demand for soybeans could thus have increased global soybean prices.

B This is not surprising, given that the weather disturbance did not severely affect the soybean-producing countries, but it does not explain how the weather disturbance could have caused soybean prices to increase.

C The rise in soybean prices after the earlier weather disturbance could easily have been a coincidence. Or, unlike last year's disturbance, the earlier disturbance could have reduced soybean production.

D This suggests that demand for soybeans should have fallen as a result of the weather disturbance, so it does not explain why soybean prices rose.

E If soybean prices were unusually low for some temporary reason when the weather disturbance occurred, they might have been likely to rise back to normal levels even without the weather disturbance.

The correct answer is A.

CR03541

642. Most of the year, the hermit thrush, a North American songbird, eats a diet consisting mainly of insects, but in autumn, as the thrushes migrate to their Central and South American wintering grounds, they feed almost exclusively on wild berries. Wild berries, however, are not as rich in calories as insects, yet thrushes need to consume plenty of calories in order to complete their migration. One possible explanation is that berries contain other nutrients that thrushes need for migration and that insects lack.

Which of the following, if true, most seriously calls into question the explanation given for the thrush's diet during migration?

(A) Hermit thrushes, if undernourished, are unable to complete their autumn migration before the onset of winter.

(B) Insect species contain certain nutrients that are not found in wild berries.

(C) For songbirds, catching insects requires the expenditure of significantly more calories than eating wild berries does.

(D) Along the hermit thrushes' migration routes, insects are abundant throughout the migration season.

(E) There are some species of wild berries that hermit thrushes generally do not eat, even though these berry species are exceptionally rich in calories.

Argument Evaluation

Situation Hermit thrushes are songbirds that usually eat insects but switch to eating berries when migrating. The thrushes need lots of calories to migrate, but berries contain fewer calories than insects do. Perhaps the berries contain nutrients that insects do not provide.

Reasoning *What would cast doubt on the claim that the thrushes switch to berries because berries contain nutrients that insects lack and that the thrushes need for their migration?* Evidence that berries do not contain such nutrients or that thrushes do not decrease their net calorie consumption by eating berries would cast doubt on the proposed explanation. So would any evidence that supported an alternative explanation for the diet change during migration—for example, seasonal or regional differences in the amount or quality of berries or insects available for the thrushes to consume.

A Even if thrushes need to be well-nourished to finish migrating before winter, extra nutrients found in berries but not insects might help provide the nourishment they need.

B Even if insects contain *certain nutrients* not found in wild berries, those specific nutrients may not be the ones the thrushes need for their migration.

C **Correct.** This suggests that the thrushes might gain more net calories from eating berries than from eating insects, which could explain why they switch to eating berries even if the berries contain no extra nutrients.

D By ruling out a lack of insects to eat while migrating as an alternative explanation for why the thrushes switch to eating berries, this would support the proposed explanation.

E The calorie-rich species of berries the thrushes do not eat might be poisonous or indigestible for them, even if the species of berries the thrushes do eat contain nutrients they need to migrate.

The correct answer is C.

CR01879

643. The kinds of hand and wrist injuries that result from extended use of a computer while maintaining an incorrect posture are common among schoolchildren in Harnville. Computers are important to the school curriculum there, so instead of reducing the amount their students use computers, teachers plan to bring about a sharp reduction in the number of these injuries by carefully monitoring their students' posture when using computers in the classroom.

Which of the following would it be most useful to know in order to assess the likelihood that the teachers' plan will be successful?

(A) Whether extended use of a computer while maintaining incorrect posture can cause injuries other than hand and wrist injuries

(B) Whether hand and wrist injuries not caused by computer use are common among schoolchildren in Harnville

(C) What proportion of schoolchildren in Harnville with hand and wrist injuries use computers extensively outside the classroom

(D) Whether changes in the curriculum could reduce the schools' dependence on computers

(E) What proportion of schoolchildren in Harnville already use correct posture while using a computer

Evaluation of a Plan

Situation Hand and wrist injuries from using computers while maintaining poor posture are common among schoolchildren in Harnville. Teachers plan to greatly reduce the number of such injuries by monitoring their students' posture while the students use computers in the classroom.

Reasoning *What would be most helpful to know to determine the likelihood that the teachers' plan will succeed?* The primary concern is the *posture* students adopt while using computers. To succeed, the teachers' plan must reduce the time students spend with poor posture while using computers and reduce it enough to greatly reduce the number of injuries. To know how likely this is, it would help to know how effectively the teachers will be able to monitor and improve their students' posture inside the classroom. But how many of the students use computers *outside of school* while maintaining poor posture and how often do they do so? If many students do so quite often, they may develop hand and wrist injuries regardless of what happens in school.

A The teachers do not plan to reduce any injuries other than hand and wrist injuries, so whether computer use with poor posture causes any such other injuries is irrelevant to the likelihood that their plan will produce its intended effect.

B The plan being discussed concerns only the reduction of hand and wrist injuries caused specifically by computer use with poor posture, so the frequency of hand and wrist injuries from other causes is irrelevant to the likelihood that the plan will produce its intended effect.

C **Correct.** If the students' school use of computers is a large part of their overall computer use, any retraining that accompanies the monitoring might have some effect on their posture and related injury rates overall. However, the greater the proportion of children with hand and wrist injuries who use computers extensively outside the classroom, the more children are likely to keep developing the injuries regardless of any monitoring at school, so the less effective the teachers' plan involving only computer use at school is likely to be.

D Knowing whether this is the case might help in developing a potential alternative to the teachers' plan, but if it did, this would not help significantly toward assessing the likelihood that the actual plan will succeed. The teachers' actual plan involves monitoring computer use in school without reducing such use. Other possible means of achieving the plan's goal are not part of the plan and are therefore irrelevant to the likelihood that the teachers' actual plan will succeed.

E The passage indicates that the proportion of the schoolchildren maintaining poor posture while using computers is high enough for many to develop hand and wrist injuries as a result. Whatever the exact proportion is, the teachers' plan may or may not succeed in reducing it.

The correct answer is C.

CR04718

644. A certain cultivated herb is one of a group of closely related plants that thrive in soil with high concentrations of metals that are toxic to most other plants. Agronomists studying the growth of this herb have discovered that it produces large amounts of histidine, an amino acid that, in test-tube solutions, renders these metals chemically inert. Hence, the herb's high histidine production must be the key feature that allows it to grow in metal-rich soils.

In evaluating the argument, it would be most important to determine which of the following?

(A) Whether the herb can thrive in soil that does not have high concentrations of the toxic metals

(B) Whether others of the closely related group of plants also produce histidine in large quantities

(C) Whether the herb's high level of histidine production is associated with an unusually low level of production of some other amino acid

(D) Whether growing the herb in soil with high concentrations of the metals will, over time, reduce their concentrations in the soil

(E) Whether the concentration of histidine in the growing herb declines as the plant approaches maturity

Argument Evaluation

Situation A certain herb and closely related species thrive in soil full of metals toxic to most plants. The herb produces much histidine, which makes those metals chemically inert. Histidine production, therefore, is largely what accounts for the herb's thriving in metal-rich soils.

Reasoning *What evidence would help determine whether the herb's histidine production is what enables it to thrive in metal-rich soils?* The argument is that since the herb's histidine chemically neutralizes the metals that are toxic to most plants, it must explain why the herb can thrive in metal-rich soils. To evaluate this argument, it would be helpful to know about the relationship between other closely related plant species' histidine production and the ability to thrive in metal-rich soils. It would also be helpful to know about any other factors that might plausibly explain why the herb can thrive in those soils.

A Whether or not the herb thrives in metal-free soils, histidine production could enable it to thrive in soils that contain toxic metals.

B **Correct.** If the closely related plants do not produce much histidine, whatever other factor allows them to thrive in metal-rich soils would likely account for why the herb thrives in those soils as well.

C The given information suggests no particular reason to suppose that a low level of some unspecified amino acid would enable a plant to thrive in metal-rich soils.

D The herb might absorb metals from any metal-rich soil it grows in, regardless of why it thrives in that soil.

E Whether or not histidine concentrations in the herb decline as it approaches maturity, there could still be enough histidine in the growing herb to neutralize the metals and explain why it can grow in metal-rich soil.

The correct answer is B.

CR01293

645. Many people suffer an allergic reaction to certain sulfites, including those that are commonly added to wine as preservatives. However, since there are several winemakers who add sulfites to none of the wines they produce, people who would like to drink wine but are allergic to sulfites can drink wines produced by these winemakers without risking an allergic reaction to sulfites.

Which of the following is an assumption on which the argument depends?

(A) These winemakers have been able to duplicate the preservative effect produced by adding sulfites by means that do not involve adding any potentially allergenic substances to their wine.

(B) Not all forms of sulfite are equally likely to produce the allergic reaction.

(C) Wine is the only beverage to which sulfites are commonly added.

(D) Apart from sulfites, there are no substances commonly present in wine that give rise to an allergic reaction.

(E) Sulfites are not naturally present in the wines produced by these winemakers in amounts large enough to produce an allergic reaction in someone who drinks these wines.

Argument Construction

Situation People who are allergic to certain sulfites can avoid risking an allergic reaction by drinking wine from one of the several producers that does not add sulfites.

Reasoning *On what assumption does the argument depend?* Drinking wine to which no sulfites have been *added* will not prevent exposure to sulfites if, for instance, sulfites occur naturally in wines. In particular, if the wines that do not have sulfites added have sulfites present naturally in quantities sufficient to produce an allergic reaction, drinking these wines will not prevent an allergic reaction. The argument therefore depends on assuming that this is not the case.

A The argument does not require this because the conclusion does not address allergic reactions to substances other than sulfites.

B The argument specifically refers to "certain sulfites" producing allergic reactions. It is entirely compatible with certain other forms of sulfites not producing allergic reactions in anyone.

C This is irrelevant. The argument does not claim that one can avoid having an allergic reaction to sulfites *from any source* just by restricting one's wine consumption to those varieties to which no sulfites have been added.

D Once again, the argument's conclusion does not address allergic reactions to substances other than sulfites in wine.

E **Correct.** The argument relies on this assumption.

The correct answer is E.

CR11447

646. A new law gives ownership of patents—documents providing exclusive right to make and sell an invention—to universities, not the government, when those patents result from government-sponsored university research. Administrators at Logos University plan to sell any patents they acquire to corporations in order to fund programs to improve undergraduate teaching.

Which of the following, if true, would cast the most doubt on the viability of the college administrators' plan described above?

(A) Profit-making corporations interested in developing products based on patents held by universities are likely to try to serve as exclusive sponsors of ongoing university research projects.

(B) Corporate sponsors of research in university facilities are entitled to tax credits under new federal tax-code guidelines.

(C) Research scientists at Logos University have few or no teaching responsibilities and participate little if at all in the undergraduate programs in their field.

(D) Government-sponsored research conducted at Logos University for the most part duplicates research already completed by several profit-making corporations.

(E) Logos University is unlikely to attract corporate sponsorship of its scientific research.

Evaluation of a Plan

Situation Universities own the patents resulting from government-sponsored research at their institutions. One university plans to sell its patents to corporations to fund a program to improve teaching.

Reasoning *Which point casts doubt on the university's plan?* The university's plan assumes there will be a market for its patents, and that the corporations will want to buy them. What might make this untrue? If some of the corporations have already done the same or similar research, they will not be prospective buyers of the university's patents.

A This point is irrelevant to the plan to sell patents in order to fund a program.

B The university plans to sell the patents to the corporations, not to invite the corporations to sponsor research.

C This point is irrelevant to the university's plan to sell off patents since the plan does not specify that the research scientists will be involved in the programs to improve undergraduate teaching.

D **Correct.** This statement properly identifies a factor that casts doubt on the university's plan to sell its patents to corporations.

E The plan concerns selling patents resulting from government-sponsored research, not attracting corporate sponsorship for research.

The correct answer is D.

CR01848

647. Since it has become known that **several of a bank's top executives have been buying shares in their own bank**, the bank's depositors, who had been worried by rumors that the bank faced impending financial collapse, have been greatly relieved. They reason that, since top executives evidently have faith in the bank's financial soundness, those worrisome rumors must be false. Such reasoning might well be overoptimistic, however, since **corporate executives have been known to buy shares in their own company in a calculated attempt to dispel negative rumors about the company's health.**

In the argument given, the two boldfaced portions play which of the following roles?

(A) The first describes evidence that has been taken as supporting a conclusion; the second gives a reason for questioning that support.

(B) The first describes evidence that has been taken as supporting a conclusion; the second states a contrary conclusion that is the main conclusion of the argument.

(C) The first provides evidence in support of the main conclusion of the argument; the second states that conclusion.

(D) The first describes the circumstance that the argument as a whole seeks to explain; the second gives the explanation that the argument seeks to establish.

(E) The first describes the circumstance that the argument as a whole seeks to explain; the second provides evidence in support of the explanation that the argument seeks to establish.

Argument Evaluation

Situation Top executives at a bank that has been rumored to be in financial trouble have been buying shares in the bank. Bank depositors see this as a good sign, because they believe that it indicates that the executives have faith in the bank. However, corporate executives sometimes do this just to dispel rumors about a company's health.

Reasoning *What is the role that the two boldfaced portions play in the argument?* The first boldfaced portion states that bank executives are buying bank shares, which the passage indicates is taken by bank depositors to be evidence of the executives' faith in the bank. The passage then tells us what some have inferred from this, and finally offers in the second boldfaced statement evidence that undermines this inference.

A **Correct.** This option correctly identifies the roles played by the boldfaced portions.

B This correctly describes the first statement's role, but the second statement is not offered as a conclusion—no evidence is given for it; rather it is evidence for something else.

C The second statement is not offered as a conclusion; no evidence is given for it.

D The second statement is not itself offered as an explanation of why these bank executives are investing in the bank; if it were, that would mean that the bank executives are doing so *because* corporate executives are known to do such things in a calculated effort to dispel worries. Furthermore the argument does not conclude that this other explanation (which the boldfaced portion points to) is correct, only that the one inferred by depositors may not be.

E The argument is not so much seeking to establish an explanation of its own as it is trying to undermine that inferred by the depositors.

The correct answer is A.

CR03814

648. Between 1980 and 2000 the sea otter population of the Aleutian Islands declined precipitously. There were no signs of disease or malnutrition, so there was probably an increase in the number of otters being eaten by predators. Orcas will eat otters when seals, their normal prey, are unavailable, and the Aleutian Islands seal population declined dramatically in the 1980s. Therefore, orcas were most likely the immediate cause of the otter population decline.

Which of the following, if true, most strengthens the argument?

(A) The population of sea urchins, the main food of sea otters, has increased since the sea otter population declined.

(B) Seals do not eat sea otters, nor do they compete with sea otters for food.

(C) Most of the surviving sea otters live in a bay that is inaccessible to orcas.

(D) The population of orcas in the Aleutian Islands has declined since the 1980s.

(E) An increase in commercial fishing near the Aleutian Islands in the 1980s caused a slight decline in the population of the fish that seals use for food.

Argument Evaluation

Situation A sea otter population declined even though there were no signs of disease or malnutrition. The local seal population also declined. Orcas eat otters when seals are unavailable, and thus are probably the cause of the decline in the otter population.

Reasoning *What would be evidence that predation by orcas reduced the sea otter population?* Disease and malnutrition are ruled out as alternative explanations of the decline in the sea otter population. The argument could be further strengthened by casting doubt on other possible explanations, such as predation by other animals, or by presenting observations that predation of otters by orcas would help to explain.

A Regardless of whether or not orcas ate the sea otters, the sea urchin population would most likely have increased when the population of sea otters preying on them decreased.

B Because the seal population declined during the initial years of the otter population decline, predation by and competition with seals were already implausible explanations of the otter population decline.

C **Correct.** Orcas eating most of the accessible otters could plausibly explain this observation, which therefore provides additional evidence that orca predation reduced the sea otter population.

D If the orca population declined at the same time as the sea otter population, it would be less likely that increasing predation by orcas reduced the otter population.

E Since the sea otters showed no signs of malnutrition, they were probably getting enough fish. But if they were not, commercial fishing rather than orcas might have caused the otter population decline.

The correct answer is C.

CR05960

649. Studies in restaurants show that the tips left by customers who pay their bill in cash tend to be larger when the bill is presented on a tray that bears a credit-card logo. Consumer psychologists hypothesize that simply seeing a credit-card logo makes many credit-card holders willing to spend more because it reminds them that their spending power exceeds the cash they have immediately available.

Which of the following, if true, most strongly supports the psychologists' interpretation of the studies?

(A) The effect noted in the studies is not limited to patrons who have credit cards.

(B) Patrons who are under financial pressure from their credit-card obligations tend to tip less when presented with a restaurant bill on a tray with a credit-card logo than when the tray has no logo.

(C) In virtually all of the cases in the studies, the patrons who paid bills in cash did not possess credit cards.

(D) In general, restaurant patrons who pay their bills in cash leave larger tips than do those who pay by credit card.

(E) The percentage of restaurant bills paid with a given brand of credit card increases when that credit card's logo is displayed on the tray with which the bill is presented.

Argument Evaluation

Situation Studies have found that restaurant customers give more generous tips when their bills are brought on trays bearing a credit-card logo. Psychologists speculate that this is because the logo reminds customers of their ability to spend more money than they have.

Reasoning *Which of the options most helps to support the psychologists' explanation of the studies?* The psychologists' hypothesis is that the credit-card logos on the trays bring to the minds of those who tip more the fact that they have more purchasing power than merely the cash that they have at hand. This explanation would not be valid even if those people who are *not* reminded of their own excess purchasing power—if in fact they have any such power—when they see such a logo nonetheless tip more in such trays. Thus, if restaurant patrons who are under financial pressure from their credit-card obligations do not tip more when their bills are presented on trays bearing credit-card logos, then the psychologists' interpretation of the studies is supported.

A This undermines the psychologists' interpretation, for it shows that the same phenomenon occurs even when the alleged cause has been removed.

B **Correct.** This option identifies the result that would most strengthen the psychologists' interpretation.

C This undermines the psychologists' interpretation by showing that the same phenomenon occurs even when the alleged cause has been removed; patrons cannot be reminded of something that is not there.

D To the extent that this bears on the interpretation of the study, it weakens it. Patrons *using* credit cards are surely aware that they have credit, and yet they spend *less* generously.

E This does not support the idea that being reminded that one has a credit card induces one to be *more generous*, only that it induces one to *use* that credit card.

The correct answer is B.

CR11633

650. In an experiment, each volunteer was allowed to choose between an easy task and a hard task and was told that another volunteer would do the other task. Each volunteer could also choose to have a computer assign the two tasks randomly. Most volunteers chose the easy task for themselves and under questioning later said they had acted fairly. But when the scenario was described to another group of volunteers, almost all said choosing the easy task would be unfair. This shows that most people apply weaker moral standards to themselves than to others.

Which of the following is an assumption required by this argument?

(A) At least some volunteers who said they had acted fairly in choosing the easy task would have said that it was unfair for someone else to do so.

(B) The most moral choice for the volunteers would have been to have the computer assign the two tasks randomly.

(C) There were at least some volunteers who were assigned to do the hard task and felt that the assignment was unfair.

(D) On average, the volunteers to whom the scenario was described were more accurate in their moral judgments than the other volunteers were.

(E) At least some volunteers given the choice between assigning the tasks themselves and having the computer assign them felt that they had made the only fair choice available to them.

Argument Construction

Situation In an experiment, most volunteers chose to do an easy task themselves and leave a hard task for someone else. They later said they had acted fairly, but almost all volunteers in another group to which the scenario was described said choosing the easy task would be unfair, indicating that most people apply weaker moral standards to themselves.

Reasoning *What must be true in order for the facts presented to support the conclusion that most people apply weaker moral standards to themselves than to others?* One set of volunteers said they had acted fairly in taking the easy task, whereas different volunteers said that doing so would be unfair. In neither case did any of the volunteers actually judge their own behavior differently from how they judged anyone else's. So the argument implicitly infers from the experimental results that most of the volunteers would judge their own behavior differently from someone else's if given the chance. This inference assumes that the volunteers in the second group would have applied the same moral standards that those in the first group did if they had been in the first group's position, and vice versa.

A **Correct.** If none of the volunteers who said their own behavior was fair would have judged someone else's similar behavior as unfair, then their relaxed moral judgment of themselves would not suggest that they applied weaker moral standards to themselves than to others.

B Even if this is so, the experimental results could still suggest that the volunteers would apply weaker moral standards to themselves than to others.

C The argument would be equally strong even if volunteers who were assigned the hard task did not know that someone else had gotten an easier task—or even if no volunteers were actually assigned the hard task at all.

D Even if the moral standards applied by the volunteers who judged themselves were as accurate as those applied by the volunteers to whom the scenario was described, the former standards were still weaker.

E Even if all the volunteers in the first group had felt that all the choices available to them would have been fair for them to make personally, they might have applied stricter moral standards to someone else in the same position.

The correct answer is A.

CR08527

651. Country X's recent stock-trading scandal should not diminish investors' confidence in the country's stock market. For one thing, **the discovery of the scandal confirms that Country X has a strong regulatory system**, as the following considerations show. In any stock market, some fraudulent activity is inevitable. If a stock market is well regulated, any significant stock-trading fraud in it will very likely be discovered. This deters potential perpetrators and facilitates improvement in regulatory processes.

In the argument, the portion in boldface plays which of the following roles?

(A) It is the argument's only conclusion.

(B) It is a conclusion for which the argument provides support and which itself is used to support the argument's main conclusion.

(C) It is the argument's main conclusion and is supported by another explicitly stated conclusion for which further support is provided.

(D) It is an assumption for which no explicit support is provided and is used to support the argument's only conclusion.

(E) It is a compound statement containing both the argument's main conclusion and an assumption used to support that conclusion.

Argument Construction

Situation Country X recently had a stock-trading scandal.

Reasoning *What role does the statement that the scandal's discovery confirms that Country X has a strong regulatory system play in the argument?* In the sentence containing the boldface statement, the phrase *For one thing* indicates that the statement is being used to justify the claim in the preceding sentence. Thus, the boldface statement must support that preceding sentence as a conclusion. Directly after the boldface statement, the phrase as *the following considerations show* indicates that the subsequent sentences are being used to support the boldface statement. Thus, the boldface statement is a conclusion supported by the sentences following it, and this statement itself supports the sentence preceding it, which must be the argument's main conclusion.

A As explained above, the boldface statement supports the claim in the preceding sentence, so it cannot be the argument's only conclusion.

B **Correct.** As explained above, the boldface statement is supported by the statements following it and in turn is used to support the argument's main conclusion in the statement preceding it.

C As explained above, the boldface statement cannot be the argument's main conclusion, because it supports a further conclusion presented in the sentence preceding it.

D As explained above, the sentences following the boldface statement are the explicit support provided for it.

E As explained above, the argument's main conclusion is stated only in the first sentence, which precedes the boldface statement. It is not repeated anywhere in the boldface statement.

The correct answer is B.

CR05644

652. **Delta Products Inc. has recently switched at least partly from older technologies using fossil fuels to new technologies powered by electricity.** The question has been raised whether it can be concluded that **for a given level of output Delta's operation now causes less fossil fuel to be consumed than it did formerly.** The answer, clearly, is yes, since the amount of fossil fuel used to generate the electricity needed to power the new technologies is less than the amount needed to power the older technologies, provided level of output is held constant.

In the argument given, the two boldfaced portions play which of the following roles?

(A) The first identifies the content of the conclusion of the argument; the second provides support for that conclusion.

(B) The first provides support for the conclusion of the argument; the second identifies the content of that conclusion.

(C) The first states the conclusion of the argument; the second calls that conclusion into question.

(D) The first provides support for the conclusion of the argument; the second calls that conclusion into question.

(E) Each provides support for the conclusion of the argument.

Argument Evaluation

Situation Delta switched from technologies using fossil fuels to ones using electricity. It has been asked whether this results in less fossil fuel used per level of output. The answer is that it does.

Reasoning *What roles do the two boldfaced portions play in the argument?* The first boldfaced statement is simply asserted by the passage. But the second boldfaced statement, when it is first introduced, is not asserted to be true, but rather is identified as something that might be inferred from the first statement. By the end of the passage the argument concludes that the second statement is true.

A This option simply reverses the roles that the statements play in the argument.

B **Correct.** This option identifies the roles the boldfaced portions play.

C Nothing in the passage is intended to support the first statement; and the second statement is not supposed to call the first into question.

D This correctly identifies the role of the first statement, but the second boldfaced portion does not call the argument's conclusion into question—it is part of a sentence that refers to the question whether that conclusion can be drawn from the first statement.

E Again, this is only half right. The second boldfaced portion is not offered as support for the conclusion; if it were offered as such support, the argument would be guilty of circular reasoning, since the second boldfaced portion states exactly what the argument concludes.

The correct answer is B.

CR00907

653. Theater Critic: The play *La Finestrina*, now at Central Theater, was written in Italy in the eighteenth century. The director claims that this production is as similar to the original production as is possible in a modern theater. Although the actor who plays Harlequin the clown gives a performance very reminiscent of the twentieth-century American comedian Groucho Marx, Marx's comic style was very much within the comic acting tradition that had begun in sixteenth-century Italy.

The considerations given best serve as part of an argument that

(A) modern audiences would find it hard to tolerate certain characteristics of a historically accurate performance of an eighteenth-century play

(B) Groucho Marx once performed the part of the character Harlequin in *La Finestrina*

(C) in the United States the training of actors in the twentieth century is based on principles that do not differ radically from those that underlay the training of actors in eighteenth-century Italy

(D) the performance of the actor who plays Harlequin in *La Finestrina* does not serve as evidence against the director's claim

(E) the director of *La Finestrina* must have advised the actor who plays Harlequin to model his performance on comic performances of Groucho Marx

Argument Construction

Situation The director of the local production of *La Finestrina* says it is as similar to the original production as is possible in a modern theater. The actor playing Harlequin gives a performance reminiscent of Groucho Marx, whose comic style falls within an acting tradition which began in sixteenth-century Italy.

Reasoning *For which of the options would the consideration given best serve as an argument?* The actor's performance was reminiscent of someone who fell within a tradition going back to sixteenth-century Italy. The play was written, and therefore was likely first performed, in eighteenth-century Italy. All of this suggests that there could be a similarity between the performances of Harlequin in the local production and in the original production. While the two performances *might* have been quite dissimilar, there is nothing *here* that supports that.

A Regardless of how plausible this option might be on its own merits, the passage provides no support for it because the passage provides no information about the characteristics of a historically accurate performance of an eighteenth-century play.

B The passage neither says this nor implies it.

C The passage says nothing about the training of actors, so this option would be supported by the passage only in a very roundabout, indirect way.

D **Correct.** This is the option that the considerations most support.

E That the performance reminded the theater critic of Groucho Marx hardly shows that the similarity was intentional, let alone that it was at the director's instruction.

The correct answer is D.

CR07257

654. Although the discount stores in Goreville's central shopping district are expected to close within five years as a result of competition from a SpendLess discount department store that just opened, those locations will not stay vacant for long. In the five years since the opening of Colson's, a nondiscount department store, a new store has opened at the location of every store in the shopping district that closed because it could not compete with Colson's.

Which of the following, if true, most seriously weakens the argument?

(A) Many customers of Colson's are expected to do less shopping there than they did before the SpendLess store opened.

(B) Increasingly, the stores that have opened in the central shopping district since Colson's opened have been discount stores.

(C) At present, the central shopping district has as many stores operating in it as it ever had.

(D) Over the course of the next five years, it is expected that Goreville's population will grow at a faster rate than it has for the past several decades.

(E) Many stores in the central shopping district sell types of merchandise that are not available at either SpendLess or Colson's.

Argument Evaluation

Situation Due to competition from a recently opened SpendLess discount department store, discount stores in Goreville's central shopping district are expected to close within five years. But those locations will not be vacant long, for new stores have replaced all those that closed because of the opening five years ago of a Colson's nondiscount department store.

Reasoning *The question is which option would most weaken the argument?* The arguer infers that stores that leave because of the SpendLess will be replaced in their locations by other stores because that is what happened after the Colson's department store came in. Since the reasoning relies on a presumed similarity between the two cases, any information that brings to light a relevant dissimilarity would weaken the argument. If the stores that were driven out by Colson's were replaced mostly by discount stores, that suggests that the stores were replaced because of a need that no longer exists after the opening of SpendLess.

A The fact that Colson's may be seeing fewer customers does not mean that the discount stores that close will not be replaced; they might be replaced by stores that in no way compete with Colson's or SpendLess.

B **Correct.** This option most seriously weakens the argument.

C If anything, this strengthens the argument by indicating that Goreville's central shopping district is thriving.

D This, too, strengthens the argument because one is more likely to open a new store in an area with a growing population.

E Because this statement does not indicate whether any of these stores that offer goods not sold at SpendLess or Colson's will be among those that are closing, it is not possible to determine what effect it has on the strength of the argument.

The correct answer is B.

CR05685

655. Last year all refuse collected by Shelbyville city services was incinerated. This incineration generated a large quantity of residual ash. In order to reduce the amount of residual ash Shelbyville generates this year to half of last year's total, the city has revamped its collection program. This year city services will separate for recycling enough refuse to reduce the number of truckloads of refuse to be incinerated to half of last year's number.

Which of the following is required for the revamped collection program to achieve its aim?

(A) This year, no materials that city services could separate for recycling will be incinerated.

(B) Separating recyclable materials from materials to be incinerated will cost Shelbyville less than half what it cost last year to dispose of the residual ash.

(C) Refuse collected by city services will contain a larger proportion of recyclable materials this year than it did last year.

(D) The refuse incinerated this year will generate no more residual ash per truckload incinerated than did the refuse incinerated last year.

(E) The total quantity of refuse collected by Shelbyville city services this year will be no greater than that collected last year.

Argument Construction

Situation To cut in half the residual ash produced at its incinerator, the city will separate for recycling enough refuse to cut in half the number of truckloads of refuse going to the incinerator.

Reasoning *Which option is required if the city's revamped collection program is to achieve its aim?* Cutting the number of truckloads of refuse in half must reduce the amount of residual ash to half last year's level. But if removal of the recycled refuse does not proportionately reduce the amount of ash, this will not happen. So if the amount of residual ash produced per truckload increases after recycling, then the amount of ash produced will not be cut in half by cutting in half the number of truckloads.

A This merely indicates that no further reduction of ash through recycling could be achieved this year; it indicates nothing about how much the ash will be reduced.

B This suggests a further benefit from recycling, but does not bear on the amount of ash that will be produced.

C Since no information is provided about how much, if any, recyclable materials were removed from the refuse last year, this does not affect the reasoning.

D **Correct.** This states a requirement for the collection program to achieve its aim.

E This is not a requirement because even if the city collects more refuse this year, it could still cut in half the amount of residual ash by cutting in half the number of truckloads going to the incinerator.

The correct answer is D.

CR01801

656. Veterinarians generally derive some of their income from selling several manufacturers' lines of pet-care products. Knowing that pet owners rarely throw away mail from their pet's veterinarian unread, one manufacturer of pet-care products offered free promotional materials on its products to veterinarians for mailing to their clients. Very few veterinarians accepted the offer, however, even though the manufacturer's products are of high quality.

Which of the following, if true, most helps to explain the veterinarians' reaction to the manufacturer's promotional scheme?

(A) Most of the veterinarians to whom the free promotional materials were offered were already selling the manufacturer's pet-care products to their clients.

(B) The special promotional materials were intended as a supplement to the manufacturer's usual promotional activities rather than as a replacement for them.

(C) The manufacturer's products, unlike most equally good competing products sold by veterinarians, are also available in pet stores and in supermarkets.

(D) Many pet owners have begun demanding quality in products they buy for their pets that is as high as that in products they buy for themselves.

(E) Veterinarians sometimes recommend that pet owners use products formulated for people when no suitable product specially formulated for animals is available.

Evaluation of a Plan

Situation Veterinarians generally derive some income from selling various manufacturers' pet-care products, but very few veterinarians accepted free promotional materials from one such manufacturer to mail to their clients.

Reasoning *What would most help explain why so few veterinarians accepted the free promotional materials to mail to their clients?* The passage says that veterinarians generally derive income from selling pet-care products, which suggests that it should have been in many veterinarians' financial interest to accept and mail out the free promotional materials to increase sales. Any evidence that mailing out these specific promotional materials from this manufacturer would not actually have been in many veterinarians' financial interest could help explain why so few veterinarians accepted the materials.

A This suggests that most of the veterinarians should have had a financial interest in accepting and mailing out the promotional materials in order to increase their sales of the manufacturer's products.

B Even if the promotional materials supplemented the manufacturer's usual promotional activities, they could still have increased the veterinarians' sales of the manufacturer's products and thus generated more income for the veterinarians.

C **Correct.** If this manufacturer's products are available in pet stores and supermarkets but most other products sold by veterinarians are not, then distributing the manufacturer's promotional materials could have encouraged customers to buy this manufacturer's products from pet stores and supermarkets rather than to buy competing products from the veterinarians. Thus, the veterinarians may have been concerned that the promotions would reduce their profits.

D The passage says the manufacturer's products are of high quality, so we have no reason to suppose that clients' demand for quality products would discourage veterinarians from accepting the manufacturer's promotional materials.

E Presumably the manufacturer's products are specially formulated for pets, so any products veterinarians recommend only when no specially formulated pet-care products are available would not reduce the veterinarians' interest in promoting the manufacturer's products.

The correct answer is C.

CR00778

657. The average hourly wage of television assemblers in Vernland has long been significantly lower than that in neighboring Borodia. Since Borodia dropped all tariffs on Vernlandian televisions three years ago, the number of televisions sold annually in Borodia has not changed. However, recent statistics show a drop in the number of television assemblers in Borodia. Therefore, updated trade statistics will probably indicate that the number of televisions Borodia imports annually from Vernland has increased.

Which of the following is an assumption on which the argument depends?

(A) The number of television assemblers in Vernland has increased by at least as much as the number of television assemblers in Borodia has decreased.

(B) Televisions assembled in Vernland have features that televisions assembled in Borodia do not have.

(C) The average number of hours it takes a Borodian television assembler to assemble a television has not decreased significantly during the past three years.

(D) The number of televisions assembled annually in Vernland has increased significantly during the past three years.

(E) The difference between the hourly wage of television assemblers in Vernland and the hourly wage of television assemblers in Borodia is likely to decrease in the next few years.

Argument Construction

Situation Television assemblers in Vernland are paid less than those in neighboring Borodia. The number of televisions sold in Borodia has not dropped since its tariffs on Vernlandian TVs were lowered three years ago, but the number of TV assemblers in Borodia has. So TV imports from Vernland have likely increased.

Reasoning *What assumption does the argument depend on?* The fact that fewer individuals in Borodia are working as TV assemblers is offered as evidence that TV imports from Vernland into Borodia have likely increased. That piece of evidence is relevant *only* as an indication that the number of TVs being produced within Borodia has decreased. But a drop in the number of TV assemblers does not indicate a drop in the number of TVs being assembled *if* the number of TVs an average assembler puts together has increased. Thus, the argument must be assuming that the average time it takes an assembler to put together a TV has not significantly decreased.

A The argument does not rely on any information about the number of television assemblers in Vernland nor for that matter on the number of TVs assembled in Vernland.

B The argument need not assume there is any difference in the features of the TVs produced in the two countries. Increased sales of Vernlandian TVs in Borodia could be due to any number of other reasons, such as price or quality.

C **Correct.** This option states an assumption on which the argument depends.

D The argument does not depend upon this being so: Vernland's domestic TV sales (or perhaps its exports to countries other than Borodia) may have decreased by more than its imports into Borodia have increased.

E The argument's conclusion addresses what *has* happened; the argument in no way relies on any assumptions about what may or may not happen in the coming years.

The correct answer is C.

CR05725

658. Guidebook writer: I have visited hotels throughout the country and have noticed that in those built before 1930 the quality of the original carpentry work is generally superior to that in hotels built afterward. Clearly carpenters working on hotels before 1930 typically worked with more skill, care, and effort than carpenters who have worked on hotels built subsequently.

Which of the following, if true, most seriously weakens the guidebook writer's argument?

(A) The quality of original carpentry in hotels is generally far superior to the quality of original carpentry in other structures, such as houses and stores.

(B) Hotels built since 1930 can generally accommodate more guests than those built before 1930.

(C) The materials available to carpenters working before 1930 were not significantly different in quality from the materials available to carpenters working after 1930.

(D) The better the quality of original carpentry in a building, the less likely that building is to fall into disuse and be demolished.

(E) The average length of apprenticeship for carpenters has declined significantly since 1930.

Argument Evaluation

Situation The original carpentry in hotels built before 1930 shows superior care, skill, and effort to that in hotels built after 1930. This leads to the conclusion that carpenters working on hotels before 1930 were superior in skill, care, and effort to those that came after.

Reasoning *Which option most seriously weakens the argument?* The argument draws an inference from a comparison between carpentry in hotels of different eras to a judgment about the carpenters working on hotels in those eras. One way to weaken this inference is by finding some way in which the carpentry in the hotels may be unrepresentative of the skill, care, and effort of the carpenters working in the eras. The comparison is between the carpentry evident in hotels of the two eras *that still exist*. Thus, if there is some reason to think that hotels with good carpentry survive longer than those with bad carpentry, then still-existing hotels from the older era will have disproportionately more good carpentry, even assuming no difference between the skill, care, and effort of the carpenters from the two eras.

A This option applies equally to both eras, so it has no bearing on the argument.

B It is not clear whether carpenters working on larger hotels would exercise more, less, or the same skill and care as those working on smaller hotels; thus this option does not weaken the argument.

C The argument does not rely, even implicitly, on there being any difference in the quality of materials used in the two eras, so it does not weaken the argument to point out that no such difference exists.

D **Correct.** This weakens the reasoning in the argument by showing a respect in which the comparison between *existing* hotels is unrepresentative.

E The longer a carpenter works as an apprentice, the more skill he or she is apt to have upon becoming a full-fledged carpenter. So this option would tend to slightly strengthen rather than weaken the argument.

The correct answer is D.

CR02997

659. Scientists typically do their most creative work before the age of forty. It is commonly thought that this happens because aging by itself brings about a loss of creative capacity. However, studies show that **of scientists who produce highly creative work beyond the age of forty, a disproportionately large number entered their field at an older age than is usual.** Since by the age of forty the large majority of scientists have been working in their field for at least fifteen years, the studies' finding strongly suggests that the real reason why scientists over forty rarely produce highly creative work is not that they have aged but rather that **scientists over forty have generally spent too long in their field.**

In the argument given, the two portions in boldface play which of the following roles?

(A) The first is a claim, the accuracy of which is at issue in the argument; the second is a conclusion drawn on the basis of that claim.

(B) The first is an objection that has been raised against a position defended in the argument; the second is that position.

(C) The first is evidence that has been used to support an explanation that the argument challenges; the second is that explanation.

(D) The first is evidence that has been used to support an explanation that the argument challenges; the second is a competing explanation that the argument favors.

(E) The first provides evidence to support an explanation that the argument favors; the second is that explanation.

Argument Evaluation

Situation It is generally thought that the reason scientists tend to do their most creative work before age forty is that creative capacity declines with age. Yet those scientists who do creative work after forty tend, disproportionately, to have started their careers in science later in life. So a better explanation is that many scientists over forty have just been at it too long.

Reasoning *What roles do the two portions of the argument that are in boldface play?* The argument describes a phenomenon and what is commonly thought to explain it. Then, the first boldfaced statement introduces evidence that suggests that there may be another explanation. After this evidence is further developed, the argument then concludes that there is indeed a better explanation for the phenomenon; that explanation is stated in the second boldfaced portion.

A The accuracy of the first statement is never called into question by the argument; rather, it is relied upon as the basis for the argument's conclusion.

B The first statement is not an objection against the position the argument defends; instead, it is a basis for that position.

C The first statement is not used to support a position the argument challenges, and the second statement is the explanation the argument supports, not the one it challenges.

D The second statement is indeed an explanation that the argument favors; but the first statement is not used to support a competing explanation that the argument challenges.

E **Correct.** This option correctly identifies the roles played by the boldfaced portions of the argument.

The correct answer is E.

CR03818

660. NorthAir charges low fares for its economy-class seats, but it provides very cramped seating and few amenities. Market research shows that economy passengers would willingly pay more for wider seating and better service, and additional revenue provided by these higher ticket prices would more than cover the additional cost of providing these amenities. Even though NorthAir is searching for ways to improve its profitability, it has decided not to make these improvements.

Which of the following, if true, would most help to explain NorthAir's decision in light of its objectives?

(A) None of NorthAir's competitors offers significantly better seating and service to economy-class passengers than NorthAir does.

(B) On many of the routes that NorthAir flies, it is the only airline to offer direct flights.

(C) A few of NorthAir's economy-class passengers are satisfied with the service they receive, given the low price they pay.

(D) Very few people avoid flying on NorthAir because of the cramped seating and poor service offered in economy class.

(E) The number of people who would be willing to pay the high fares NorthAir charges for its business-class seats would decrease if its economy-class seating were more acceptable.

Evaluation of a Plan

Situation Market research shows that improving some amenities for economy-class passengers would allow NorthAir to raise its economy ticket prices more than enough to cover the additional cost of providing those amenities. But NorthAir has decided not to improve those amenities, even though it is looking for ways to improve its profitability.

Reasoning *What would most help explain why NorthAir decided not to improve the seating and other amenities, even though the resulting increase in economy-class ticket prices would more than cover the expense?* NorthAir is looking for ways to improve its profitability. Making improvements that would increase ticket prices enough to generate more revenue than they cost should improve profitability, other things being equal. But if improving the amenities would generate side effects that reduced profitability, those side effects would provide a good reason for NorthAir's decision not to improve the amenities and hence would help explain why NorthAir made that decision.

A The passage says that for NorthAir, the cost of providing better economy seating and other amenities would be more than met by the increased revenue from the higher ticket prices that passengers would be willing to pay. This could give NorthAir a competitive edge, with improved profitability.

B Even if NorthAir faces little or no competition on certain routes, offering extra amenities might increase passengers' interest in flying those routes. It might also lead passengers to choose NorthAir on other routes that competing airlines also serve. Both of these effects could improve NorthAir's profitability.

C Even if a few NorthAir economy passengers would not pay more for extra amenities, the market research indicates that most of them would, so offering the amenities could still improve NorthAir's profits attributable to economy-class seating.

D This suggests that improving the amenities would not increase the total number of NorthAir passengers. But improving the amenities might still enable the airline to increase its ticket prices per passenger enough to improve its profitability.

E **Correct.** This suggests that improving the economy-class amenities would reduce NorthAir's revenue from sales of business-class tickets, which are likely much more expensive than economy-class tickets. This reduction in revenue could be enough to reduce NorthAir's total profitability despite the increased revenue from economy-class ticket sales.

The correct answer is E.

CR00774

661. Which of the following most logically completes the argument given?

Asthma, a chronic breathing disorder, is significantly more common today among adult competitive swimmers than it is among competitive athletes who specialize in other sports. Although chlorine is now known to be a lung irritant and swimming pool water is generally chlorinated, it would be rash to assume that frequent exposure to chlorine is the explanation of the high incidence of asthma among these swimmers, since _____.

(A) young people who have asthma are no more likely to become competitive athletes than are young people who do not have asthma

(B) competitive athletes who specialize in sports other than swimming are rarely exposed to chlorine

(C) competitive athletes as a group have a significantly lower incidence of asthma than do people who do not participate in competitive athletics

(D) until a few years ago, physicians routinely recommended competitive swimming to children with asthma, in the belief that this form of exercise could alleviate asthma symptoms

(E) many people have asthma without knowing they have it and thus are not diagnosed with the condition until they begin engaging in very strenuous activities, such as competitive athletics

Argument Construction

Situation Asthma is more common among competitive swimmers than among other competitive athletes. Chlorine is a lung irritant generally present in swimming pool water.

Reasoning *What would cast doubt on the hypothesis that exposure to chlorine in swimming pools accounts for the high incidence of asthma among adult competitive swimmers?* Evidence of any other factor that would provide an alternative explanation of why asthma is more common among adult competitive swimmers than among other competitive athletes would make it rash to assume that frequent exposure to chlorine explains the high incidence of asthma among these swimmers, so a statement providing such evidence would logically fill in the blank at the end of the passage to complete the argument.

A This might help explain why competitive athletes in general are not especially likely to have asthma, but it does not explain why adult competitive swimmers are more likely to have asthma than other competitive athletes are.

B This provides additional evidence that exposure to chlorine explains why adult competitive swimmers are more likely to have asthma than other competitive athletes are, so it does not cast doubt on that hypothesis.

C A lower incidence of asthma among competitive athletes than among nonathletes does not help explain the higher incidence of asthma among adult competitive swimmers than among other competitive athletes.

D **Correct.** Routinely encouraging children with asthma to take up competitive swimming would likely have made the proportion of adult competitive swimmers with asthma exceed the proportion of other competitive athletes with asthma, even if chlorine in swimming pool water never causes asthma in swimmers.

E This might help explain why people with asthma are just as likely as other people to become competitive athletes, but it does not help explain why adult competitive swimmers are more likely to have asthma than other competitive athletes are.

The correct answer is D.

CR01289

662. In the country of Marut, the Foreign Trade Agency's records were reviewed in 1994 in light of information then newly available about neighboring Goro. The review revealed that in every year since 1963, the agency's projection of what Goro's gross national product (GNP) would be five years later was a serious underestimate. The review also revealed that in every year since 1963, the agency estimated Goro's GNP for the previous year—a Goro state secret—very accurately.

Of the following claims, which is most strongly supported by the statements given?

(A) Goro's GNP fluctuated greatly between 1963 and 1994.

(B) Prior to 1995, Goro had not released data intended to mislead the agency in making its five-year projections.

(C) The amount by which the agency underestimated the GNP it projected for Goro tended to increase over time.

(D) Even before the new information came to light, the agency had reason to think that at least some of the five-year projections it had made were inaccurate.

(E) The agency's five-year projections of Goro's GNP had no impact on economic planning in Marut.

Argument Construction

Situation A review in 1994 revealed that every year since 1963, Marut's Foreign Trade Agency had seriously underestimated what Goro's GNP would be five years later, but accurately estimated what Goro's GNP had been the previous year.

Reasoning *What conclusion do the stated facts most strongly support?* Goro's GNP in each year at least from 1969 through 1993 had been seriously underestimated by the agency five years in advance, yet was then accurately estimated by the agency one year after the fact. It follows that for each of these years, the agency's earlier projection of Goro's GNP must have been much lower than its later estimate.

A This is not supported by the information given. The fact that the agency consistently underestimated each year's GNP in its five-year projections and then correctly estimated it after the fact does not indicate that Goro's GNP fluctuated greatly.

B This is not supported by the information given. The reason the agency's five-year projections were inaccurate might well have been that Goro deliberately released data intended to mislead the agency in making those projections.

C This is not supported by the information given. The fact that the underestimates remained large throughout the years in question does not indicate that the underestimates increased over time.

D **Correct.** As explained above, for many years there were serious discrepancies between the agency's five-year projections of Goro's GNP and its retrospective estimates of each previous year's trade. In any year at least from 1970 through 1993, these discrepancies, if noticed, would have given the agency reason to doubt some of the five-year projections.

E This is not supported by the information given. Even though at least some of the five-year projections were eventually known to be serious underestimates, they could still have affected Marut's economic planning. The economic planners might have retained an unreasonable faith in the accuracy of the most recent projections.

The correct answer is D.

CR05082

663. Vargonia has just introduced a legal requirement that student-teacher ratios in government-funded schools not exceed a certain limit. All Vargonian children are entitled to education, free of charge, in these schools. When a recession occurs and average incomes fall, the number of children enrolled in government-funded schools tends to increase. Therefore, though most employment opportunities contract in economic recessions, getting a teaching job in Vargonia's government-funded schools will not be made more difficult by a recession.

Which of the following would be most important to determine in order to evaluate the argument?

(A) Whether in Vargonia there are any schools not funded by the government that offer children an education free of charge

(B) Whether the number of qualified applicants for teaching positions in government-funded schools increases significantly during economic recessions

(C) What the current student-teacher ratio in Vargonia's government-funded schools is

(D) What proportion of Vargonia's workers currently hold jobs as teachers in government-funded schools

(E) Whether in the past a number of government-funded schools in Vargonia have had student-teacher ratios well in excess of the new limit

Argument Evaluation

Situation During a recession, the number of children in government-funded schools in Vargonia tends to increase. Vargonian children are entitled to a free education in these schools. A new law requires student-teacher ratios in these schools to remain below a certain limit.

Reasoning *Which of the five questions would provide us with the best information for evaluating the argument?* The argument's conclusion is that recessions do not make teaching jobs in Vargonia's government-funded schools harder to get. During recessions, the reasoning goes, more students will enroll in Vargonia's government-funded schools than in nonrecession times. Implicit in the argument is the thought that, because the new law sets an upper limit on the average number of students per teacher, schools that get an influx of new students would have to hire more teachers. During a recession, however, there might be much more competition in the labor market for teachers because many more qualified people are applying for teaching jobs.

A This information is not significant in the context of the argument, which does not need to assume that only government-funded schools provide free education.

B **Correct.** Getting an answer to this question would provide us with specific information useful in evaluating the argument. A "yes" answer to this question would suggest that competition for teaching jobs in Vargonian government-funded schools would be keener during recessions. A "no" answer would suggest that the level of competition would decrease during recessions.

C Discovering the current student-teacher ratio in Vargonia's schools would be of no value, by itself, in evaluating the argument. We do not know what the new upper limit on the student-teacher ratio is, and we do not know whether Vargonia is currently in a recession.

D Finding out whether the proportion this refers to is 1 percent, for example, or 4 percent, would tell us nothing about whether getting teaching jobs at government-funded schools in Vargonia becomes more difficult during a recession. Among other things, we do not know whether Vargonia is currently in a recession, and we do not know what proportion of Vargonia's workers would be qualified candidates for teaching jobs.

E This is of no relevance in evaluating the argument because, presumably, the new limit on student-teacher ratios will be complied with. Thus, even if student-teacher ratios in the past would have exceeded the new limit, the argument concerns whether, *in the future*, getting a teaching job in Vargonia's government-funded schools will be made more difficult by a recession.

The correct answer is B.

CR09951

664. In Colorado subalpine meadows, nonnative dandelions co-occur with a native flower, the larkspur. Bumblebees visit both species, creating the potential for interactions between the two species with respect to pollination. In a recent study, researchers selected 16 plots containing both species; all dandelions were removed from eight plots; the remaining eight control plots were left undisturbed. The control plots yielded significantly more larkspur seeds than the dandelion-free plots, leading the researchers to conclude that the presence of dandelions facilitates pollination (and hence seed production) in the native species by attracting more pollinators to the mixed plots.

Which of the following, if true, most seriously undermines the researchers' reasoning?

(A) Bumblebees preferentially visit dandelions over larkspurs in mixed plots.

(B) In mixed plots, pollinators can transfer pollen from one species to another to augment seed production.

(C) If left unchecked, nonnative species like dandelions quickly crowd out native species.

(D) Seed germination is a more reliable measure of a species' fitness than seed production.

(E) Soil disturbances can result in fewer blooms, and hence lower seed production.

Argument Evaluation

Situation Bumblebees visit both larkspur and dandelions in certain meadows. A study found that more larkspur seeds were produced in meadow plots in which both larkspur and dandelions grew than in similar plots from which all dandelions had been removed. The researchers inferred that dandelions facilitate larkspur pollination.

Reasoning *What evidence would cast the most doubt on the inference from the study's findings to the conclusion that dandelions facilitate larkspur pollination by attracting more pollinators?* The argument assumes that the only relevant difference between the two types of plots was whether dandelions were present. Evidence that the plots differed in some other way that could provide a plausible alternative explanation of why more larkspur seeds were produced in the plots with dandelions would weaken the argument.

A This would suggest that the larkspur pollination should have been lower in the plots with dandelions, so it does not provide a plausible alternative explanation for the study's findings.

B This is fully compatible with the claim that the dandelions attracted more pollinators to the mixed plots, and it would also help to support the argument's conclusion that dandelions facilitated larkspur pollination in those plots.

C Although this suggests that the mixed plots won't remain mixed for long, it does not provide a plausible alternative explanation for the study's finding that larkspur seed production was higher in the mixed plots.

D The argument is not about how fit larkspurs are as a species, but about why they produced different numbers of seeds in the different plots.

E **Correct.** This provides a plausible alternative explanation for why larkspur seed production was lower in the plots from which dandelions had been removed, since digging them out would have disturbed the soil.

The correct answer is E.

CR11453

665. An experiment was done in which human subjects recognize a pattern within a matrix of abstract designs and then select another design that completes that pattern. The results of the experiment were surprising. The lowest expenditure of energy in neurons in the brain was found in those subjects who performed most successfully in the experiments.

Which of the following hypotheses best accounts for the findings of the experiment?

(A) The neurons of the brain react less when a subject is trying to recognize patterns than when the subject is doing other kinds of reasoning.

(B) Those who performed best in the experiment experienced more satisfaction when working with abstract patterns than did those who performed less well.

(C) People who are better at abstract pattern recognition have more energy-efficient neural connections.

(D) The energy expenditure of the subjects' brains increases when a design that completes the initially recognized pattern is determined.

(E) The task of completing a given design is more capably performed by athletes, whose energy expenditure is lower when they are at rest.

Argument Construction

Situation Experimental subjects worked with pattern recognition and completion. The subjects who performed best showed the lowest expenditure of energy in neurons in the brain.

Reasoning *Which hypothesis best accounts for the findings?* In order to account for the findings, the hypothesis must suggest a plausible link between successful performance and the energy expenditure of neurons in the brain. Consider each answer choice, and evaluate its plausibility and logic. Where is there a reasonably direct relationship between the given factors and the conclusion that is drawn? Understand that hypotheses based on factors not included in the experiment cannot be used to account for the findings.

A The experiment did not compare types of reasoning so this hypothesis does not account for the results.

B No information is provided about subjects' satisfaction, so this hypothesis is not warranted.

C **Correct.** This statement properly identifies a hypothesis that connects subjects' performance with their energy expenditure and so could account for the experiment's results.

D The most successful subjects would presumably not have completed fewer patterns than average, so the posited increase in energy would likely lead to higher energy expenditures for them, not lower.

E No information is offered on the subjects, so no hypothesis about athletes is warranted.

The correct answer is C.

CR01202

666. With seventeen casinos, Moneyland operates the most casinos in a certain state. Although intent on expanding, it was outmaneuvered by Apex Casinos in negotiations to acquire the Eldorado chain. To complete its acquisition of Eldorado, Apex must sell five casinos to comply with a state law forbidding any owner to operate more than one casino per county. Since Apex will still be left operating twenty casinos in the state, it will then have the most casinos in the state.

Which of the following, if true, most seriously undermines the prediction?

(A) Apex, Eldorado, and Moneyland are the only organizations licensed to operate casinos in the state.

(B) The majority of Eldorado's casinos in the state will need extensive renovations if they are to continue to operate profitably.

(C) Some of the state's counties do not permit casinos.

(D) Moneyland already operates casinos in the majority of the state's counties.

(E) Apex will use funds it obtains from the sale of the five casinos to help fund its acquisition of the Eldorado chain.

Argument Evaluation

Situation Moneyland operates seventeen casinos, the most in a certain state, and is intent on expanding. Another operator, Apex Casinos, is acquiring the Eldorado casino chain, but must sell five casinos to comply with a state law forbidding any owner to operate more than one casino per county. After these transactions, Apex will operate twenty casinos in the state.

Reasoning *What observation would cast the most doubt on the prediction that Apex will have the most casinos in the state after the transactions?* Apex will operate twenty casinos, whereas Moneyland now operates just seventeen, and no one else operates even that many. It follows that Apex will operate more casinos after its transactions than Moneyland or any other one owner now operates. However, if Moneyland also acquires three or more casinos during the transactions, then Apex will not have the most casinos in the state afterward. Thus, any observation suggesting that Moneyland is about to acquire several casinos would undermine the prediction.

A **Correct.** Since Apex is acquiring Eldorado, Moneyland and Apex will be the only remaining licensed casino operators in the state. Therefore, Moneyland is the only likely buyer for the five casinos Apex needs to sell. So Moneyland is likely to acquire the five casinos during the sale and end up with twenty-two casinos—more than Apex.

B This does not undermine the prediction. Even if the Eldorado casinos cannot operate profitably for long without extensive renovations, Apex will still have twenty casinos immediately after its transactions.

C This supports rather than undermines the prediction. If fewer counties permit casinos, there will be fewer opportunities for Moneyland or any other operator to acquire more casinos to surpass the twenty Apex will own.

D This supports rather than undermines the prediction. If Moneyland's seventeen casinos are in most of the state's counties already, then there are fewer counties in which Moneyland could acquire additional casinos to surpass the twenty Apex will own.

E This supports rather than undermines the prediction. Apex's use of the funds from selling the five casinos to acquire the Eldorado chain will not help anyone else to acquire more casinos to surpass the twenty Apex will own.

The correct answer is A.

CR05093

667. It is widely assumed that people need to engage in intellectual activities such as solving crossword puzzles or mathematics problems in order to maintain mental sharpness as they age. In fact, however, simply talking to other people—that is, participating in social interaction, which engages many mental and perceptual skills—suffices. Evidence to this effect comes from a study showing that the more social contact people report, the better their mental skills.

Which of the following, if true, most seriously weakens the force of the evidence cited?

(A) As people grow older, they are often advised to keep exercising their physical and mental capacities in order to maintain or improve them.

(B) Many medical conditions and treatments that adversely affect a person's mental sharpness also tend to increase that person's social isolation.

(C) Many people are proficient both in social interactions and in solving mathematical problems.

(D) The study did not itself collect data but analyzed data bearing on the issue from prior studies.

(E) The tasks evaluating mental sharpness for which data were compiled by the study were more akin to mathematics problems than to conversation.

Argument Evaluation

Situation A study shows that the more social contact people report, the better their mental skills are, so engaging in social interaction is sufficient for maintaining mental sharpness.

Reasoning *What would suggest that the study does not establish the truth of the conclusion?* The study shows a correlation between mental sharpness and social interaction but does not indicate why this correlation exists. Evidence that mental sharpness contributes to social interaction or that some third factor affects both mental sharpness and social interaction, could provide an alternative explanation for the correlation and thus cast doubt on the explanation that social interaction contributes to mental sharpness.

A People are often wrongly advised to do things that are not actually beneficial. And even if exercising mental capacities does help to maintain them, the passage says that social interaction provides such exercise.

B **Correct.** This provides evidence that the correlation observed in the study results from mental sharpness facilitating social interaction, in which case the study results do not indicate that social interaction facilitates mental sharpness.

C This would be expected, given the argument's conclusion that social interaction helps to maintain better mental skills overall.

D A study that analyzes data from prior studies can provide evidence just as well as a study that collects its own data can.

E The argument's conclusion would be compatible with this observation, and would then suggest that social interaction contributes to the mental sharpness needed for tasks similar to math problems.

The correct answer is B.